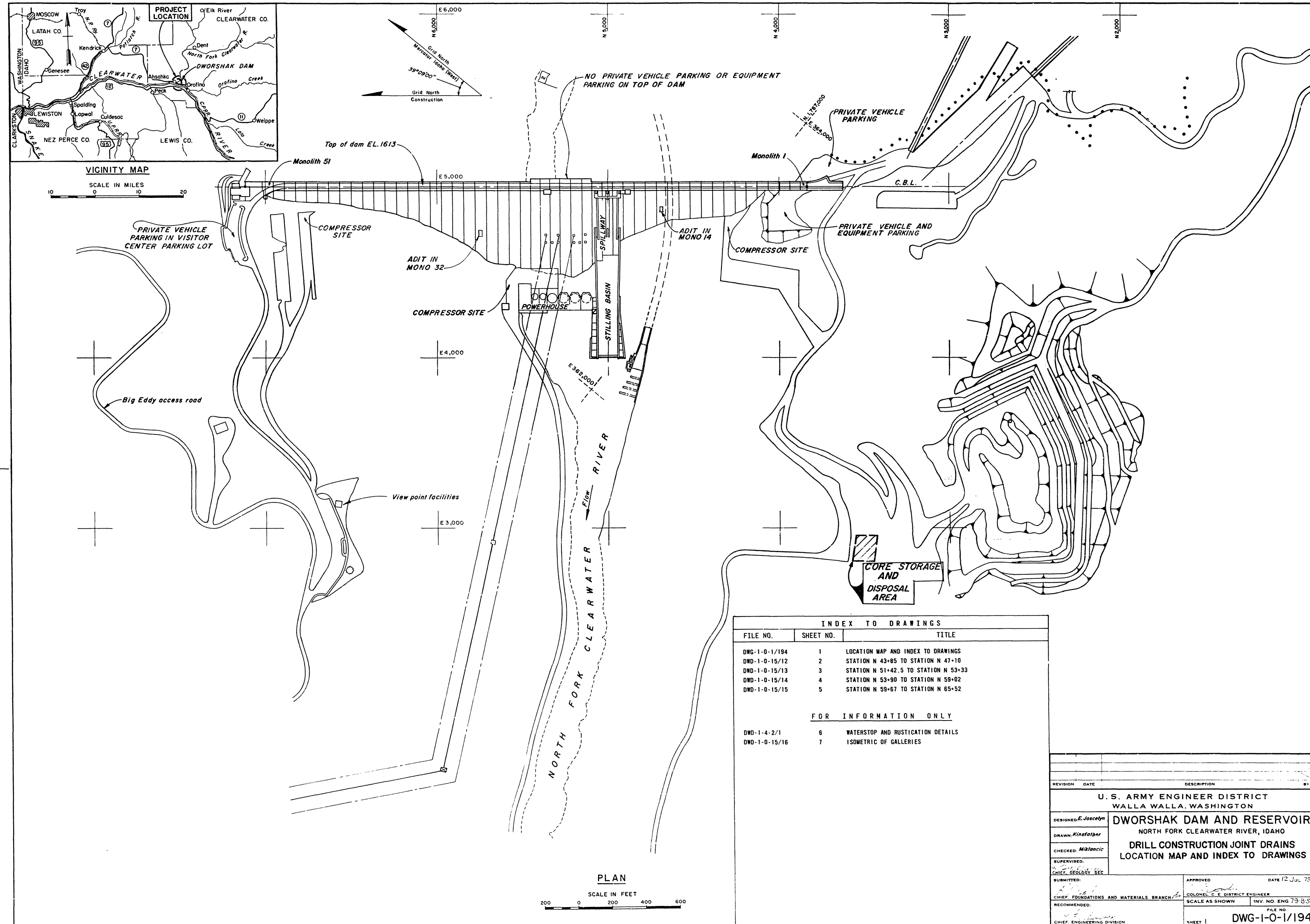
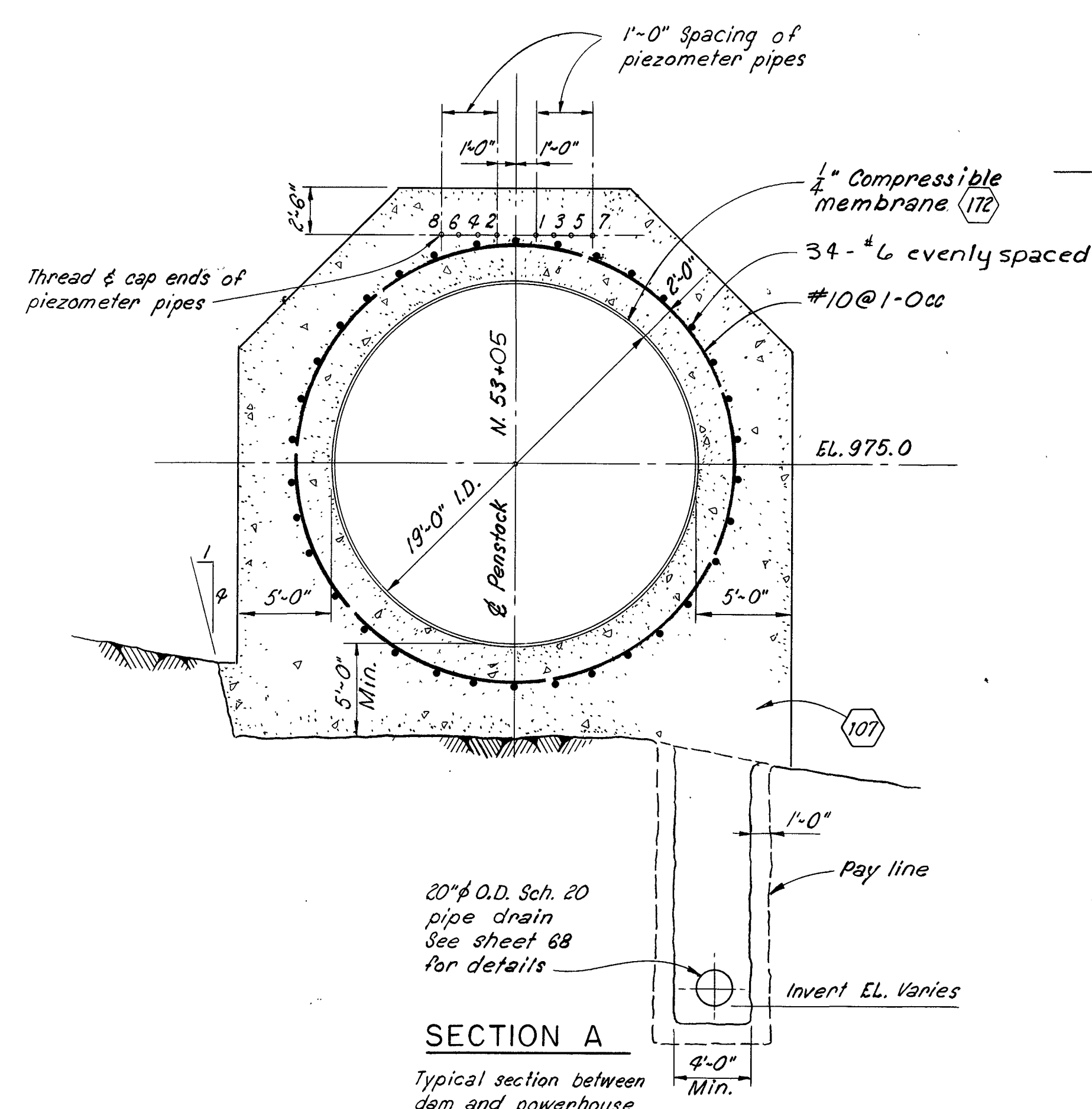
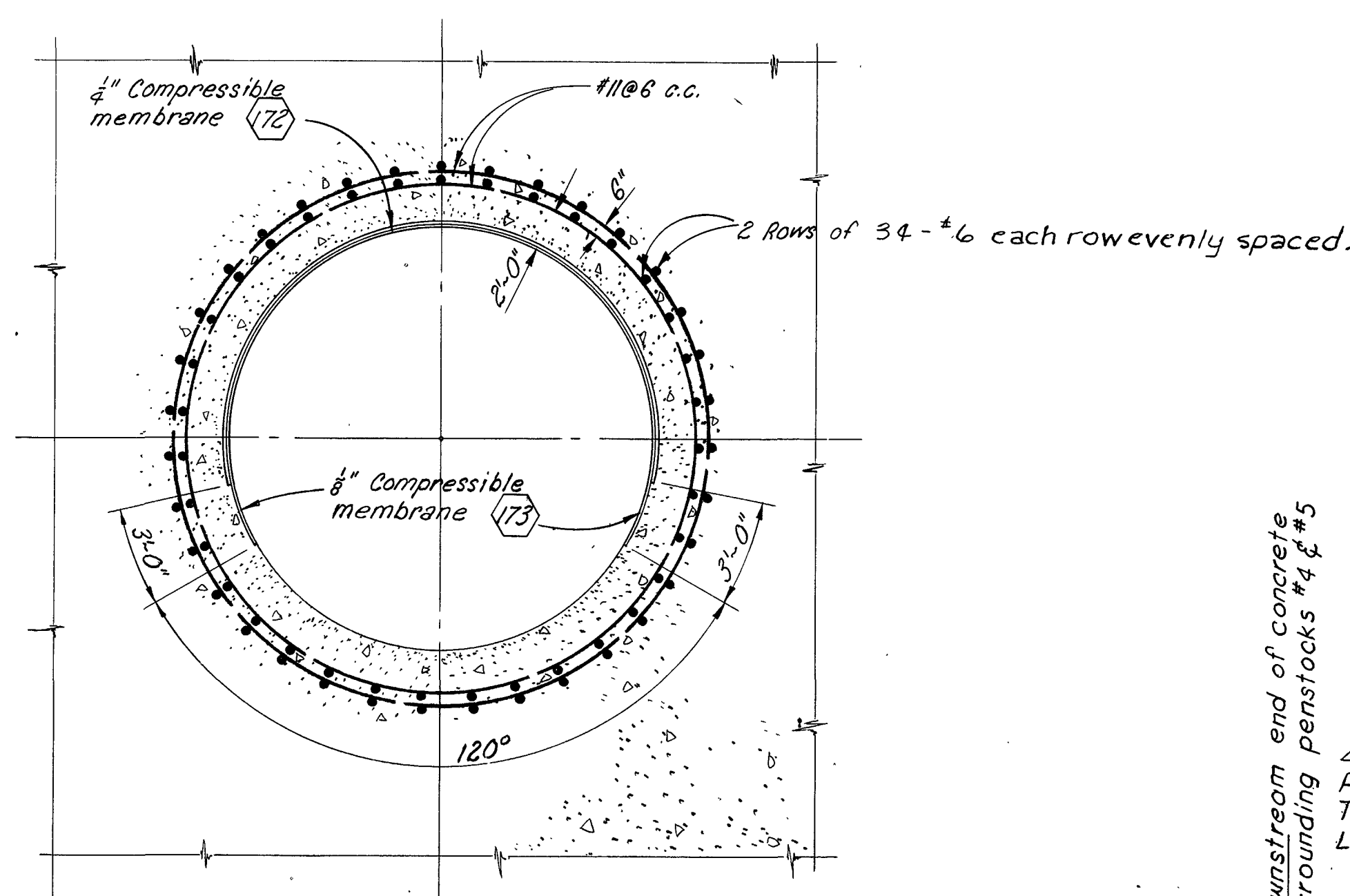
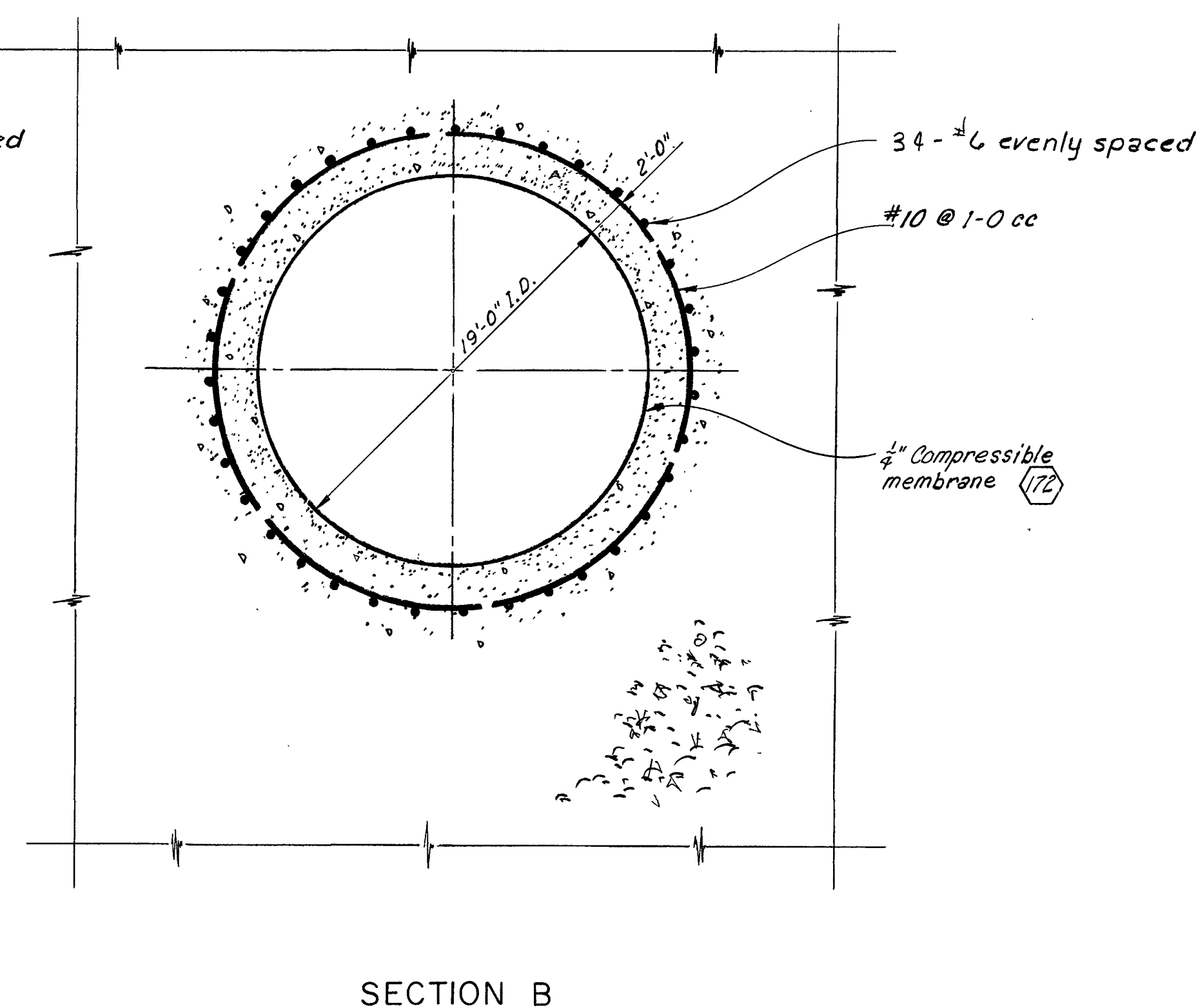






SCALE IN FEET

0 6 12 18

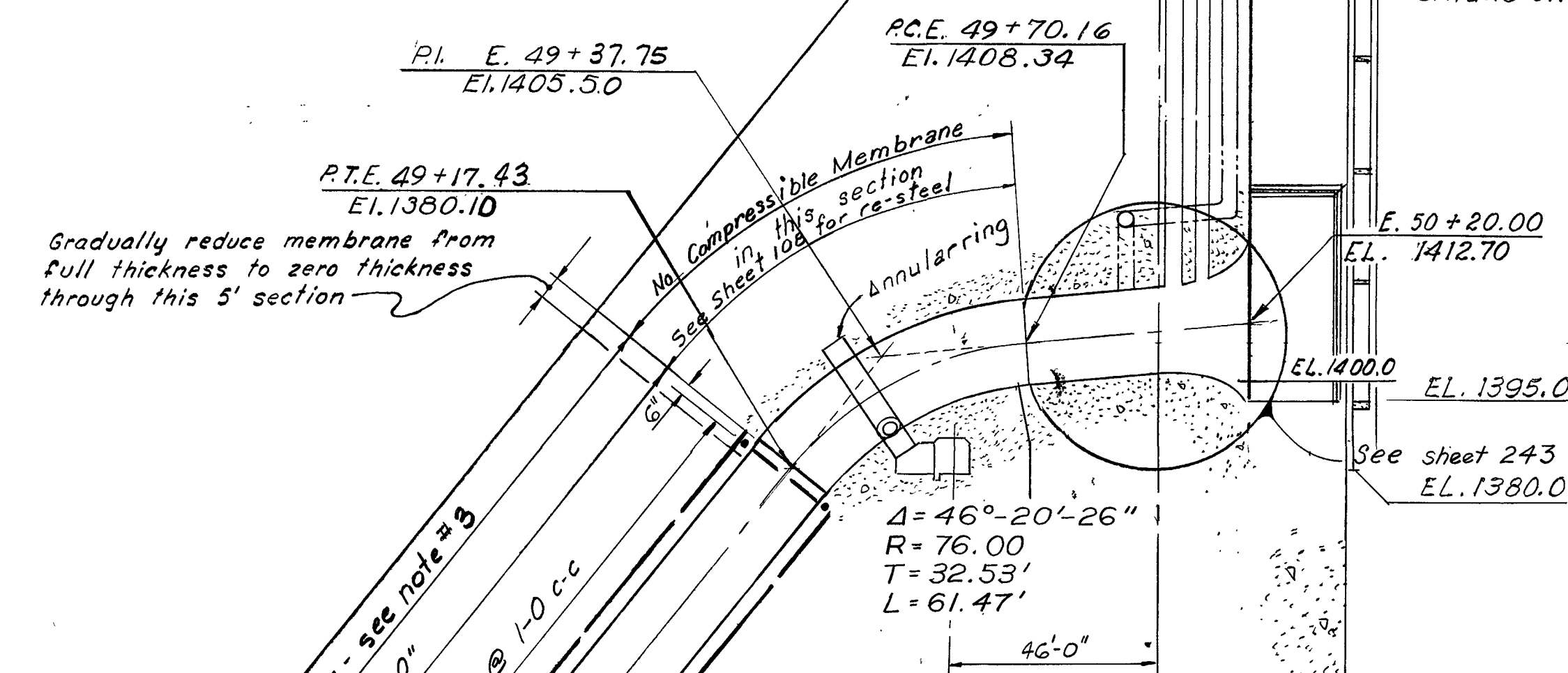
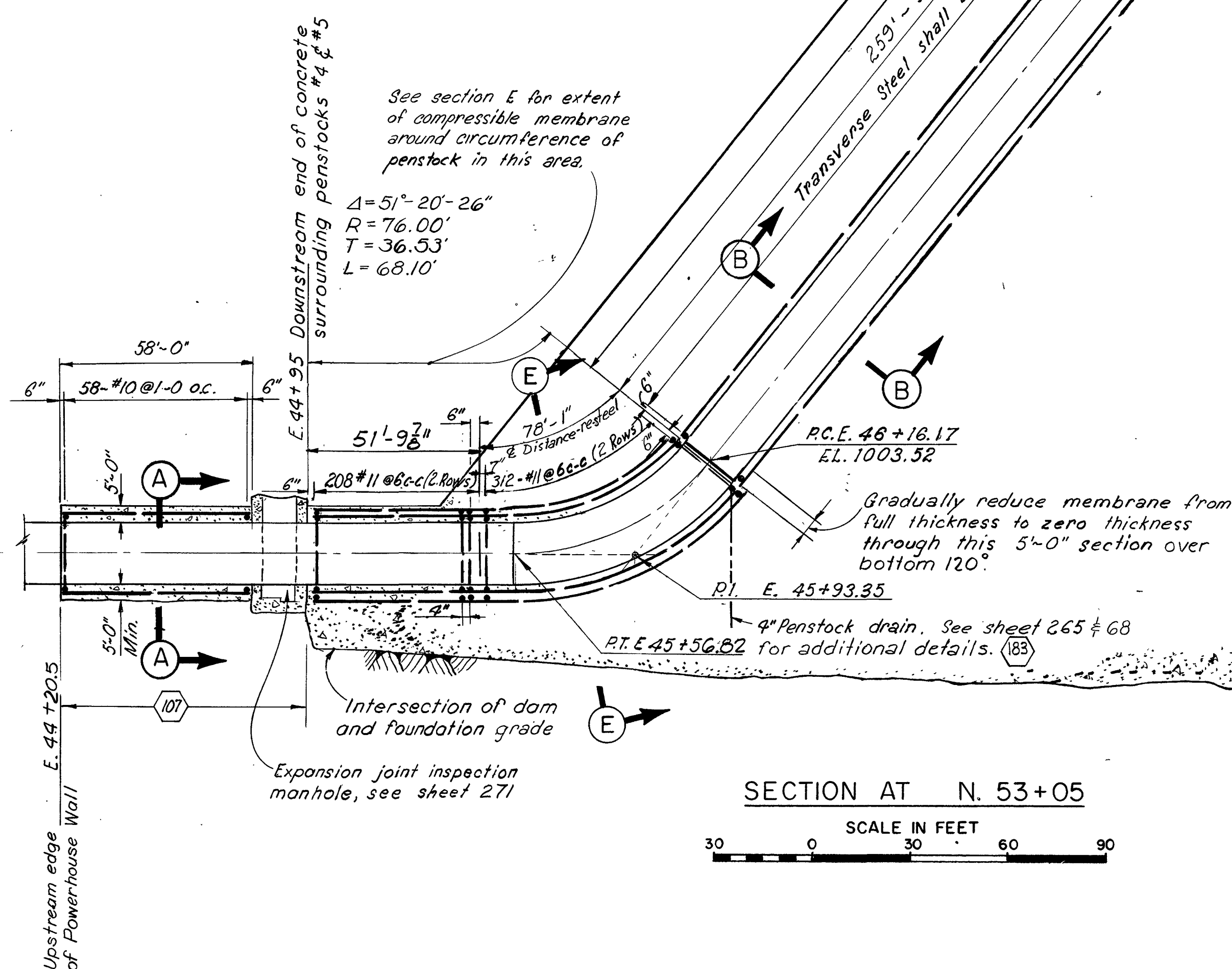


EL. 975

Upstream edge of Powerhouse Wall

E. 43 + 58

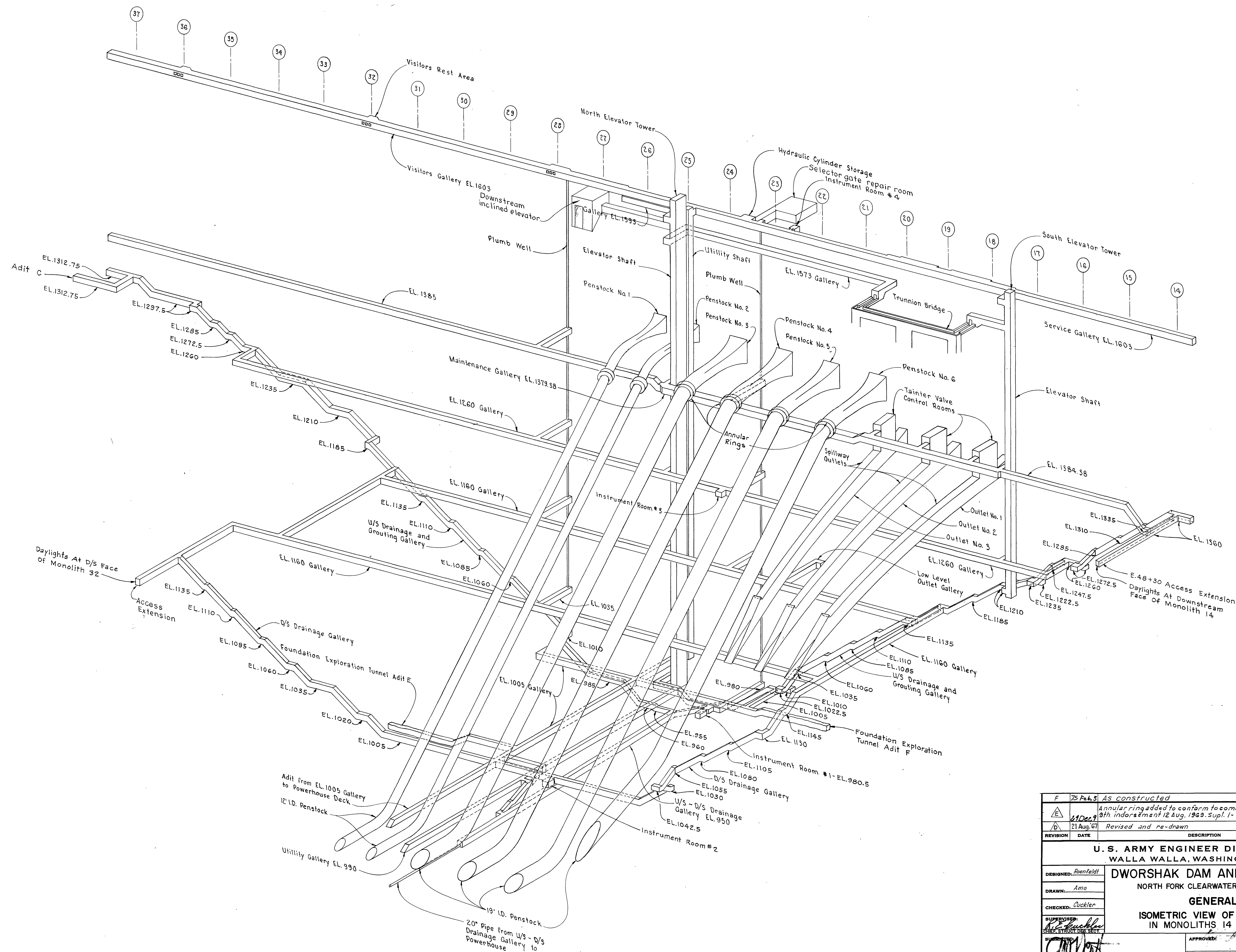
19' Dia Turbine



- NOTE:
1. 19 Foot diameter Penstocks of N. 52+40 and 51+75 shall be similarly reinforced.
 2. All resteel shown on this drawing shall conform to intermediate strength Siller or Axle steel, (40,000 yield)
 3. Penstock between bends and from inspection manhole at toe of dam to Powerhouse shall be completely wrapped in a 1/2" compressible membrane including stiffener rings. See sheet 264. See specifications for physical requirements for membrane.

REVISION	DATE	DESCRIPTION	BY
G	75 MAY 01	As Constructed	ADA
F	70 Nov 25	Added Multi-Level Power Intake Structure	ADA
E	41 Dec 21	Annular ring added to conform to comments as per N.P.D.	ADA
D	21 Aug 67	Relocated drain gallery.	ADA
C	9 Jan 67	Revised penstock reducer, minor revisions.	ADA
B	6 June 66	Added compressible membrane on sections. Added bid items	ADA
A	9 May 66	Revised section and note	ADA

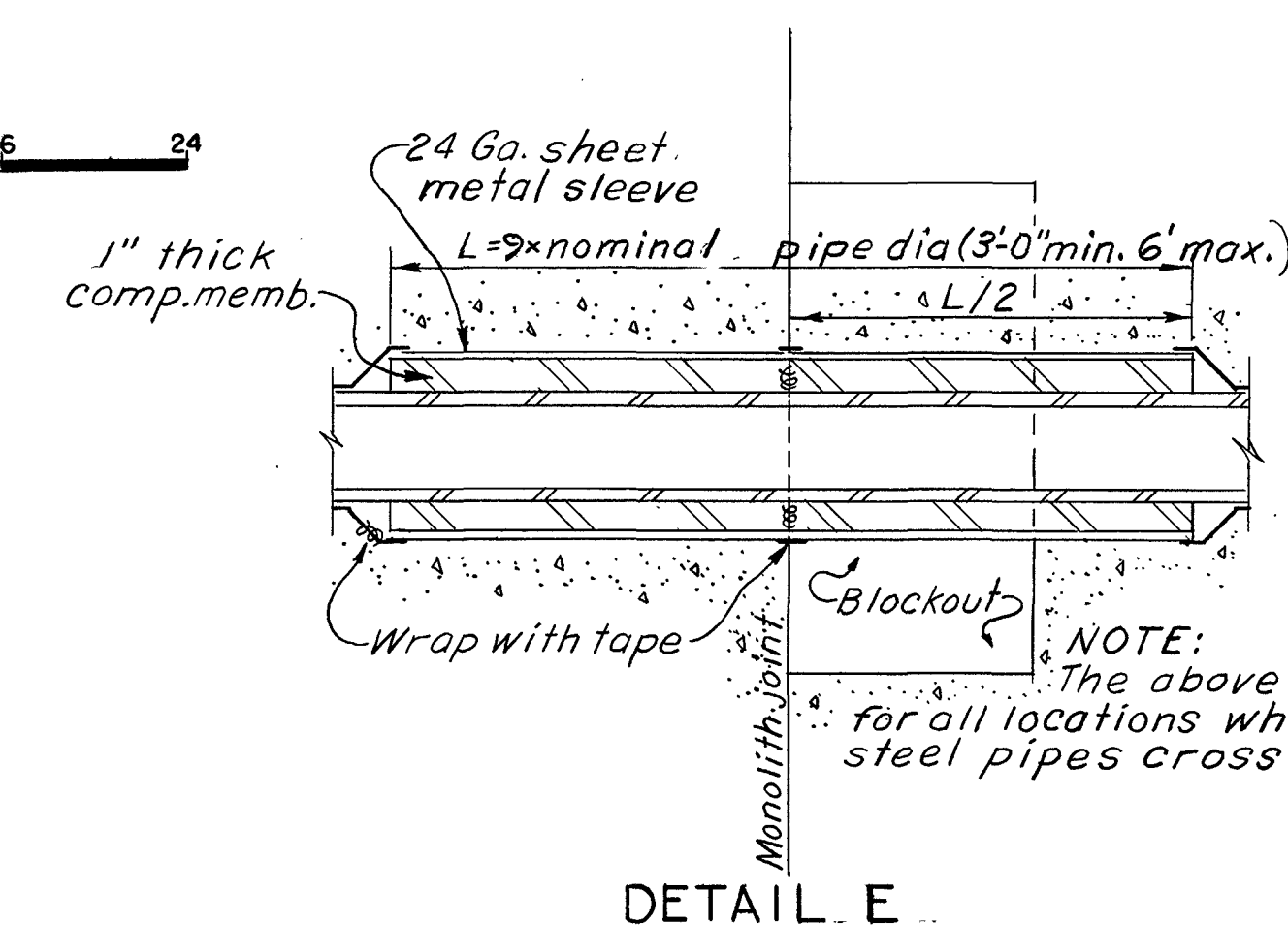
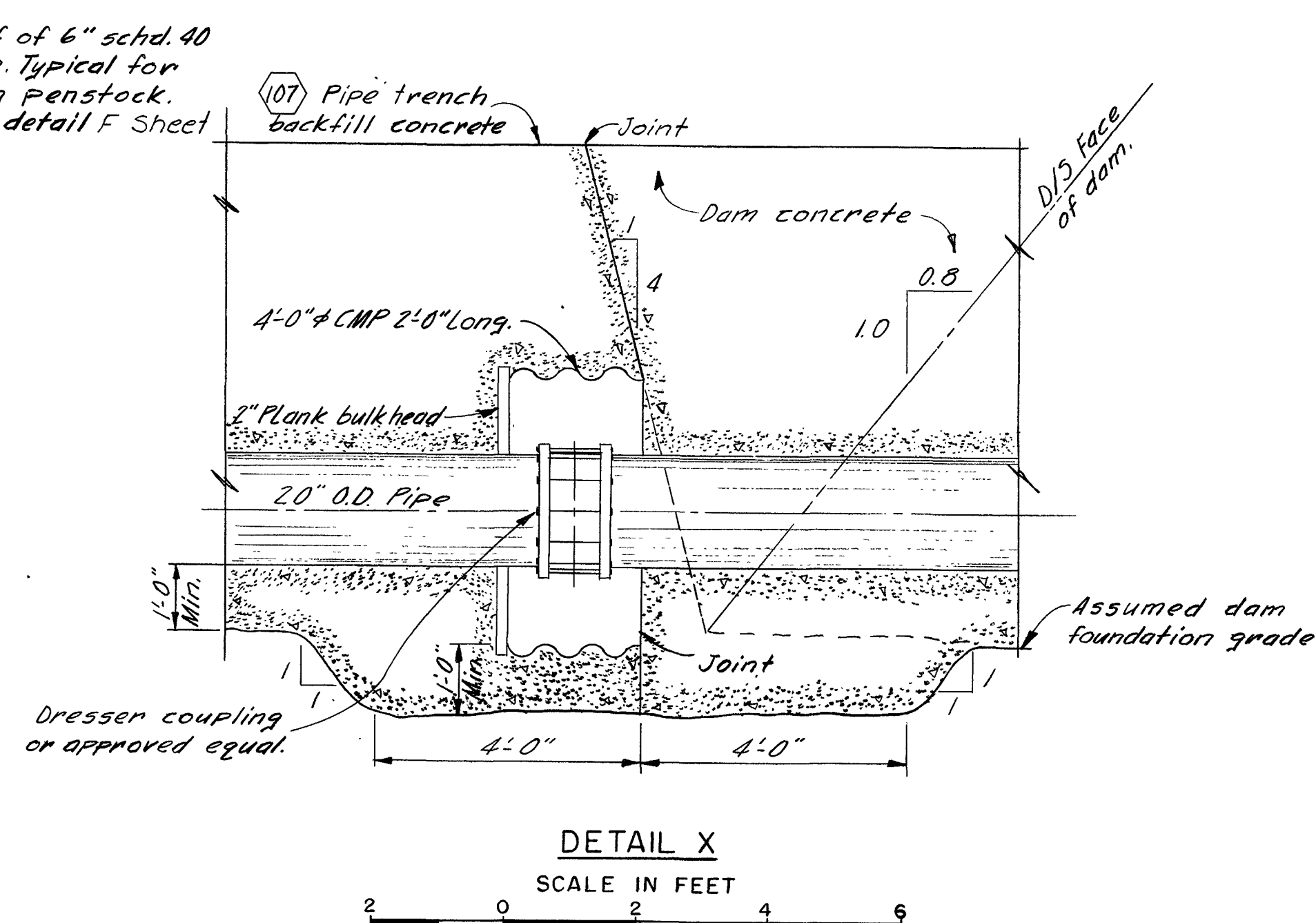
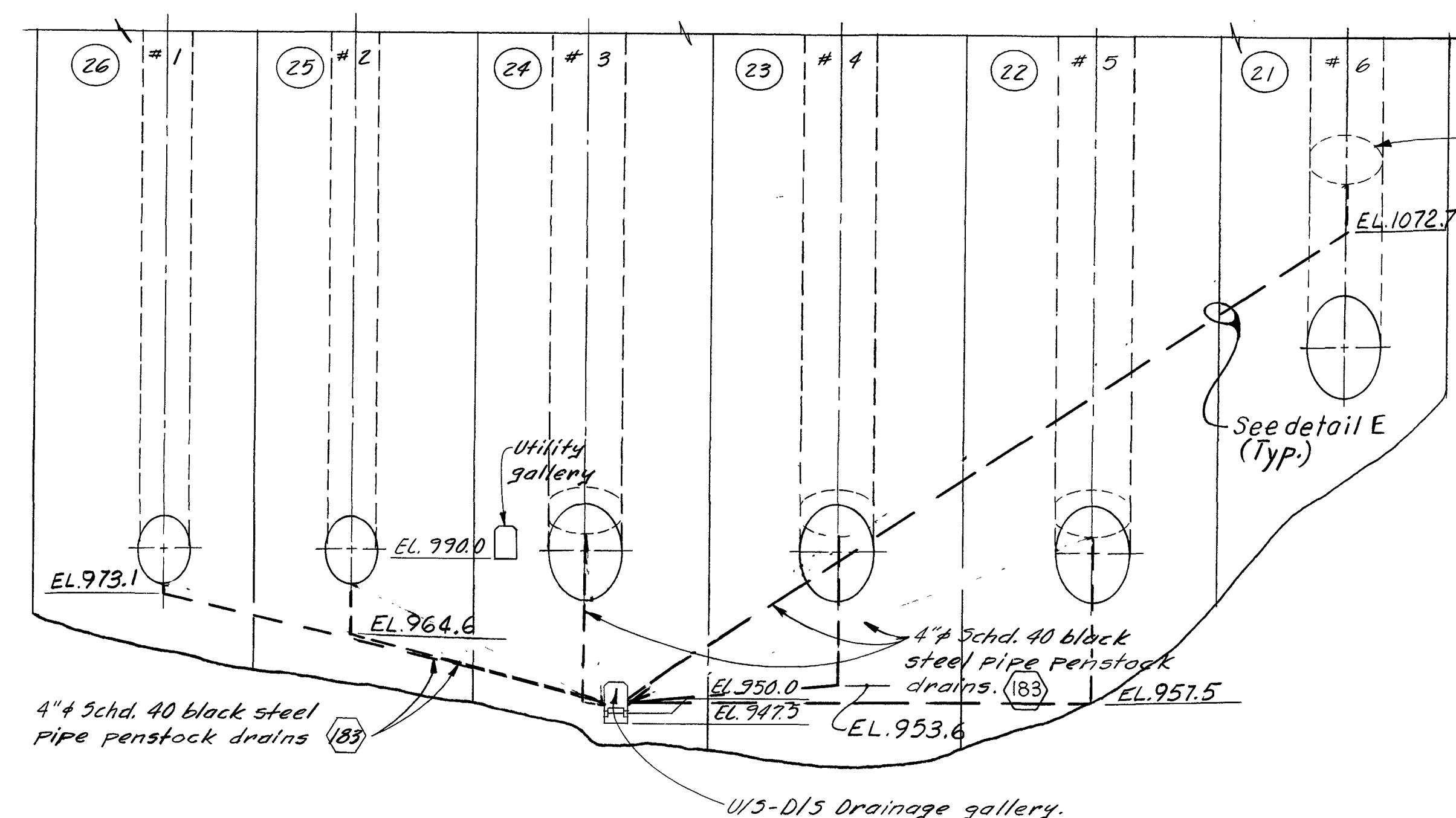
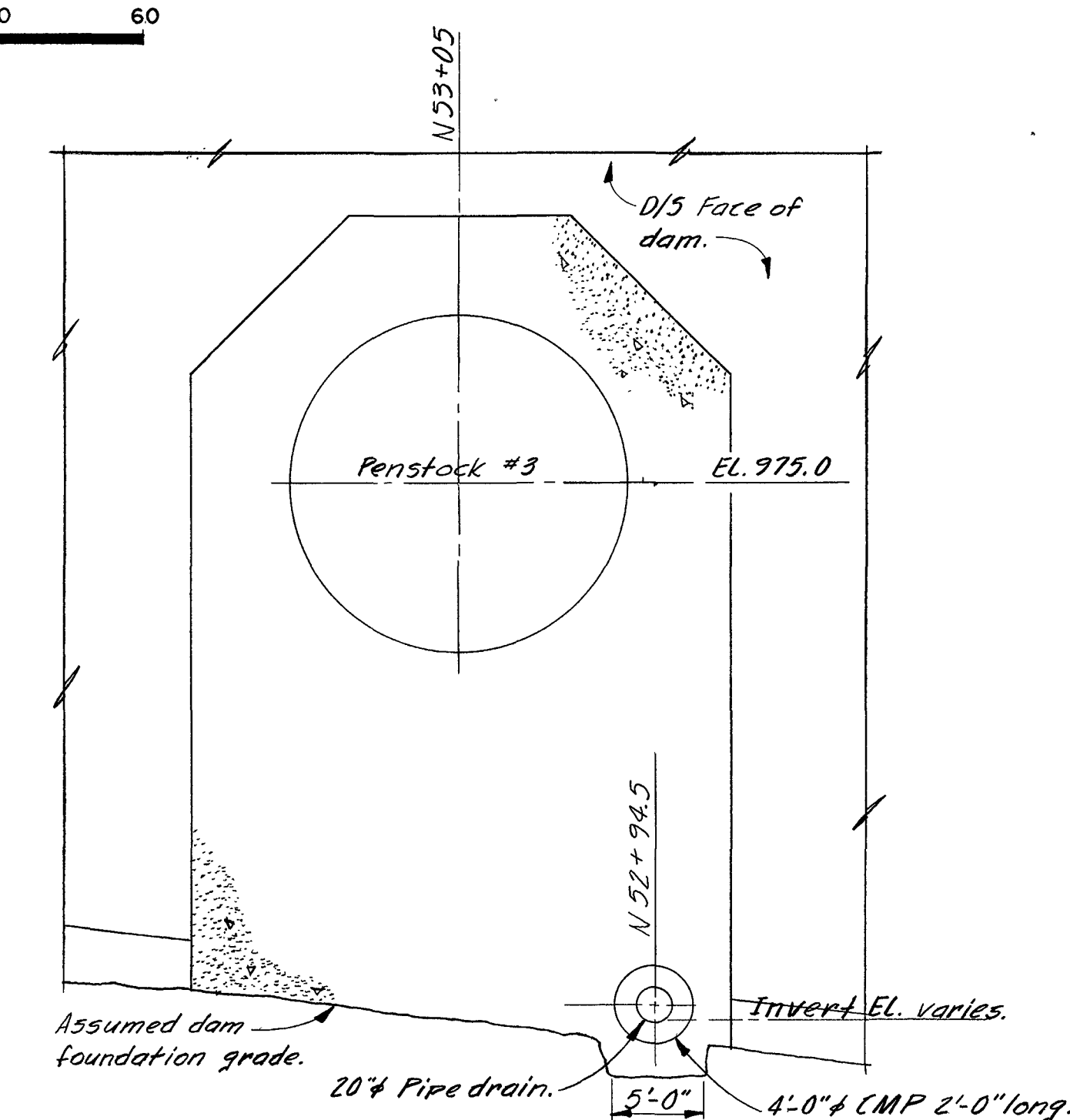
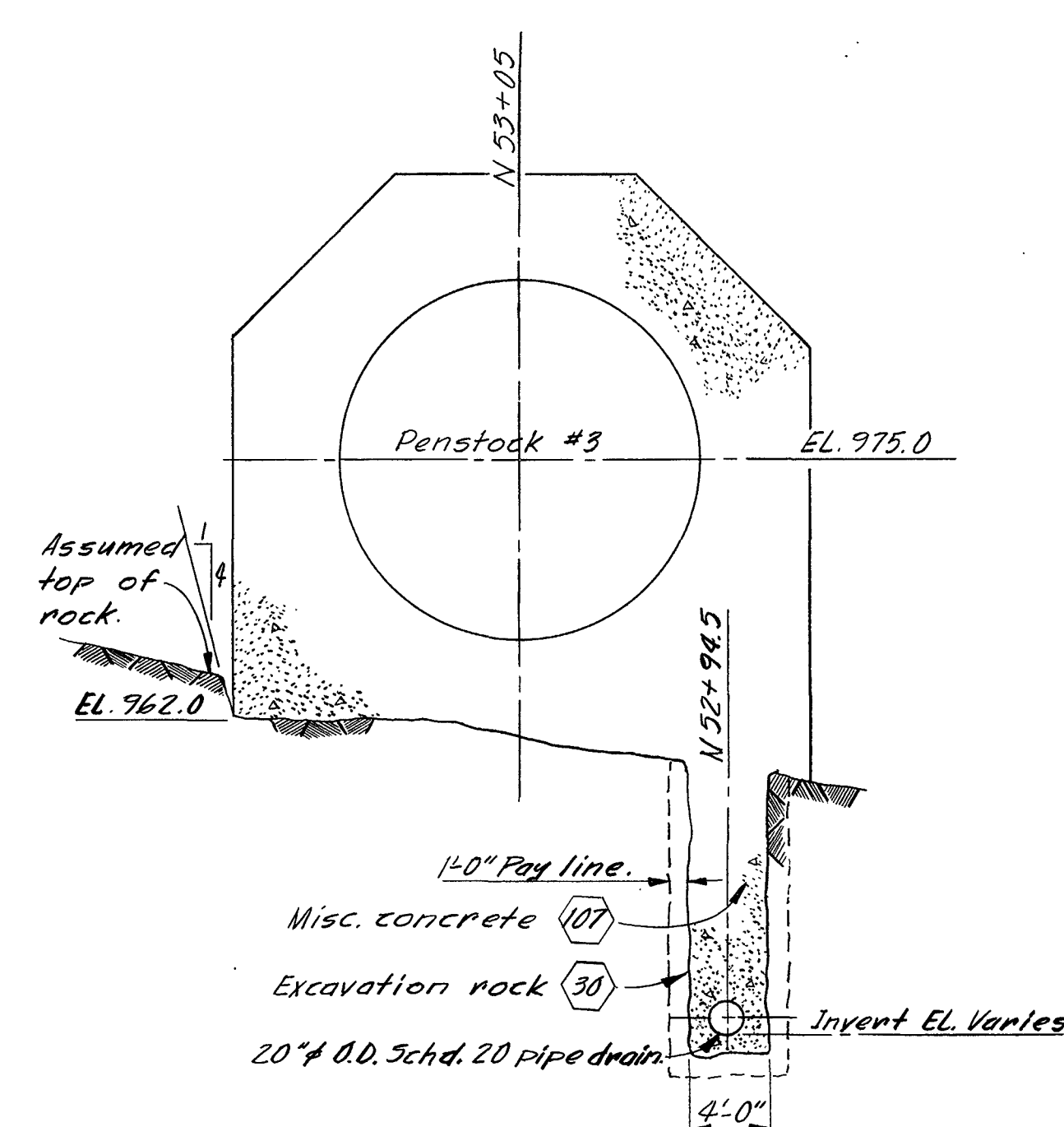
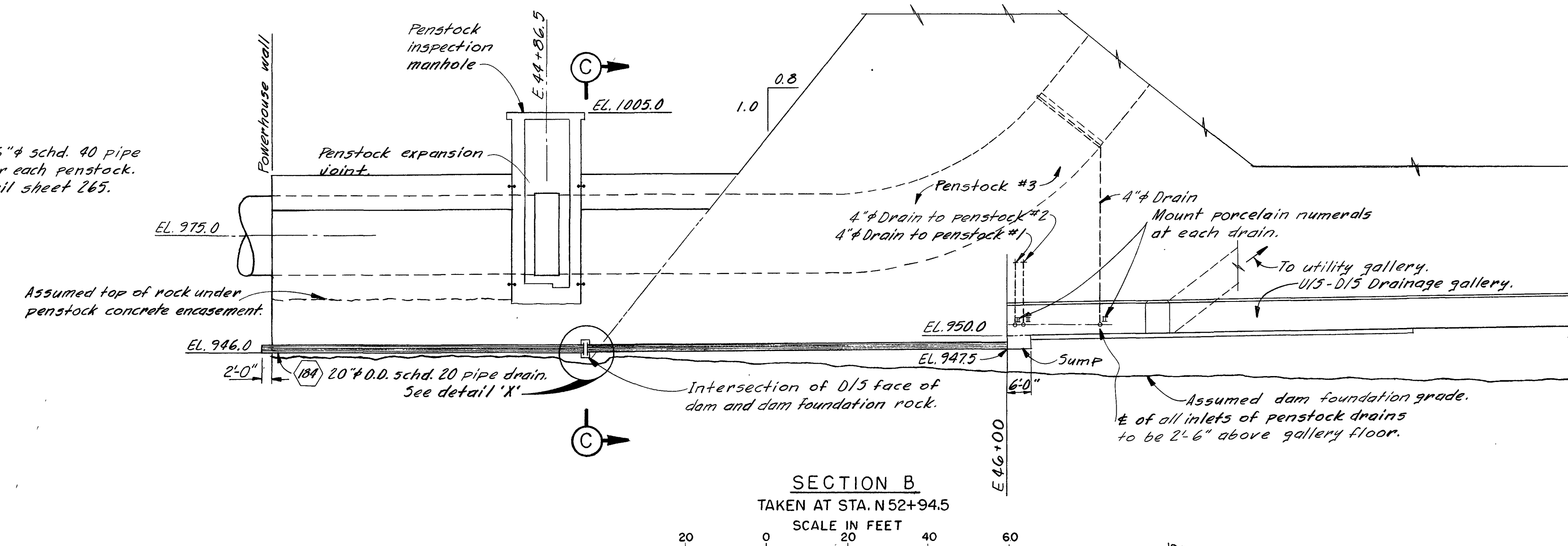
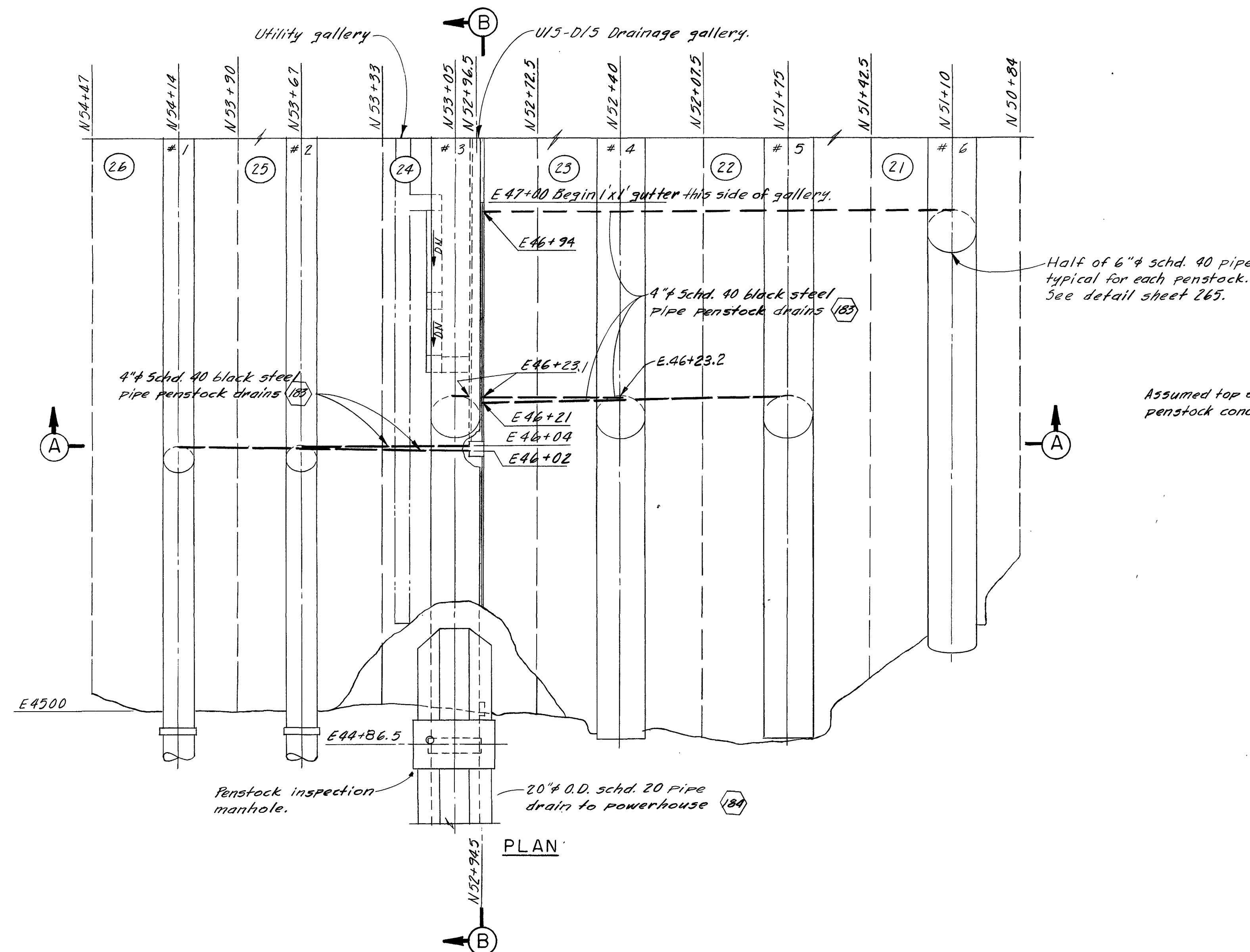
U. S. ARMY ENGINEER DISTRICT WALLA WALLA, WASHINGTON	
DESIGNED: LANKER	DWORSHAK DAM AND RESERVOIR
DRAWN: S.A.D.	NORTH FORK CLEARWATER RIVER, IDAHO
CHECKED: CUCKLER	PENSTOCKS
SUPERVISED: S.H. HAN	UNITS 3, 4 AND 5
SUBMITTED: S.H. HAN	REINFORCEMENT AND SECTIONS
CHIEF, DESIGN BRANCH	APPROVED: J.M. Mc
RECOMMENDED: J.M. Mc	DATE: 14 MAR 66
CHIEF, ENGINEERING DIVISION	COLONEL, C. E. DISTRICT ENGINEER
	SCALE AS SHOWN
	INV. NO. ENG. 66-45
	FILE NO.
	SHEET 269
	DWD-1-4-21/2



DESIGNED	25 Feb 63	AS constructed	BY	
DRAWN	21 Dec 67	Annual ring added to conform to comments as per N.R.D. 3th indorsement 12 Aug. 1963. Supl. I- Des. Memo. G.1	BY	
CHECKED	21 Aug. 67	Revised and re-drawn	BY	
U. S. ARMY ENGINEER DISTRICT WALLA WALLA, WASHINGTON DWORSHAK DAM AND RESERVOIR NORTH FORK CLEARWATER RIVER, IDAHO GENERAL ISOMETRIC VIEW OF GALLERIES IN MONOLITHS 14 THRU 37				
DESIGNED	Roanfield	DATE 21 Aug. 67		
DRAWN	Amo	COLONEL C. E. DISTRICT ENGINEER		
CHECKED	Cuckler	SCALE AS SHOWN INV. NO. ENG. 66-45		
SUPERVISOR	CHIEF, STRUCT. DES. SECT.	FILE NO.		
RECOMMENDED	CHIEF, ENGINEERING DIVISION	SHEET 50 DWD-1-0-1/8		

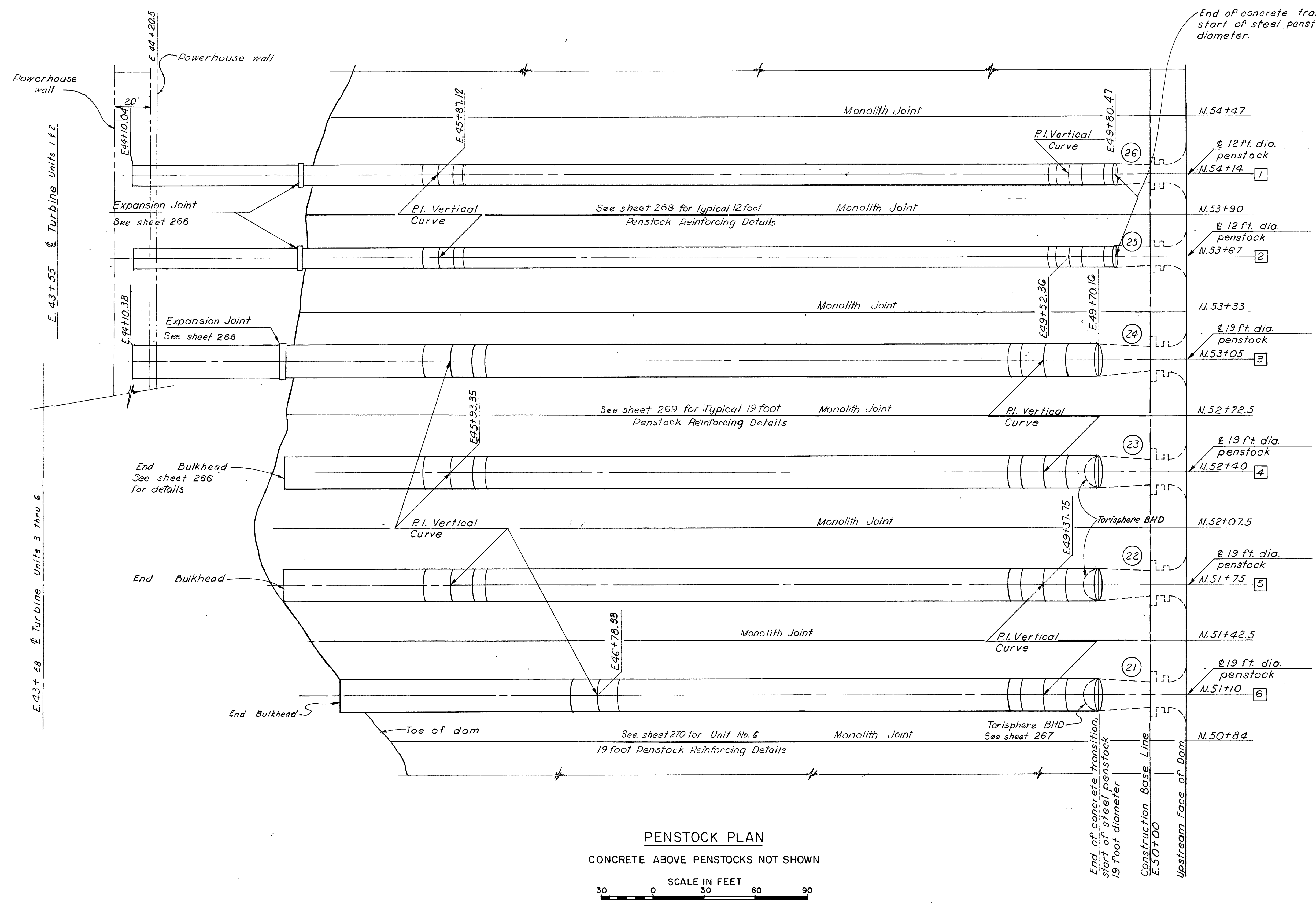
CONT. NO. 67-C-5

VOL. NO. 1



PENSTOCK DRAINS

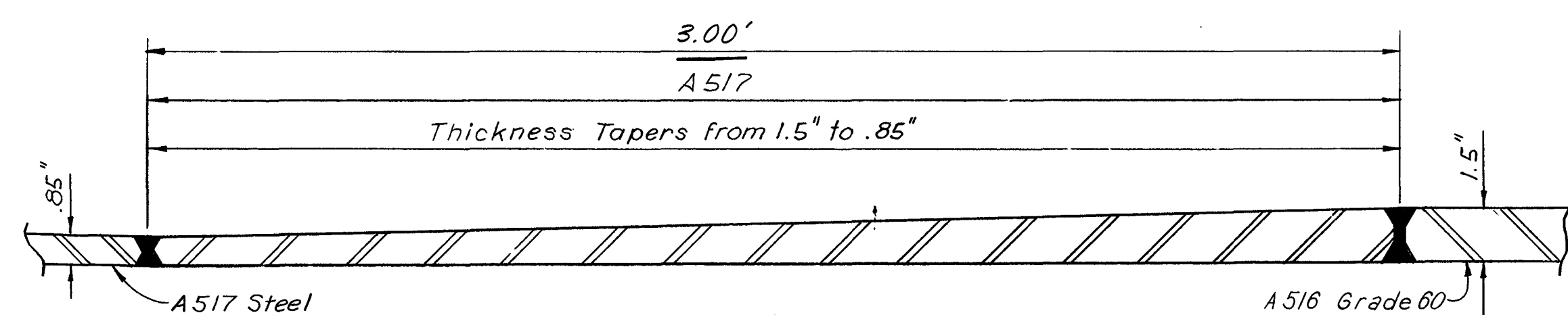
DESIGNED: Lanker	DATE: 74 Dec 9	DESCRIPTION: As constructed
DRAWN: G. Fine	DATE: 21 Aug 67	DESCRIPTION: Revised, redrawn & title change.
CHECKED: Cuckler		
SUPVISED: [Signature]		
RECOMMENDED: [Signature]		
CHIEF DESIGN BRANCH	DATE: 21 Aug 67	
CHIEF ENGINEERING DIVISION	COLONEL G. W. DISTRICT ENGINEER	
	SCALE AS SHOWN	INV. NO. ENG.
		FILE NO.
	SHEET 68	DWD-1-0-15/4



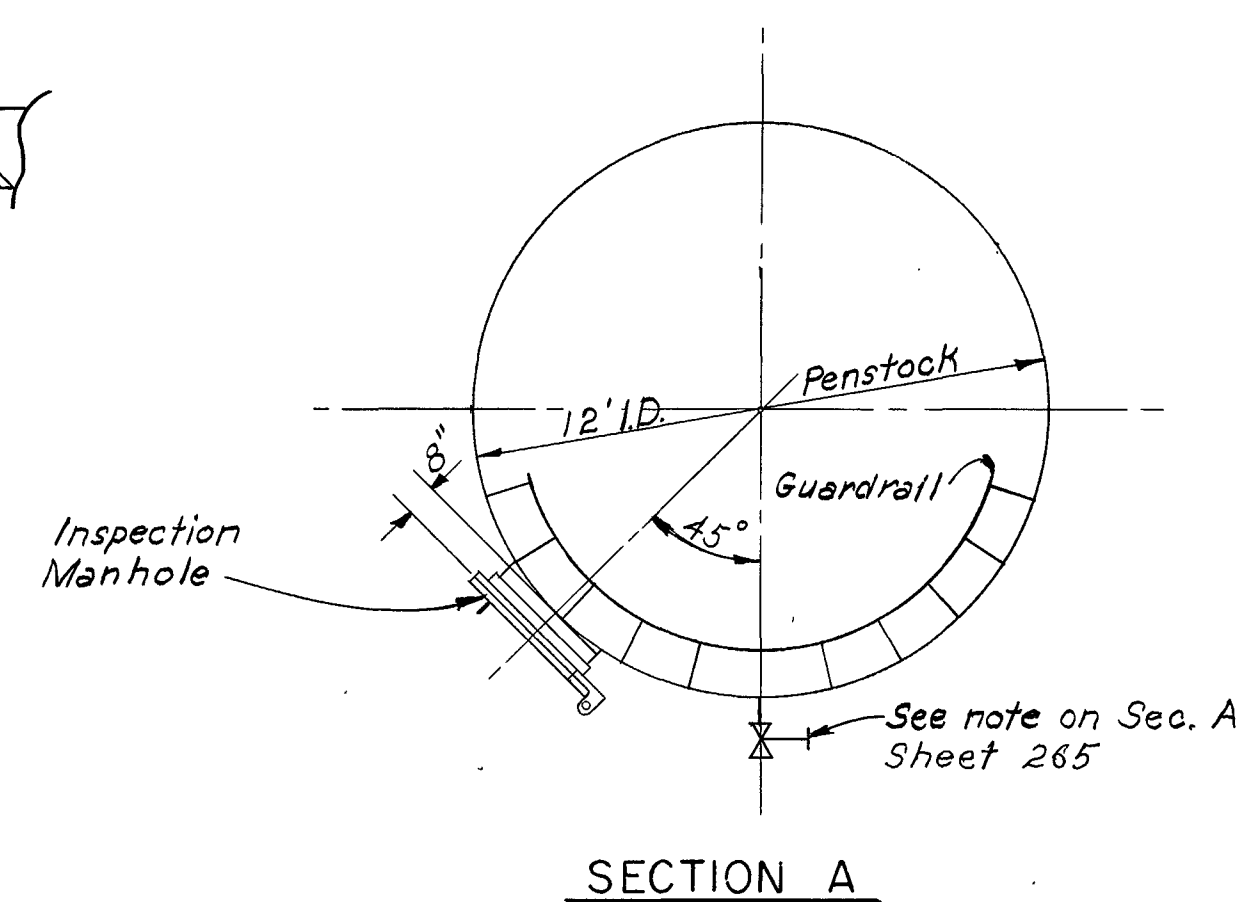
NOTES:
1. indicates monolith number.
2. indicates Penstock number.
3. For steel penstock dimensions See sheet 264.

PENSTOCK PLAN
CONCRETE ABOVE PENSTOCKS NOT SHOWN
SCALE IN FEET
0 30 60 90

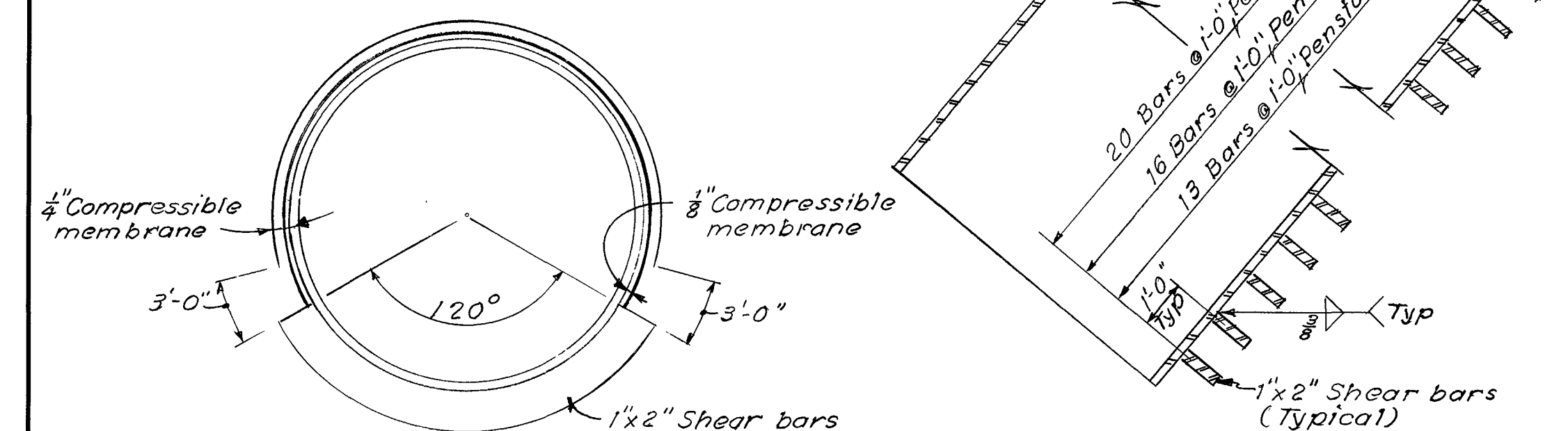
REVISION	DATE	DESCRIPTION	BY
C	75 MAY 11	As constructed	ADA
A	69 DEC 19	Deleted inspection manholes from penstocks.	120
A	5 JAN 67	Corrected penstock co-ordinates, Reducers removed	120
U. S. ARMY ENGINEER DISTRICT WALLA WALLA, WASHINGTON			
DESIGNED: Lanker			
DRAWN: J. Leigh			
CHECKED: CUCKLER			
SUPERVISED:			
SUBMITTED:			
RECOMMENDED:			
APPROVED: DATE 14 MAR 66			
COLONEL C. E. DISTRICT ENGINEER			
SCALE AS SHOWN INV. NO. ENG. 66-45			
SHEET 263 DWD-1-0-21/1			



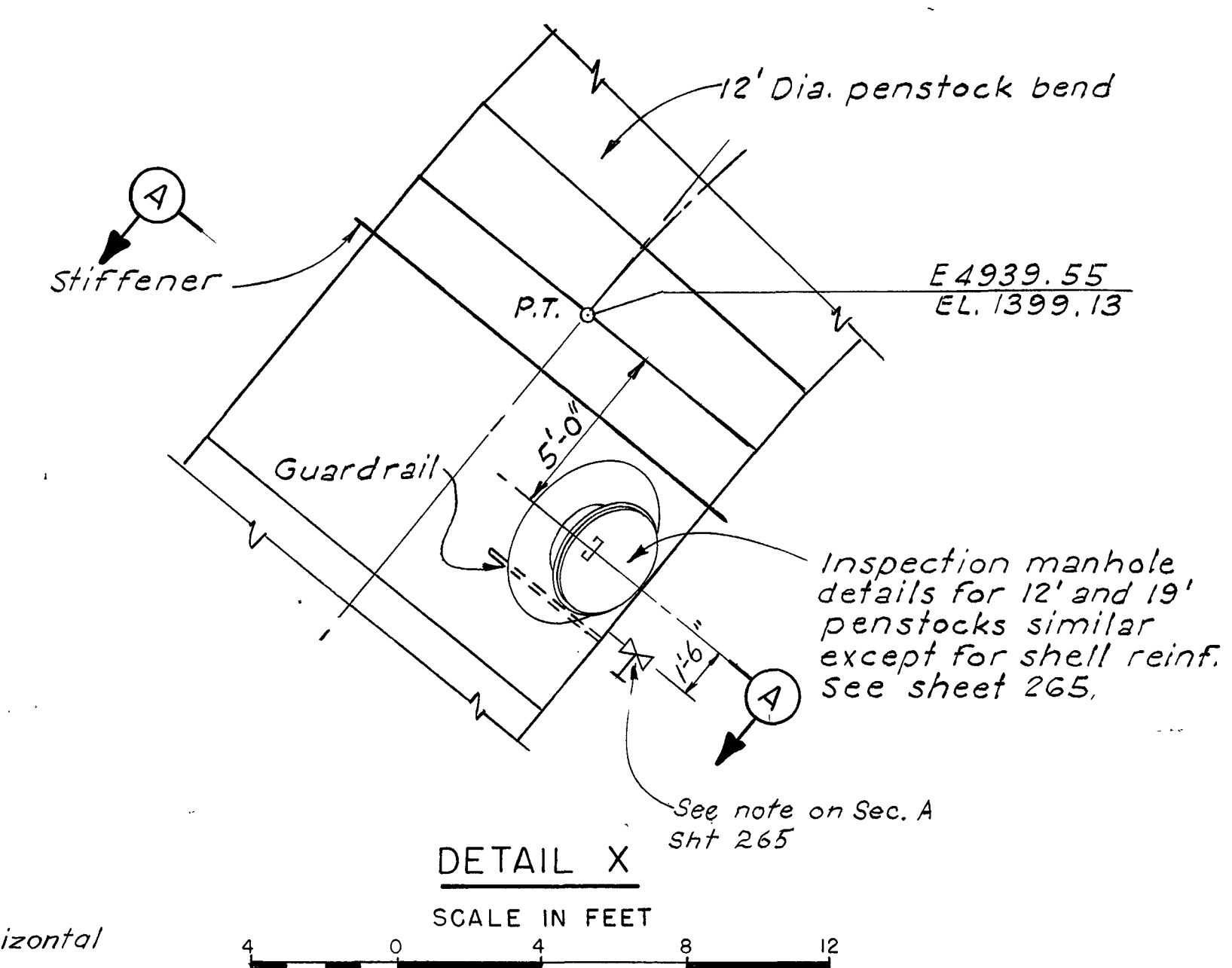
DETAIL A
TRANSITION PIECE BETWEEN
A516 AND A517 STEEL
NOT TO SCALE



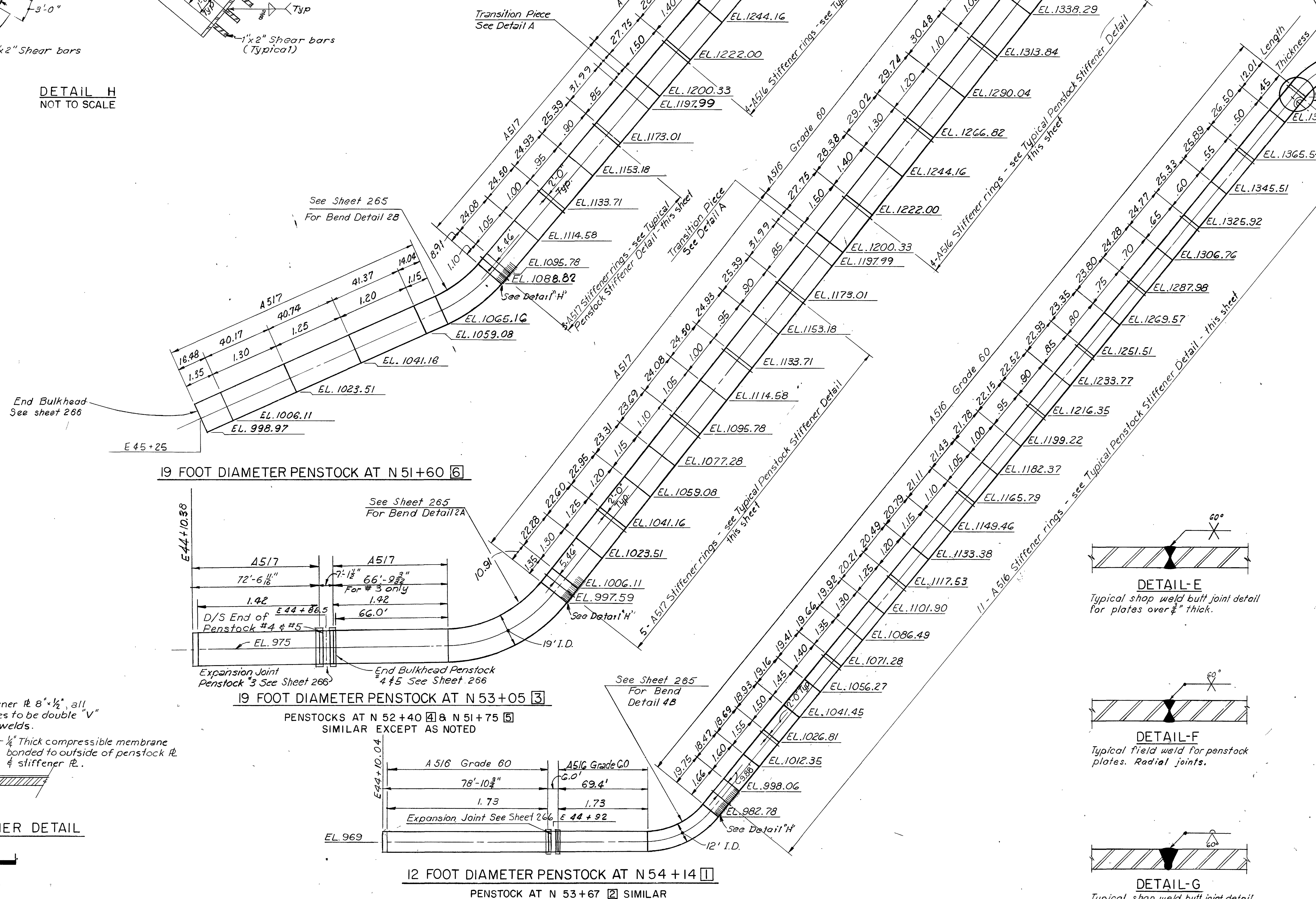
SECTION A



DETAIL H
NOT TO SCALE



DETAIL X

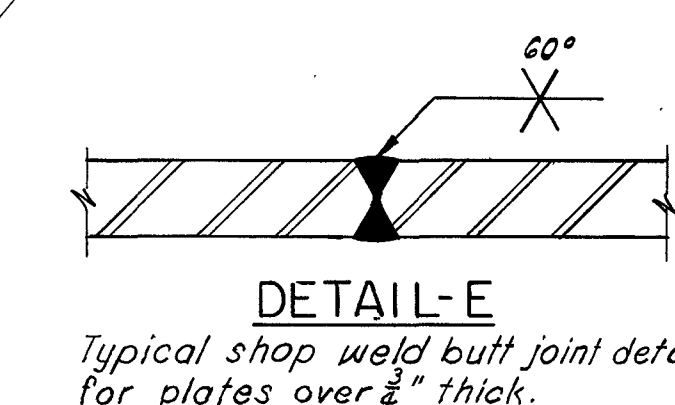


NOTES:

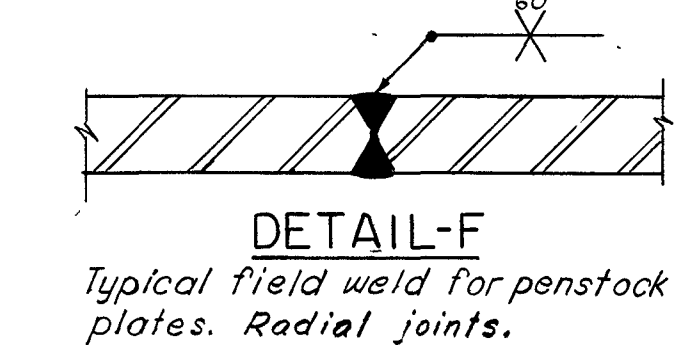
1. All steel except cutoff rings is to be of firebox quality. Cutoff rings may be either firebox or flange quality. A516 steel.
2. ☐ indicates penstock number
3. Cutoff rings shall be of the same thickness as the plate they are being welded to.
4. All interior ferrous surfaces of penstocks shall be painted system #3.

103 Steel penstocks, A516 modified.

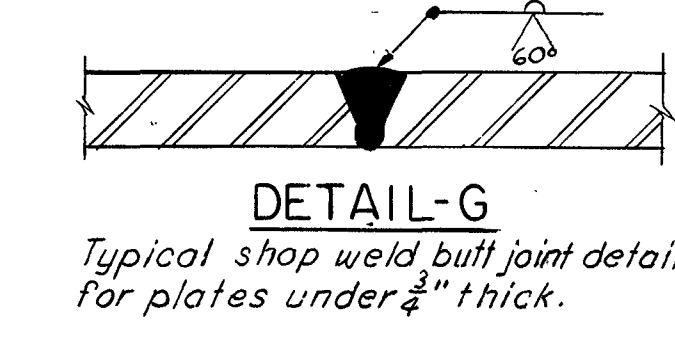
170 Steel penstocks, A517 modified.



DETAIL E
Typical shop weld butt joint detail
for plates over 1/2\"/>



DETAIL F
Typical field weld for penstock
plates. Radial joints.



DETAIL G
Typical shop weld butt joint detail
for plates under 1/2\"/>

TYPICAL PENSTOCK STIFFENER DETAIL

SCALE IN FEET

SCALE IN FEET

REVISION	DATE	DESCRIPTION	BY
1	75 MAY 01	As constructed	
2	69 JUL 12	Moved Inspection Manhole 45' away from 12' ID Penstock Invert	
3	1972 13	Changed location of revised manhole, added Detail X, as per N.P.D. 9th indorsement 12 Aug 1969, Supl. 1 D.M.G.I	
4	68 SEP 16	Added Shear Bars and Detail H	
5	9 JAN 67	Revised penstock reducer; added stiffener rings & penstock stiffener detail; deleted details A, B, C; minor rev.	
6	28 OCT 66	Corrected penstock steel and minor revisions	
7	22 AUG 66	Corrected penstock steel as per specs. addendum #3	
8	20 JUN 66	Minor revision	
9	6 JUN 66	Added bid items	
10	7 MAY 66	Revised dimensions and notes	

U. S. ARMY ENGINEER DISTRICT
WALLA WALLA, WASHINGTON

DESIGNED: Lanker

DRAWN: Platterle

CHECKED: CUCKLER

SUBMITTED: [Signature]

CHIEF, STRUCT. DES. SEC.

RECOMMENDED: [Signature]

CHIEF, DESIGN BRANCH

RECOMMENDED: [Signature]

CHIEF, ENGINEERING DIVISION

APPROVED: [Signature] DATE: 14 MAR 66

COLONEL C. E. DISTRICT ENGINEER

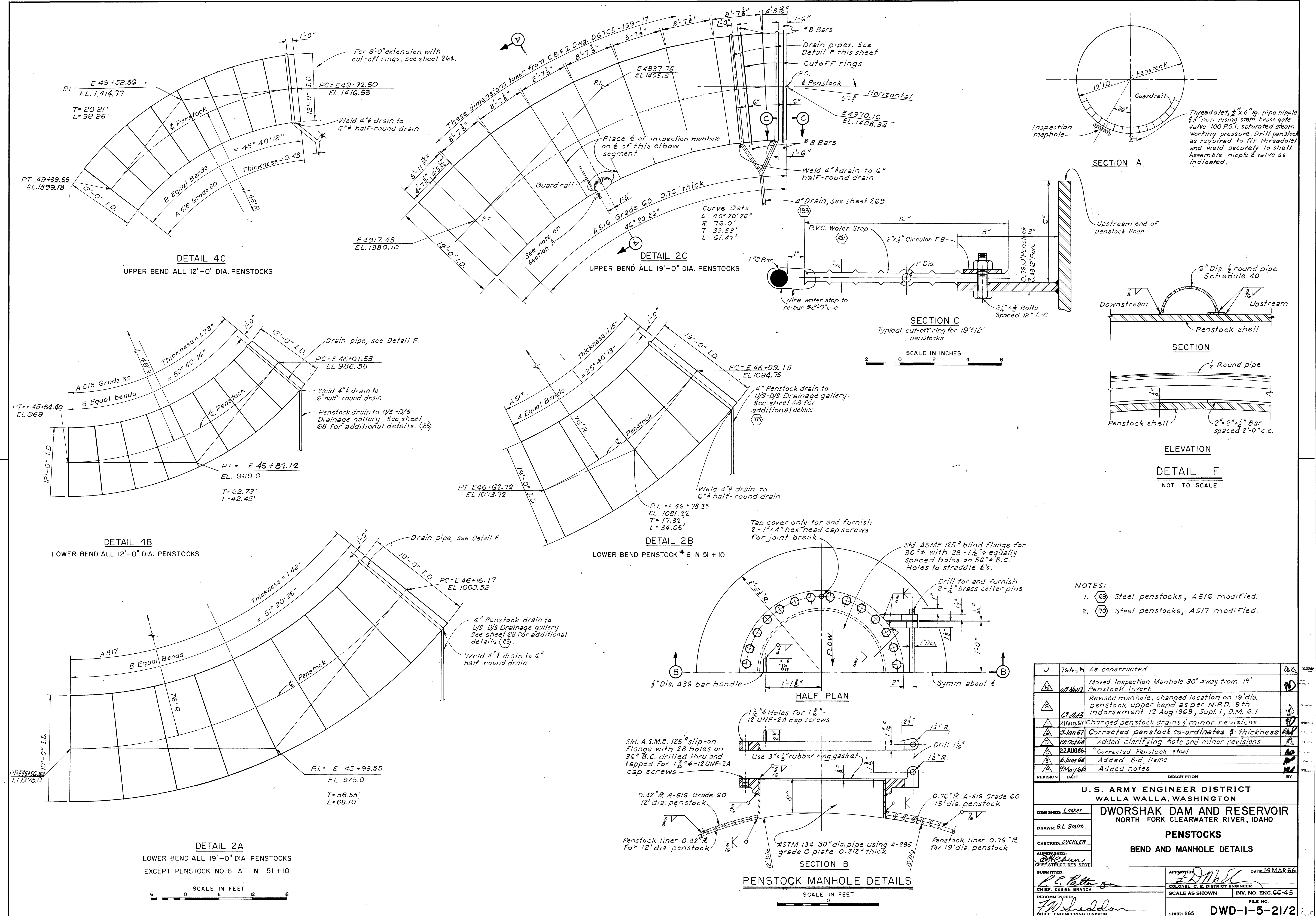
SCALE AS SHOWN INV. NO. ENG. 66-45

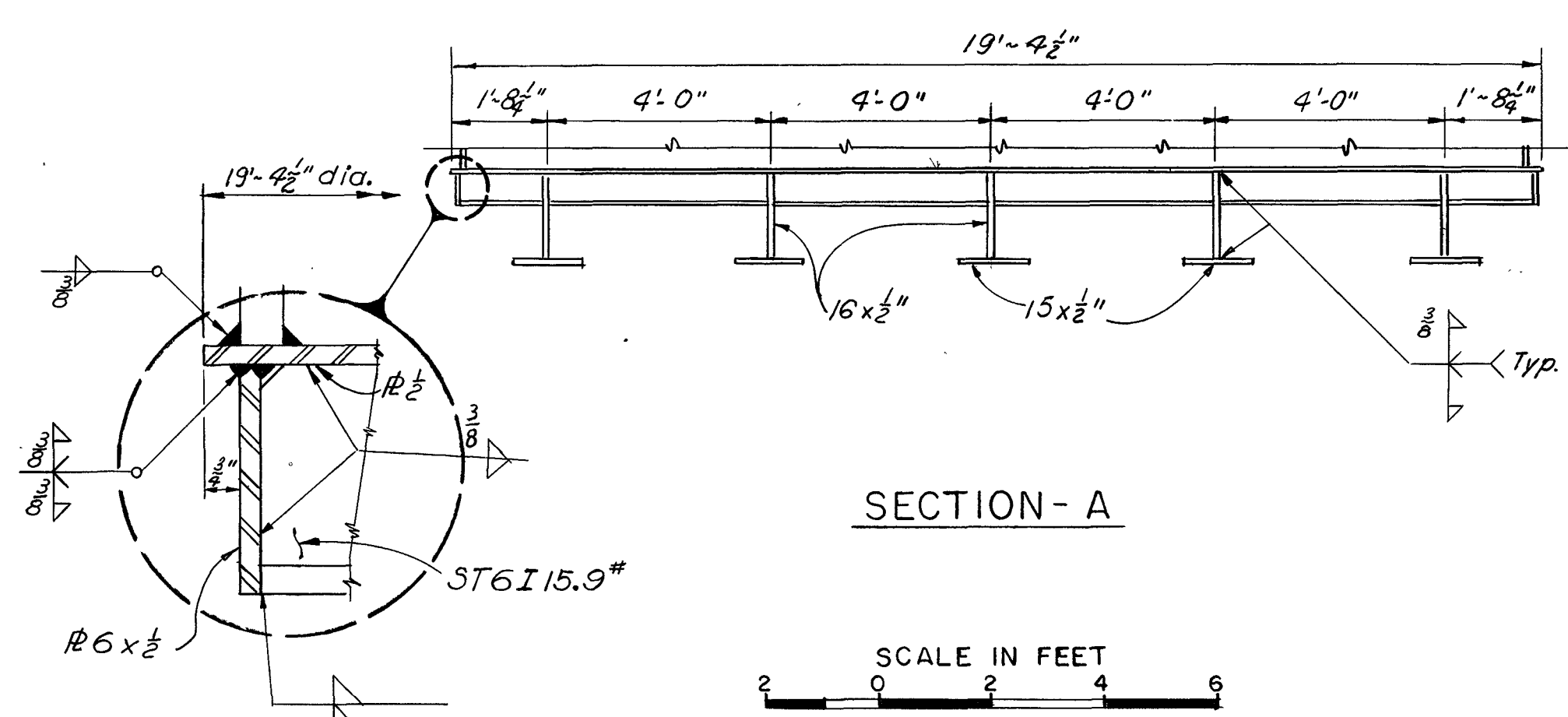
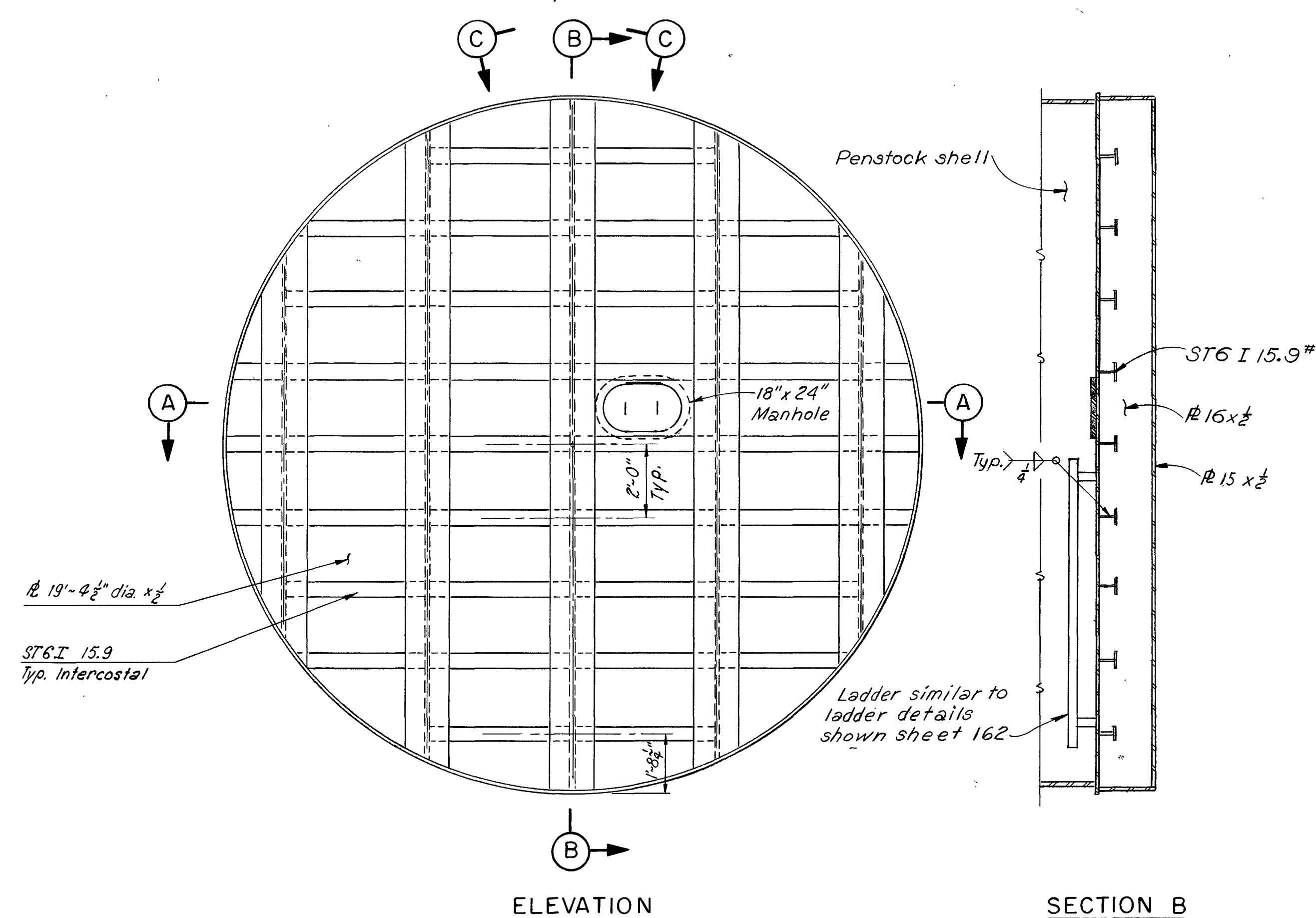
FILE NO.

SHEET 264 DWD-1-5-21/1

CONT. NO. 67-C-5

VOL. NO. 1



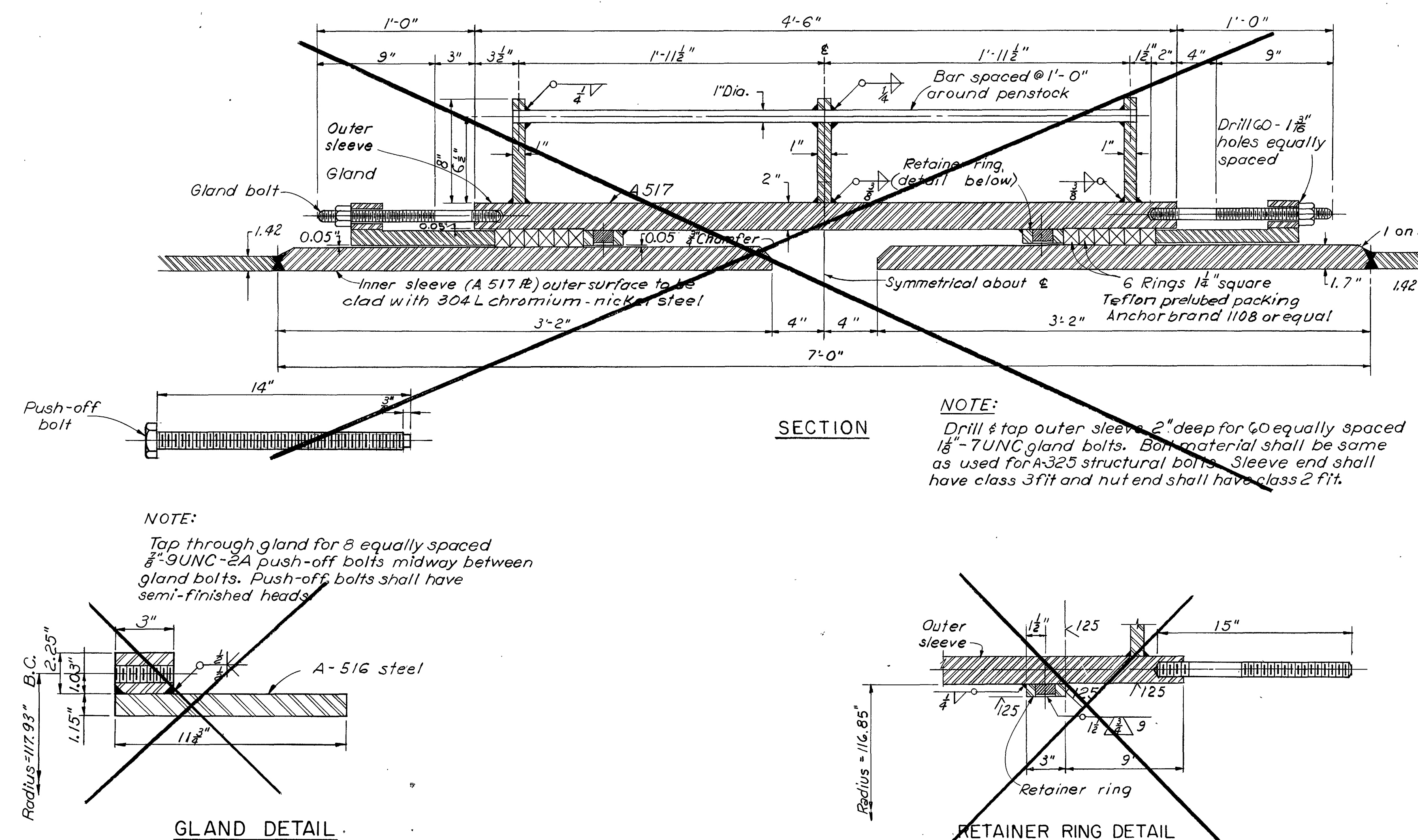


TYPICAL END DETAIL OF ALL 16" STIFFENERS

Bulkhead - A-36 Steel

BULKHEAD FOR DOWNSTREAM END OF PENSTOCKS 4, 5 & 6

FOR MANHOLE AND LADDER DETAILS ON
DOWNSTREAM BULKHEAD, SEE NORTHERN
STEEL AND SUPPLY COMPANY, SHOP
DRAWINGS NO. 11 & 17.



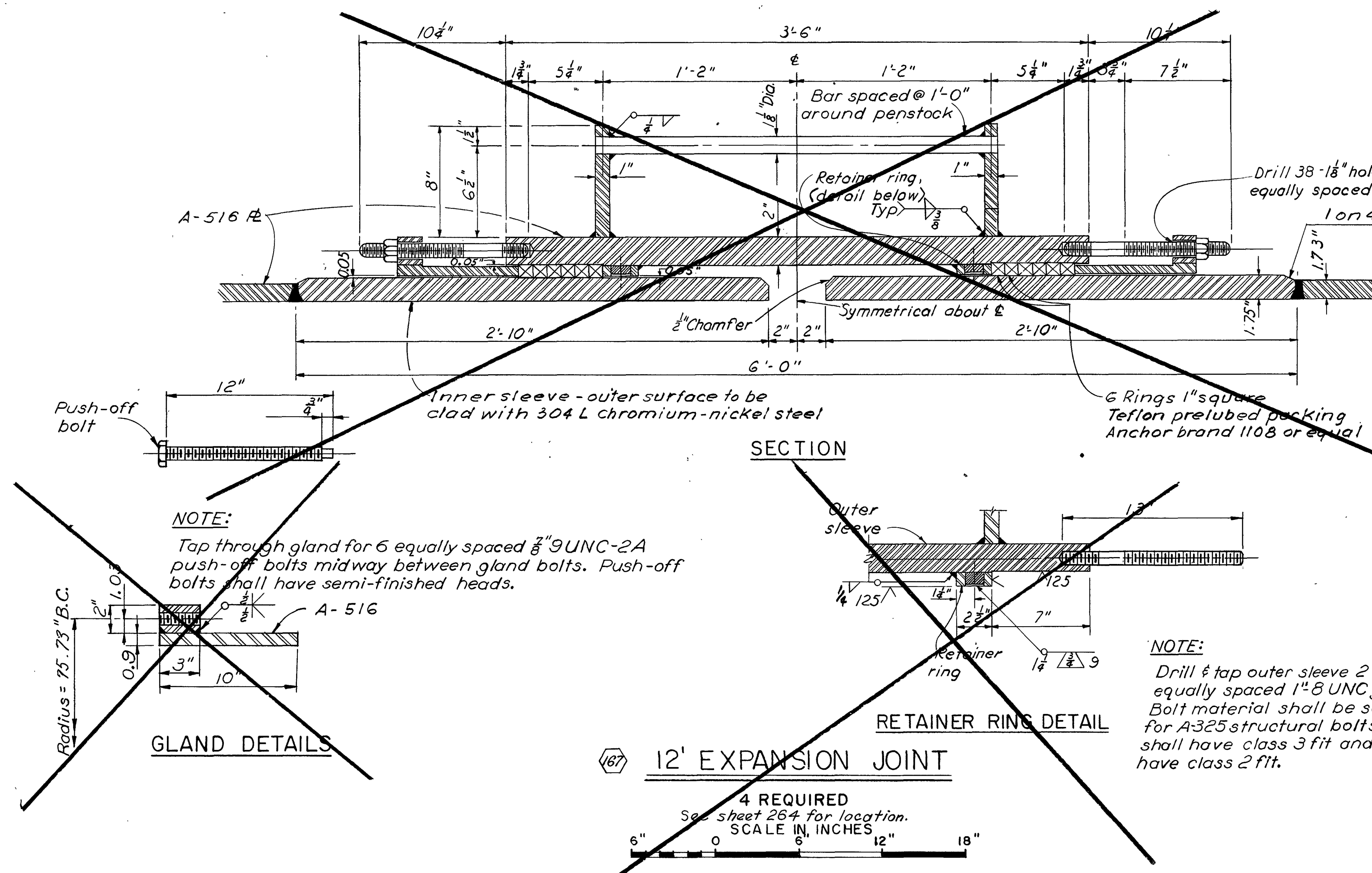
19' EXPANSION JOINT

2 REQUIRED

NOTES:

- Contractor selected the sleeve type coupling option. Reference shop drawings: Chicago Bridge and Iron Company 9-6461-1 9-6463-1

Dresser Industries Style 38 couplings
Dresser Industries reference shop drawings:
45362
45347-1
45347-2
45347-3



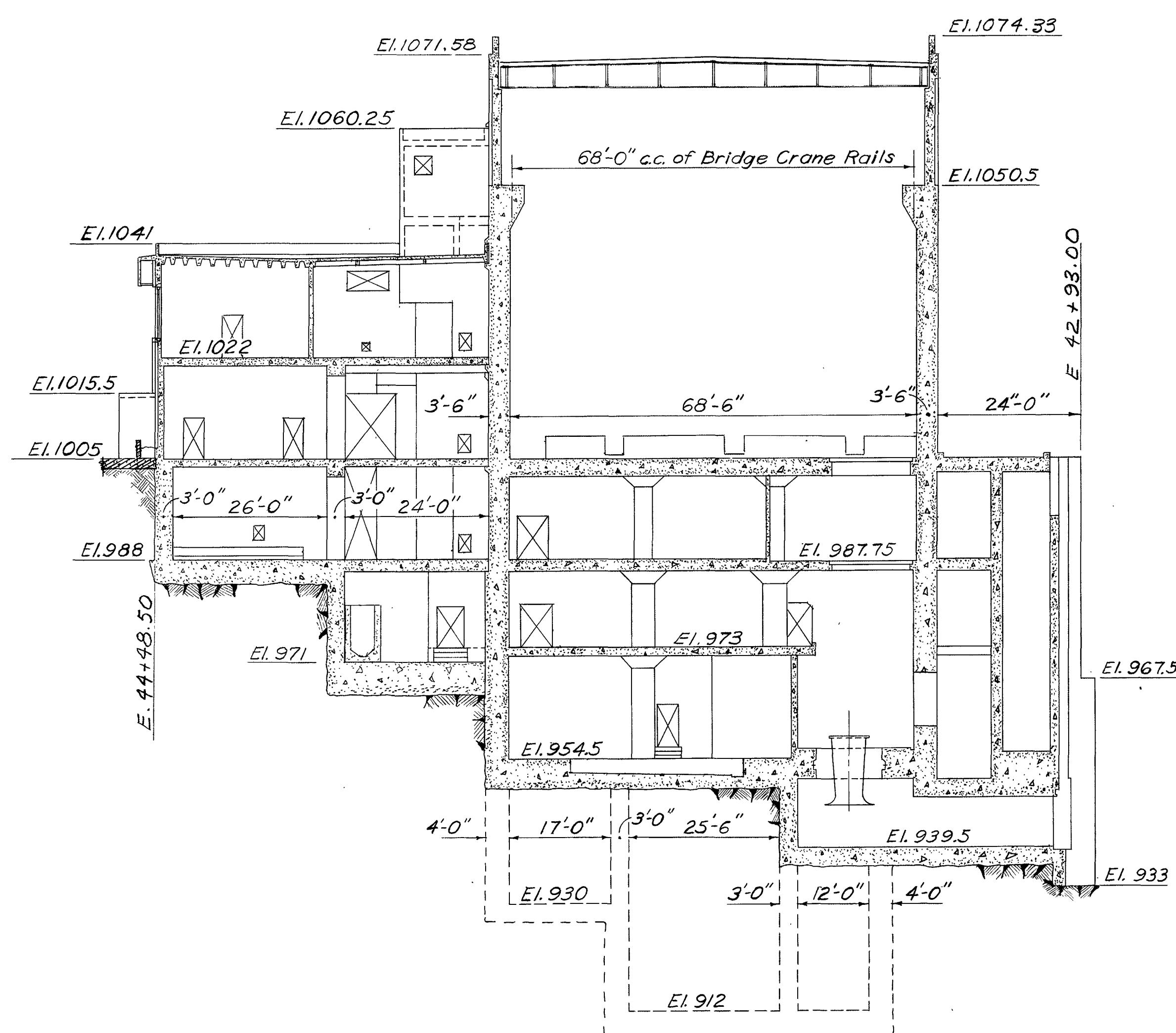
REVISION	DATE	DESCRIPTION	BY
G	76Aug19	As constructed	
A	3Jan67	Revised Bulkhead and Expansion Joint Size	
B	28Oct66	Corrected penstock steel	
C	22AUG66	Corrected Penstock steel.	
D	20June66	Minor revisions	
E	6June66	Added Bid Items	
F	7May66	Revised exp. jt. sizes	

DESIGNED: LANKER		APPROVED: <i>[Signature]</i> DATE: 14 MAR 66	
DRAWN: J. Leigh		COLONEL C. E. DISTRICT ENGINEER	
CHECKED: CUCKLER		SCALE AS SHOWN INV. NO. ENG. 66-45	
SUPERVISED: <i>[Signature]</i>		FILE NO.	
SUBMITTED: <i>[Signature]</i>		SHEET 266 DWD-1-5-21/3	
RECOMMENDED: <i>[Signature]</i>		CHIEF, ENGINEERING DIVISION	

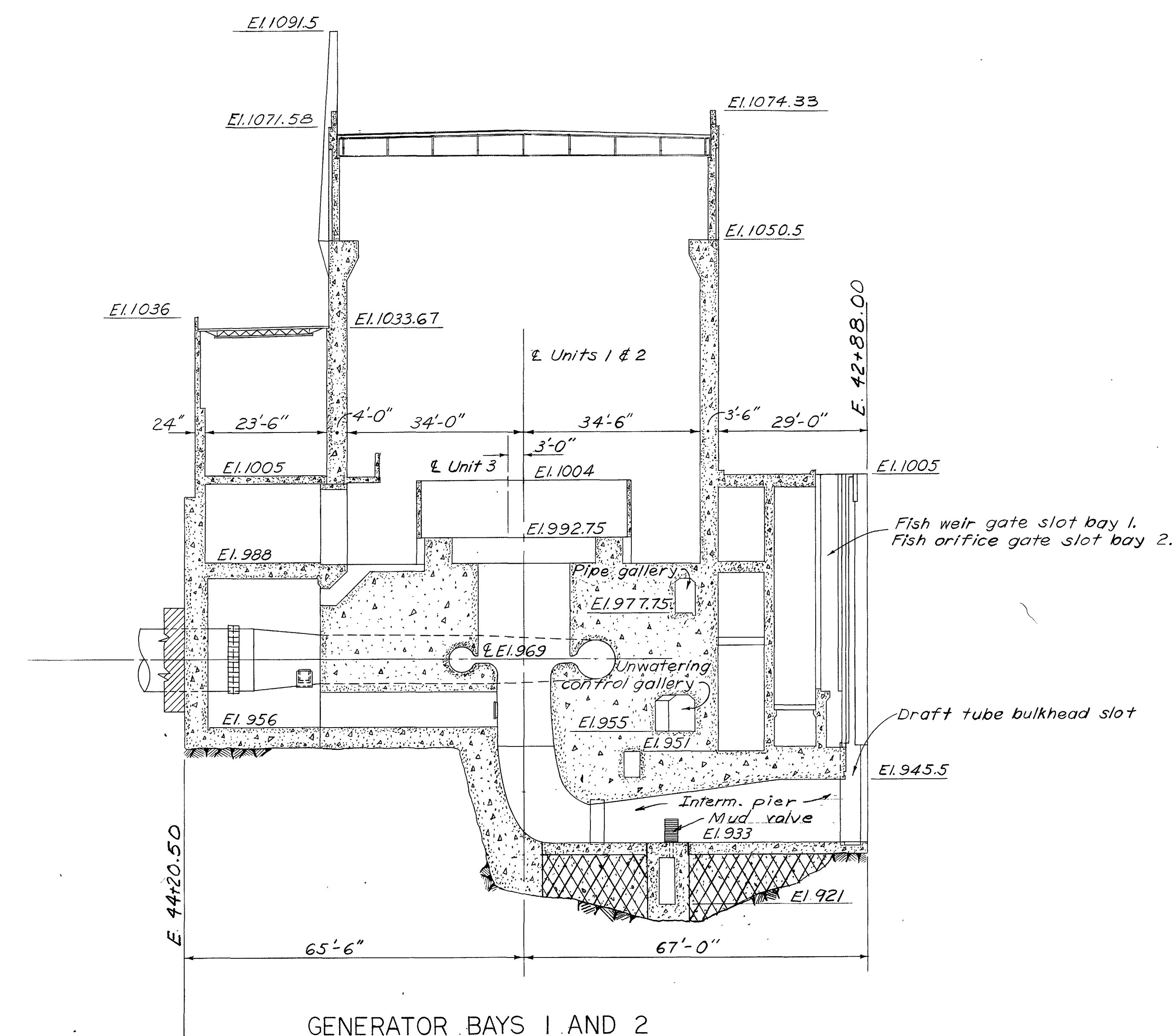
CONT. NO. 67-C-5

VOL. NO. I

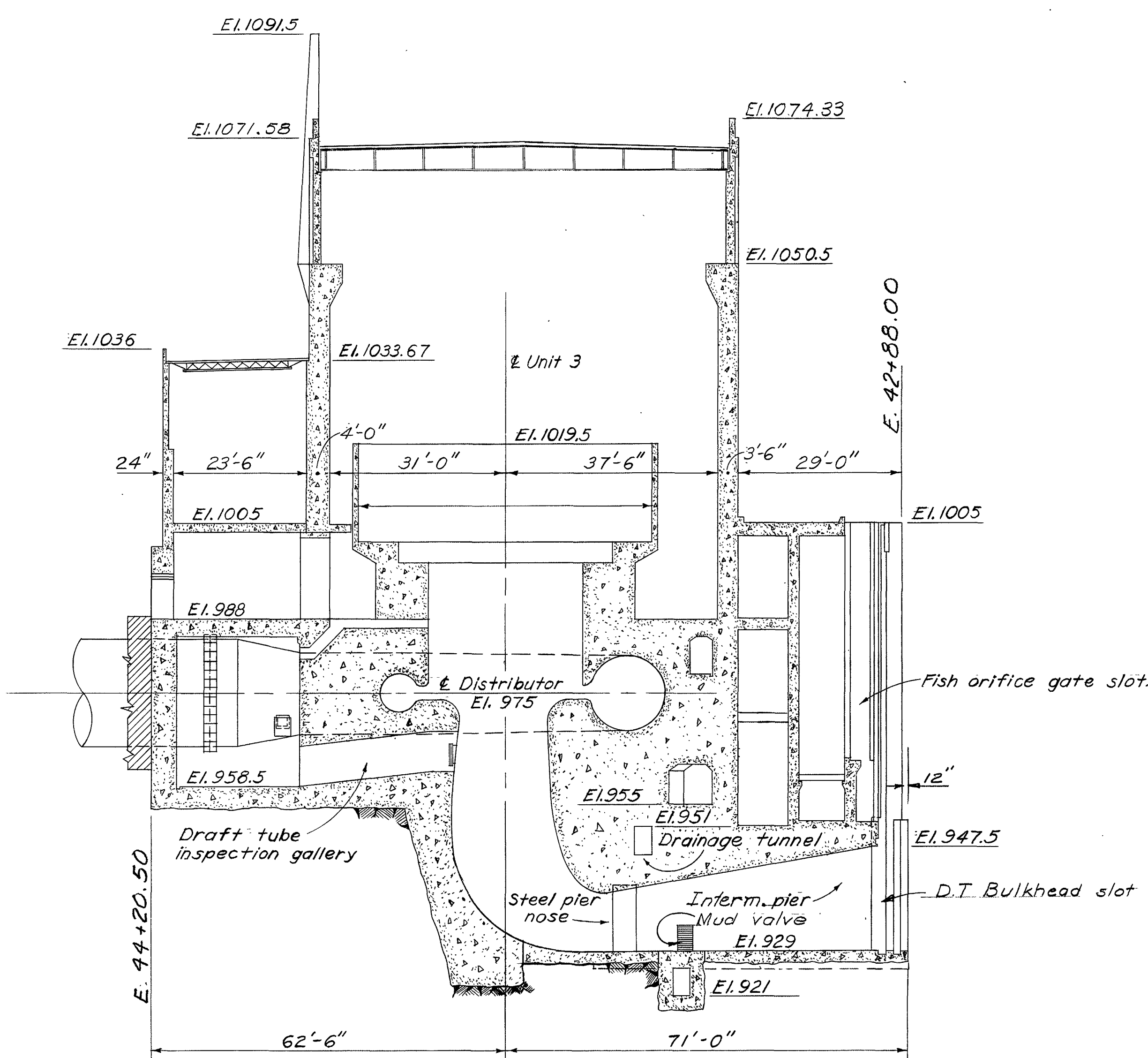
PENSTOCKS
EXPANSION JOINTS



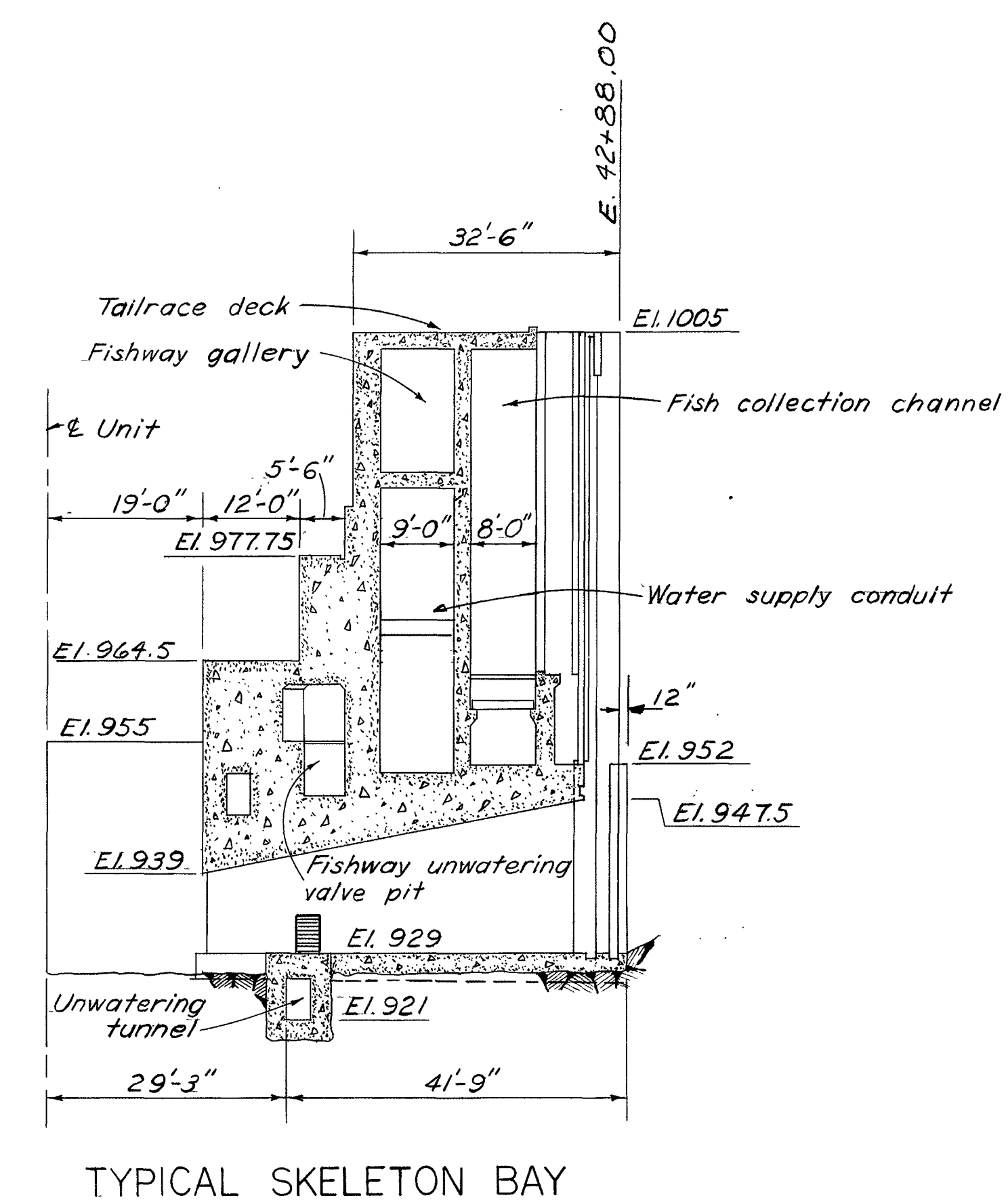
ERECTION BAY AND CONTROL STRUCTURE



GENERATOR BAYS 1 AND 2



GENERATOR BAY 3



TYPICAL SKELETON BAY

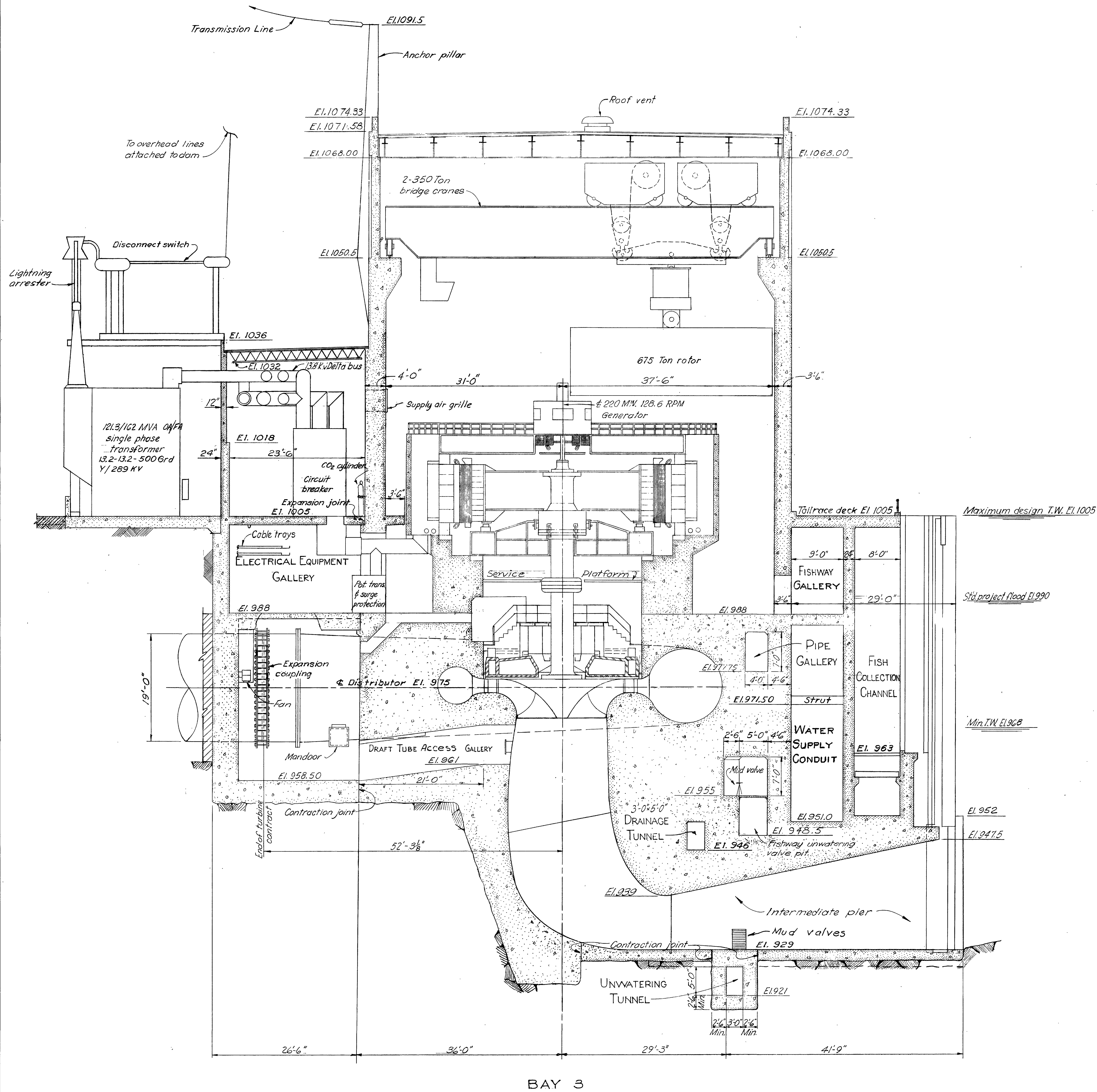
NOTE
Cross-hatched areas indicate limits
of 2,000 pounds per square inch strength concrete

AS CONSTRUCTED
HYDRO ELECTRIC DESIGN BRANCH, NPD

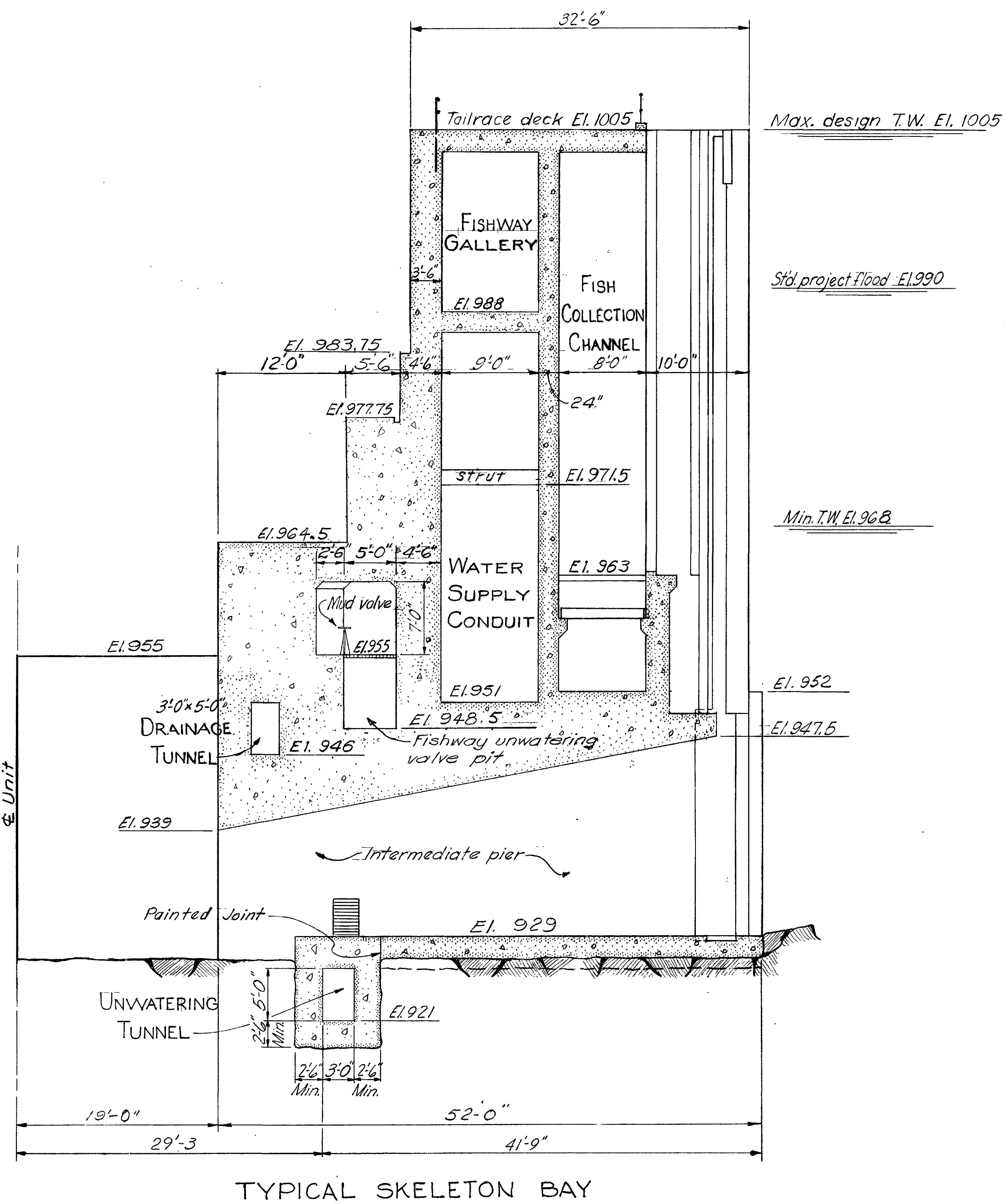
DATE 1975 JUL 31 BY *CJM*

REVISION	DATE	DESCRIPTION	BY
B	75 JUL 31	As Constructed	CJM
A	70 JAN 30	Removed roof corbel	CJM

DESIGNED BY M.J.S.	DWORSHAK DAM AND RESERVOIR NORTH FORK CLEARWATER RIVER, IDAHO POWERHOUSE GENERAL TRANSVERSE SECTIONS
DRAWN BY KUC	
CHECKED BY C.J.M.	
PREPARED BY <i>Paul C. Drake</i>	
U. S. ARMY ENGINEER DIVISION, N. P. PORTLAND, OREGON	
APPROVED FOR DIV. ENGINEER <i>CJM</i> CHIEF, HYDRO-ELECTRIC DESIGN BRANCH	
SCALE: 1/16" = 1'-0" SPEC. NO.	
SHEET 7 DWP-1-0-1/7	



BAY 3

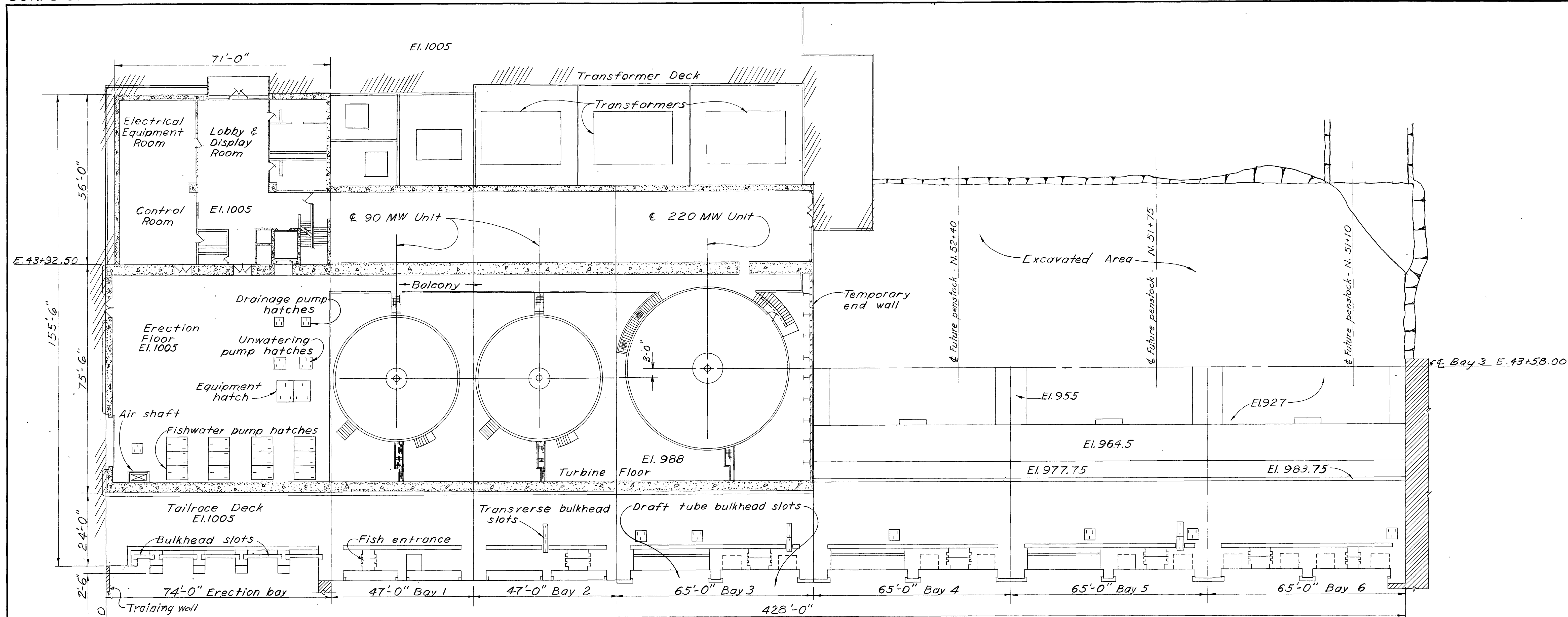


TYPICAL SKELETON BAY

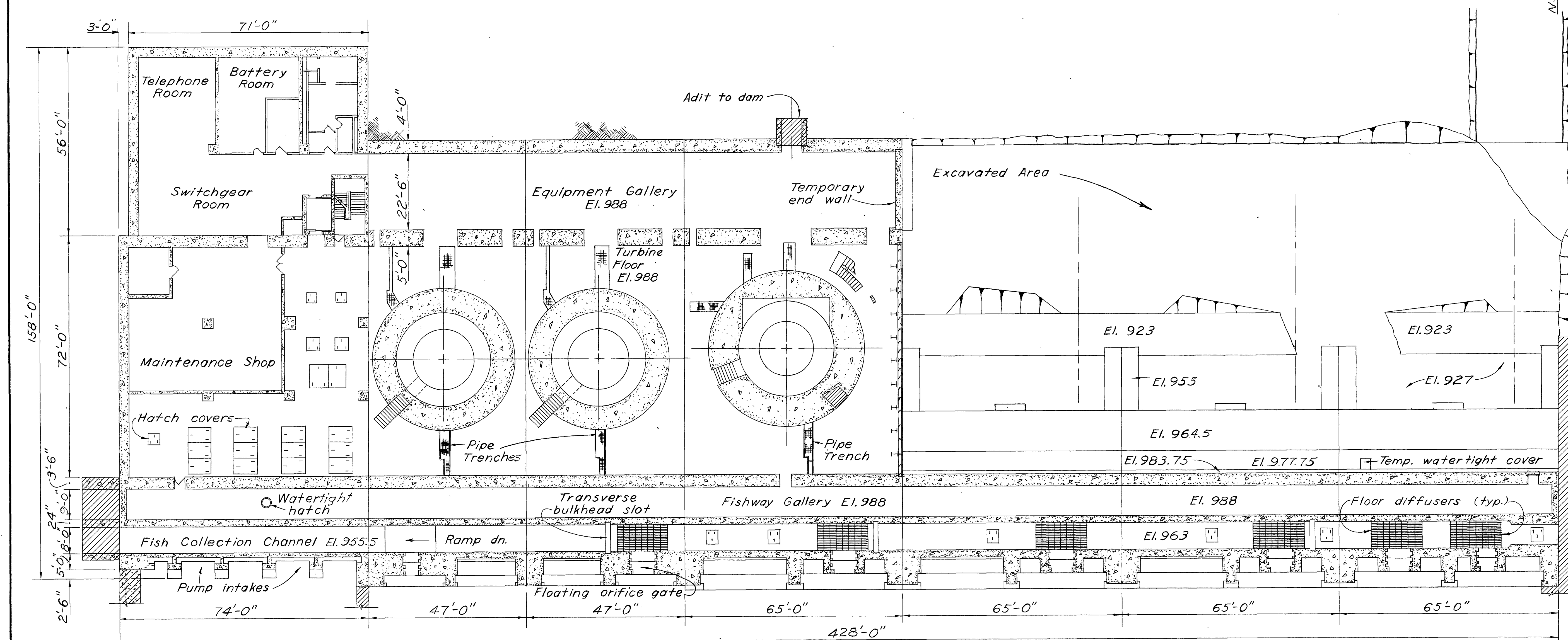
AS CONSTRUCTED
HYDRO ELECTRIC DESIGN BRANCH, NPD

DATE 1975 JUL 31 BY *C. J. M.*

C	75 July 31	As Constructed	C.J.M.
B	71 Jan 29	corrected concrete outline	C.J.M.
A	70 Jan 30	Removed roof corbels	C.J.M.
REVISION	DATE	DESCRIPTION	BY
U. S. ARMY ENGINEER DIVISION, N. P. PORTLAND, OREGON			
DESIGNED BY	M.J.S.	DWORSHAK DAM AND RESERVOIR NORTH FORK CLEARWATER RIVER, IDAHO	
DRAWN BY	B.A.W.	POWERHOUSE ARRANGEMENT	
CHECKED BY	C.J.M.	GENERATOR BAY 3 & SKELETON BAY TRANSVERSE SECTIONS	
PREPARED BY	<i>Paul J. M...</i>	APPROVED FOR DIV. ENGINEER DATE 1968 NOV 1	
SUBMITTED BY	<i>Paul J. M...</i>	CHIEF, HYDRO-ELECTRIC DESIGN BRANCH	
SCALE	1/8"=1'-0"	SPEC. NO.	DWP-6-0-0/9
SHEET 16		VOL. NO. II	



PLAN EL. 1005

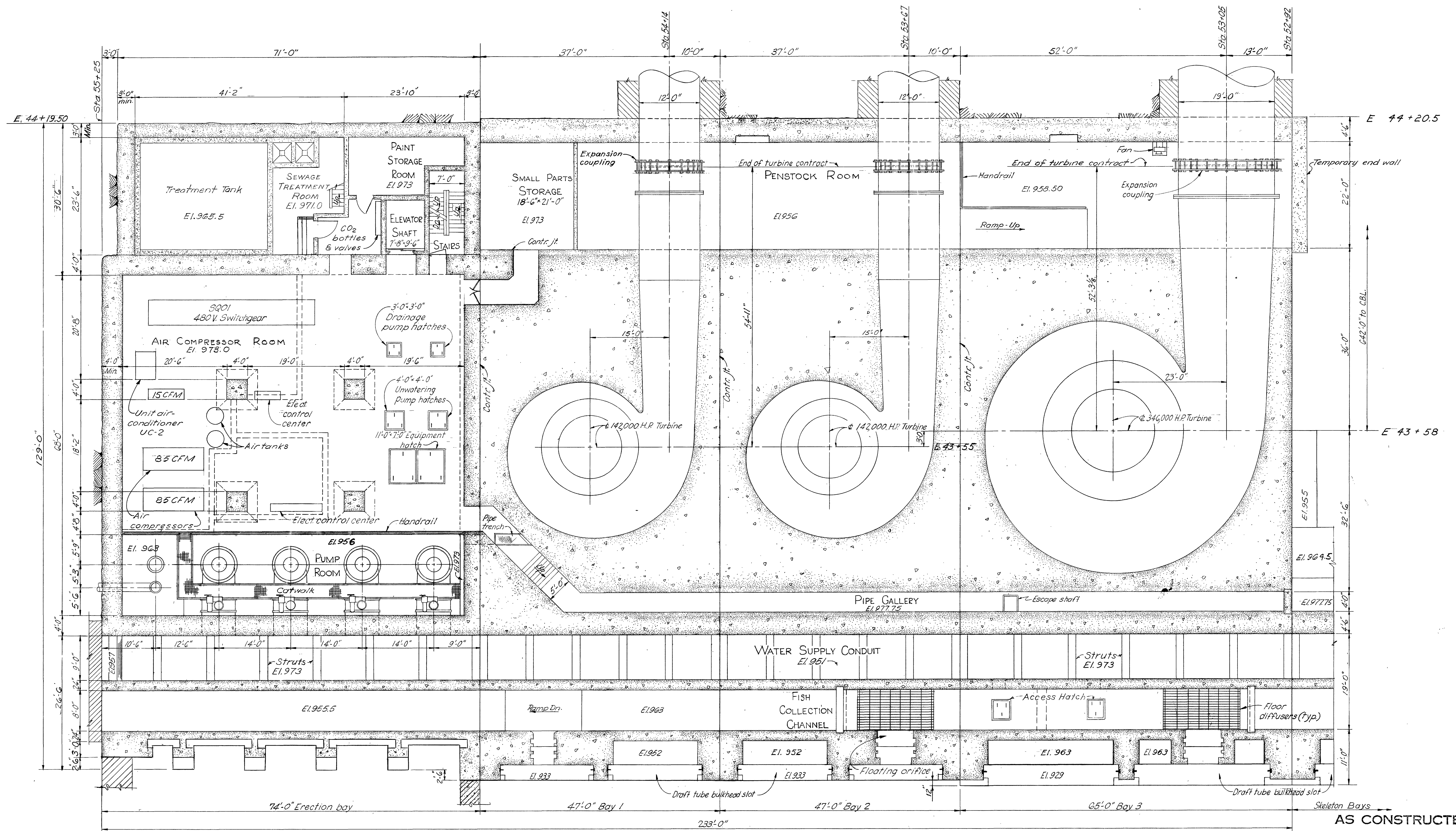


PLAN EL. 988

AS CONSTRUCTED
HYDRO ELECTRIC DESIGN BRANCH, N.P.D.

DATE 1975 JUL 31 BY C. J. M. G. L.

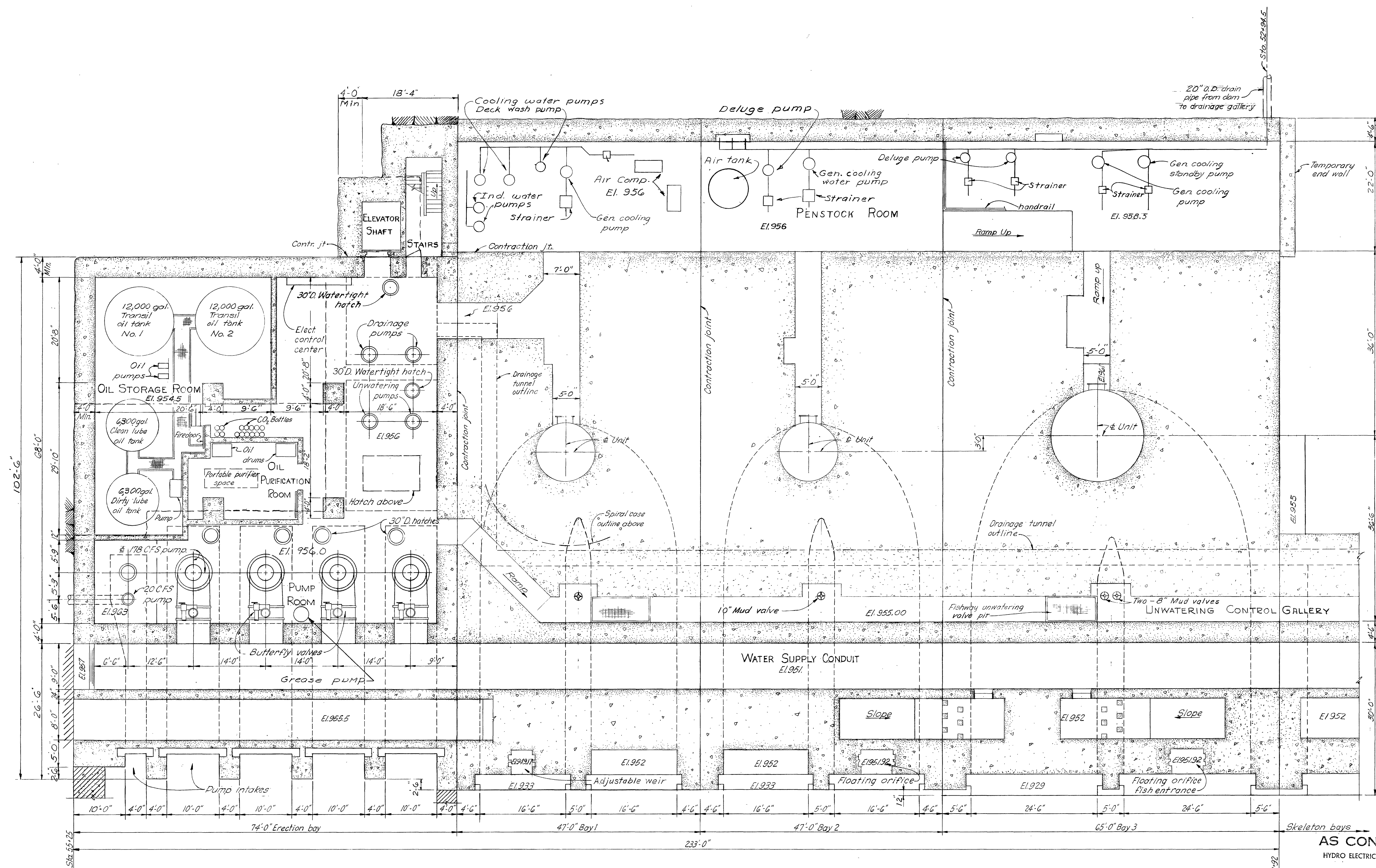
DESIGNED BY M.S.S.	15 July 31	As Constructed	CJM
DRAWN BY KJC	10 Jan 30	Minor revision	CJM
CHECKED BY C.J.M.			
PREPARED BY K. J. M.			
U. S. ARMY ENGINEER DIVISION, N. P. PORTLAND, OREGON			
DWORSHAK DAM AND RESERVOIR NORTH FORK CLEARWATER RIVER, IDAHO POWERHOUSE GENERAL PLANS-EL.1005 AND EL.988			
SUBMITTED BY C. J. M.	1975 NOV 14	APPROVED FOR DIV. ENGINEER R. S. BARNES	
CHIEF, HYDRO-ELECTRIC DESIGN BRANCH		SCALE: 1/16" = 1'-0"	SPEC. NO.
		DWP-1-0-1/5	



AS CONSTRUCTED
HYDRO ELECTRIC DESIGN BRANCH, NPD

DATE 1975 JUL 31 BY C. J. McNamee

DESIGNED BY: M.J.S.	DATE: 75 July 31	DESCRIPTION: As Constructed	BY: C.J.M.
DRAWN BY: P.E.R.	DATE: 70 Jan 30	DESCRIPTION: Changed compressor capacity	BY: C.J.M.
CHECKED BY: C.J.M.			
PREPARED BY: R. E. Wilson			
HEAD, HYDRO-ELECTRIC SECTION			
U. S. ARMY ENGINEER DIVISION, N. P. PORTLAND, OREGON			
DWORSHAK DAM AND RESERVOIR NORTH FORK CLEARWATER RIVER, IDAHO POWERHOUSE ARRANGEMENT PLAN EL. 973			
SUBMITTED: C. J. McNamee		APPROVED: FOR DIV. ENGINEER DATE 1969 NOV. 14 R. E. Wilson CHIEF, HYDRO-ELECTRIC DESIGN BRANCH	
SCALE: 1/8" = 1'-0"		SPEC. NO. DWP-6-O-0/4	



AS CONSTRUCTED
HYDRO ELECTRIC DESIGN BRANCH, NPD

DATE 1975 JUL 31 BY *C. J. McLean*

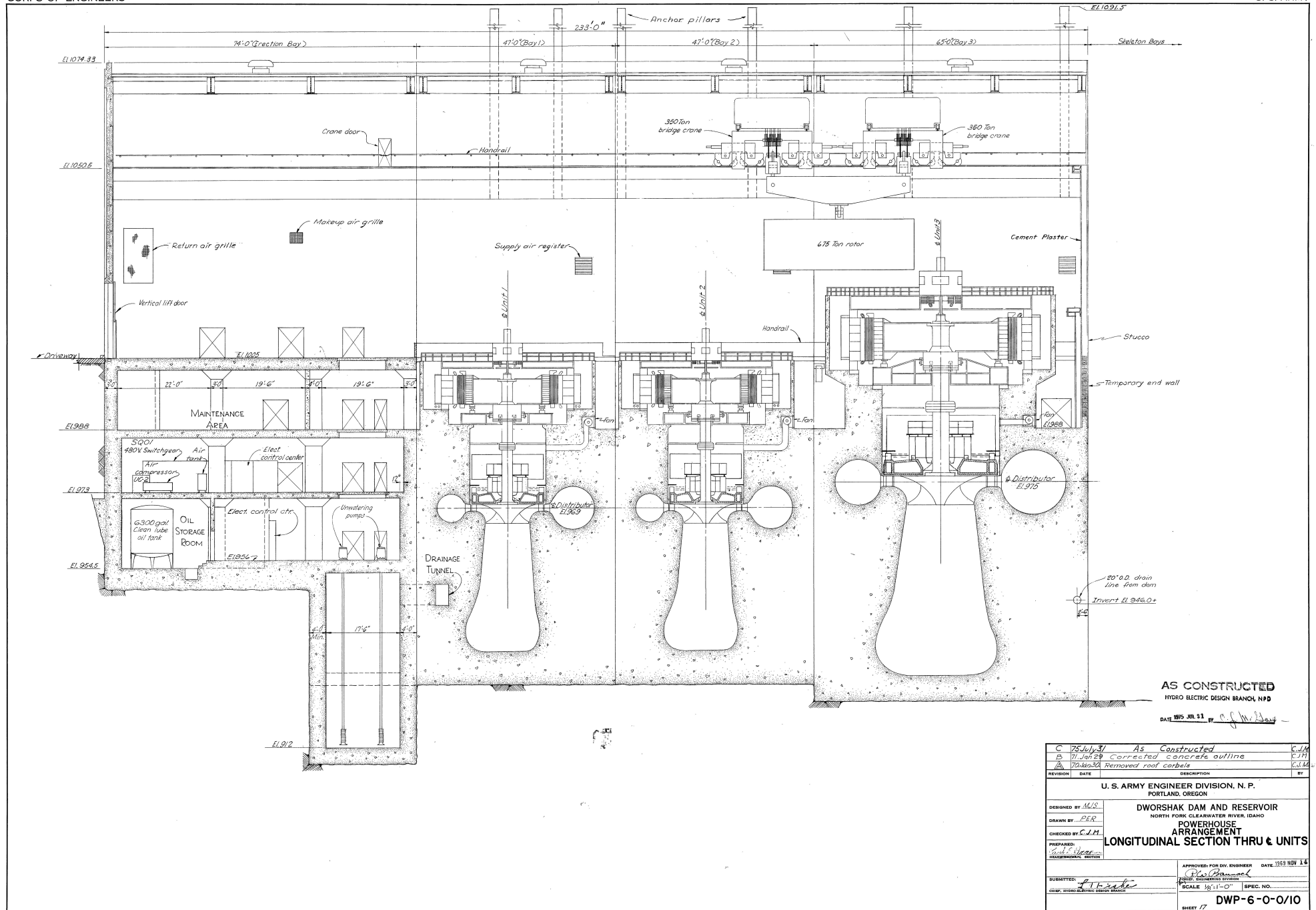
REVISION	DATE	DESCRIPTION	BY
C	75 July 31	As Constructed	CJM
B	71 Jan 29	Located grease pump	CJM
A	70 Jan 30	Revised pump locations in Penstock Rm	CJM

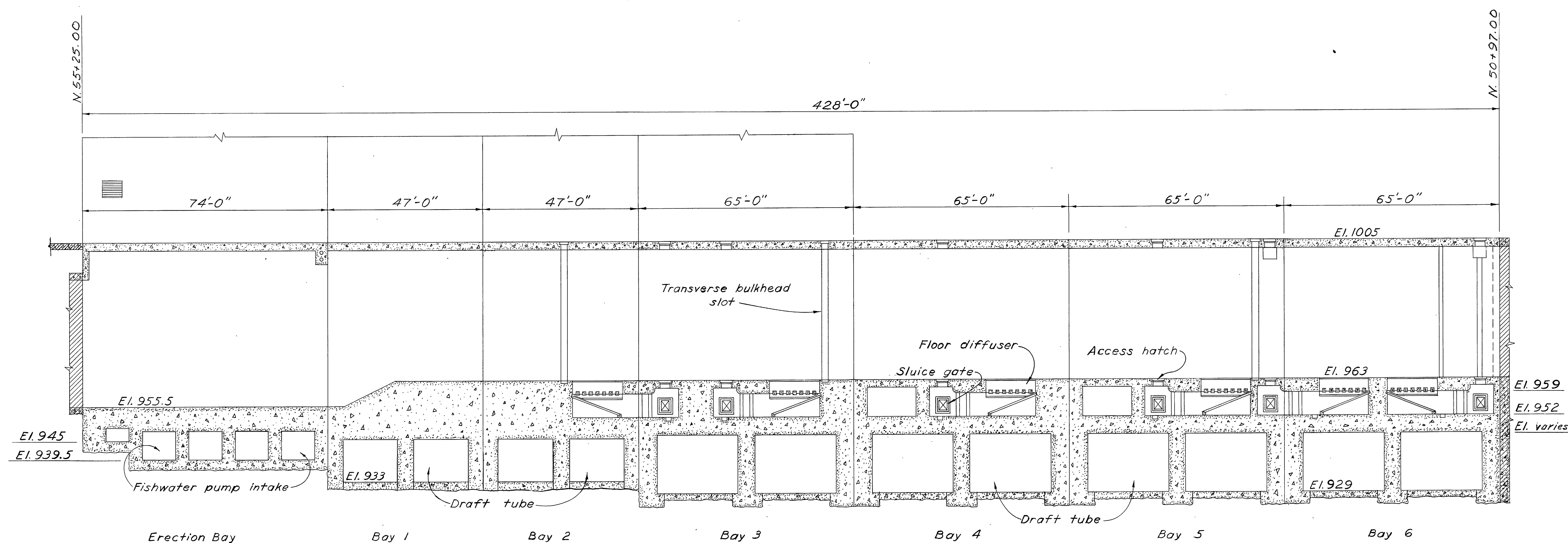
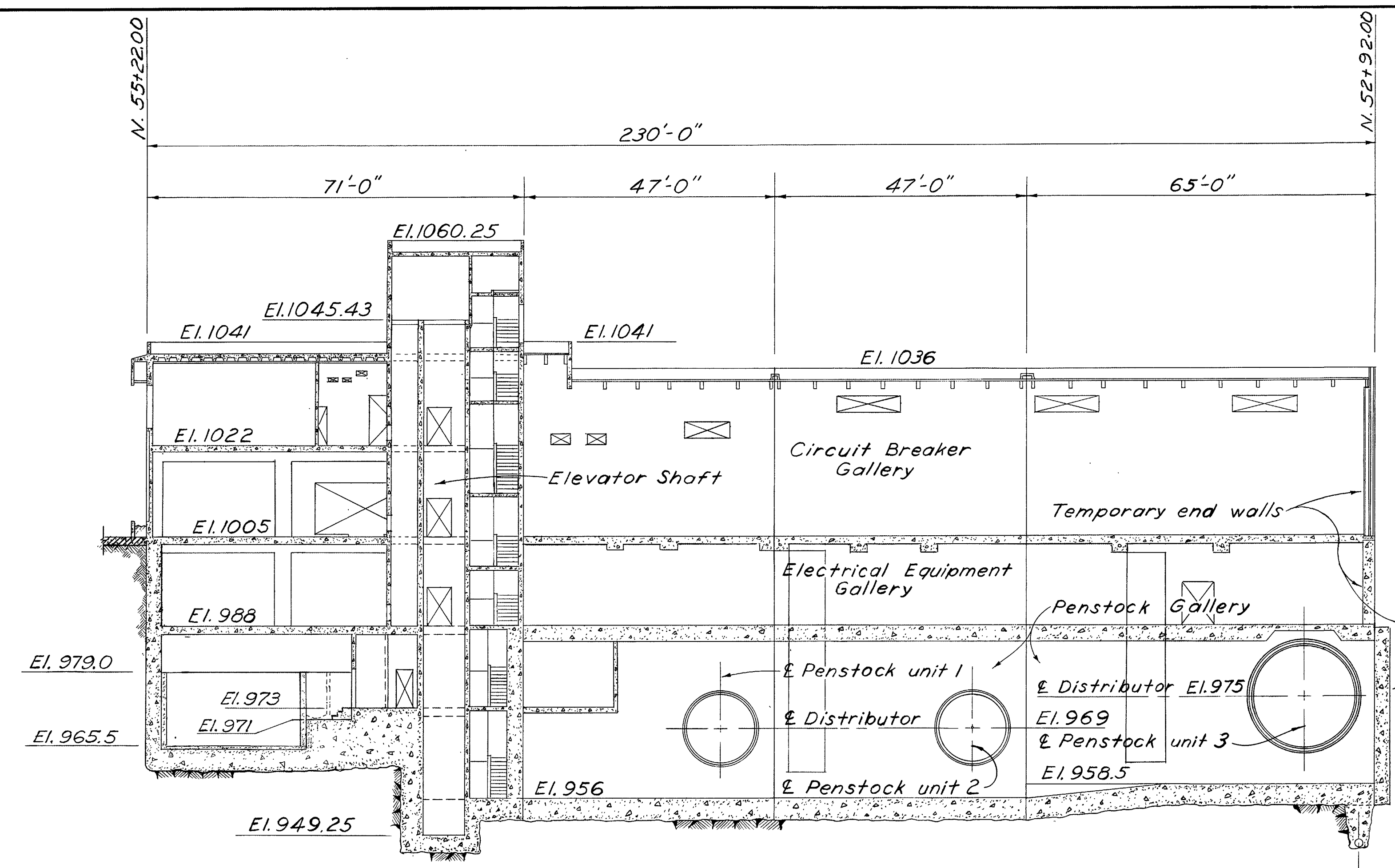
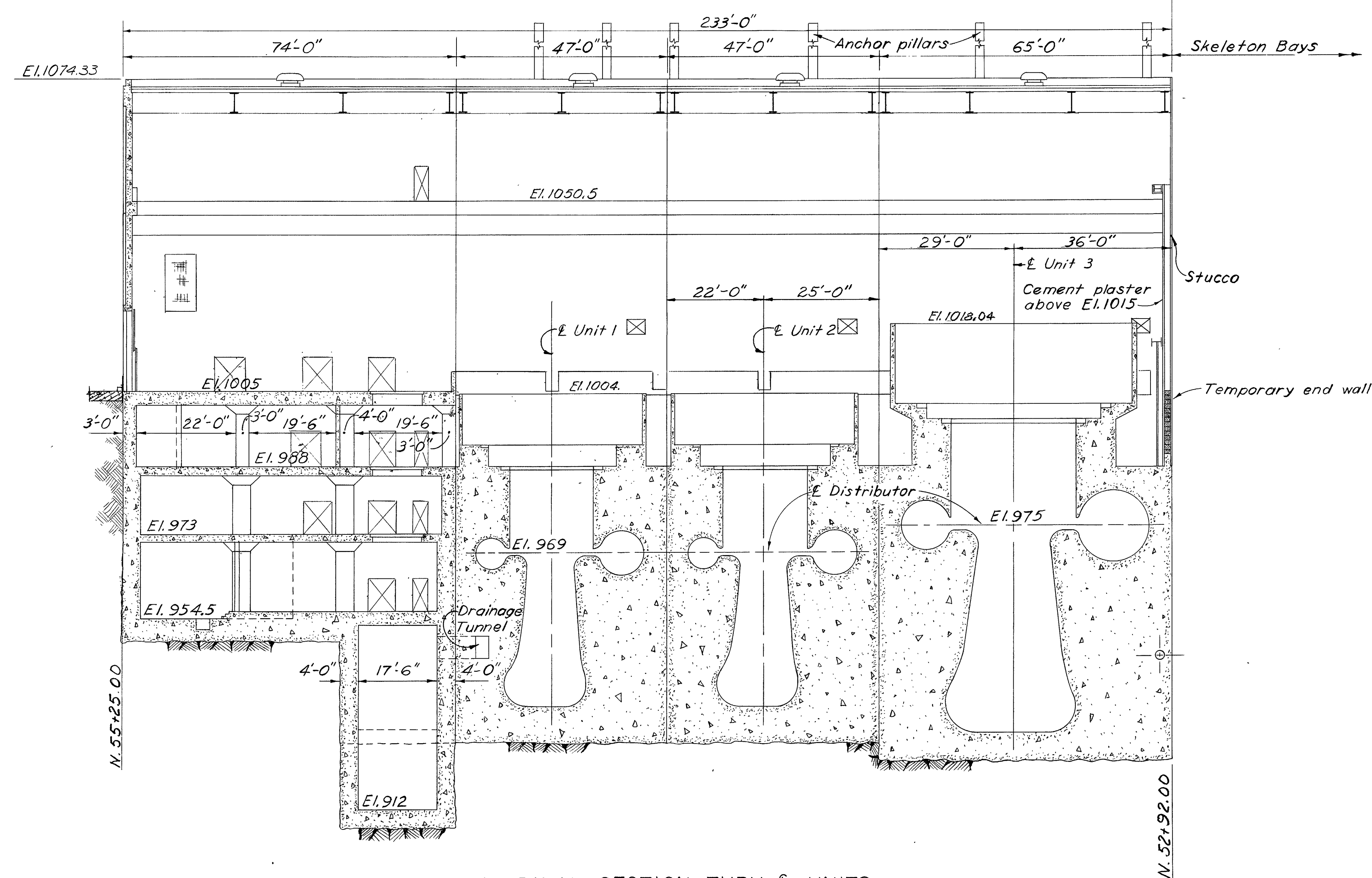
U. S. ARMY ENGINEER DIVISION, N. P.
PORTLAND, OREGON

DESIGNED BY MJS
DRAWN BY PER
CHECKED BY C.J.M.
PREPARED BY *C. J. McLean*
HEAD, HYDRO-ELECTRIC DESIGN BRANCH

APPROVED FOR DIV. ENGINEER DATE 1969 NOV 13
C. J. McLean
CHIEF, HYDRO-ELECTRIC DESIGN BRANCH

SCALE: 1/8"=1'-0"
SPEC. NO.
DWP-6-0-0/5
SHEET 12



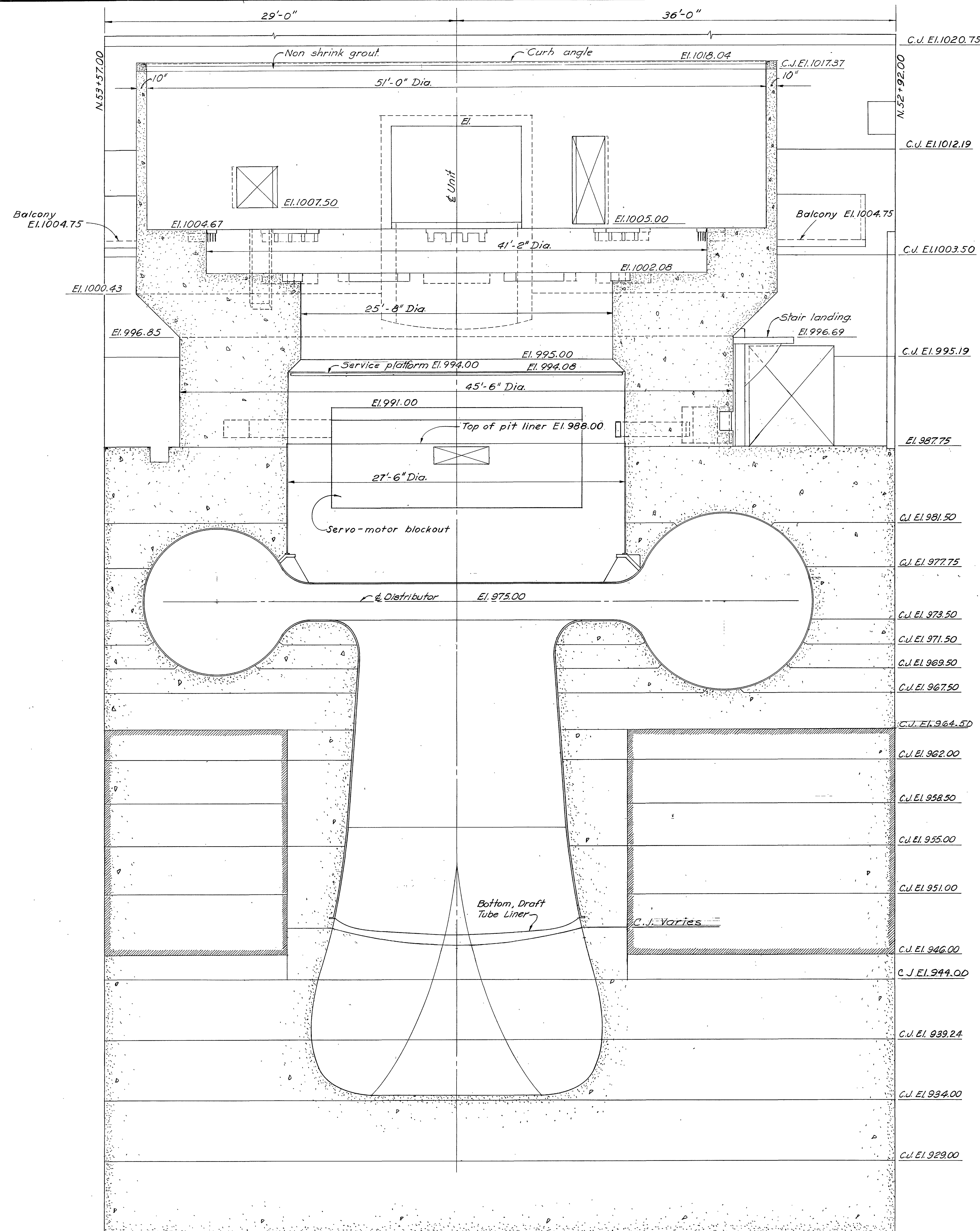


AS CONSTRUCTED
HYDRO ELECTRIC DESIGN BRANCH, NPD

HYDRO ELECTRIC DESIGN BRANCH, NPD

DATE 1975 JUL 31 BY C. J. McGowan

A 75 July 31 As Constructed B 70 Jan 30 Removed roof corbels.		C.J.M. C.J.M.	
REVISION	DATE	DESCRIPTION	BY
U. S. ARMY ENGINEER DIVISION, N. P. PORTLAND, OREGON			
DESIGNED BY <i>M.L.S.</i>		DWORSHAK DAM AND RESERVOIR NORTH FORK CLEARWATER RIVER, IDAHO	
DRAWN BY <i>K.J.C.</i>		POWERHOUSE GENERAL	
CHECKED BY <i>C.J.M.</i>		LONGITUDINAL SECTIONS	
PREPARED BY <i>Karl Drake</i> HEAD, STRUCTURAL SECTION		APPROVED FOR DIV. ENGINEER DATE 1959 NOV 14 <i>Rice</i> CHIEF, ENGINEERING DIVISION SCALE: 1/16" = 1'-0" SPEC. NO.	
SUBMITTED: <i>L. L. Fisher</i> CHIEF, HYDRO-ELECTRIC DESIGN BRANCH		SHEET 6 DWP-1-O-1/6	



AS CONSTRUCTED

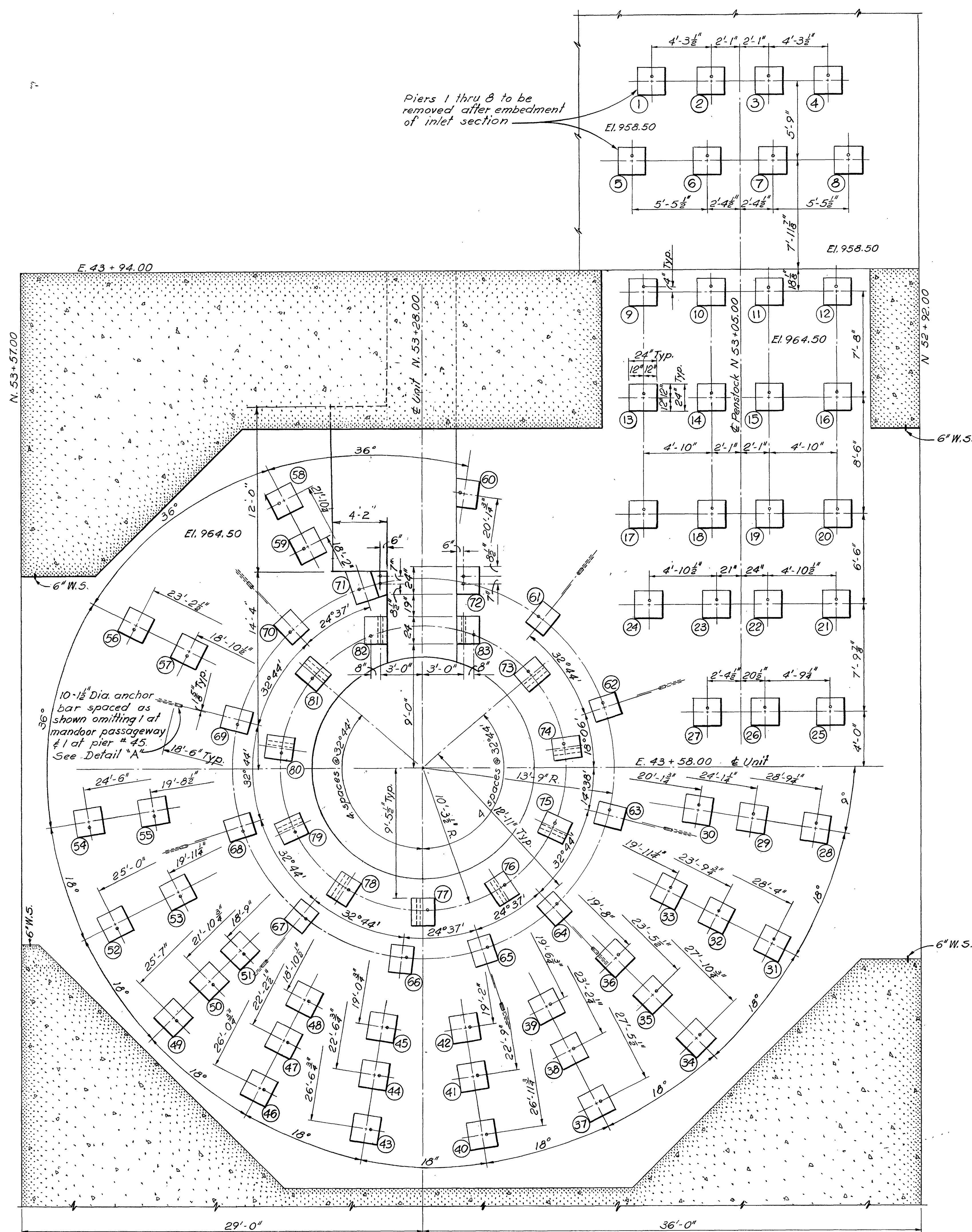
HYDRO ELECTRIC DESIGN BRANCH, NPD

DATE 1975 JUL 31 BY *C. J. M.*

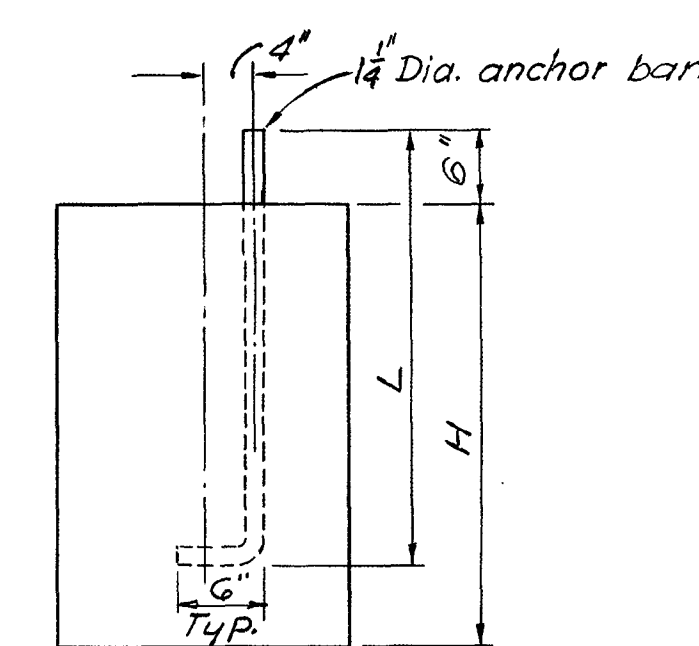
NOTES

For general notes see Dwg. DWP-1-4-2/1
 Additional concrete required because of over excavation
 and the areas enclosed by the symbol will be
 paid for under item (54) or (55); all other concrete will be paid
 for under item (56).

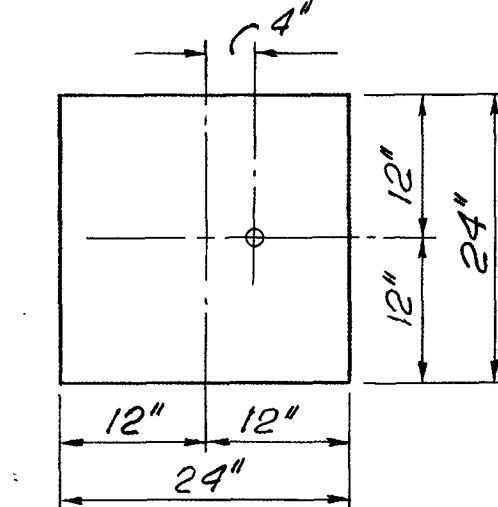
DESIGNED BY	M.S.	DATE	75 July 31	DESCRIPTION	As Constructed	BY	C.J.M.
DRAWN BY	J.W.	DATE	75 Nov 10	DESCRIPTION	Revised top of generator barrel	BY	C.J.M.
CHECKED BY	C.J.M.	DATE	75 Jan 30	DESCRIPTION	Revised generator barrel	BY	C.J.M.
PREPARED BY	<i>Paul C. Brown</i>	DATE	75 Jan 30	DESCRIPTION	Minor revision	BY	C.J.M.
U. S. ARMY ENGINEER DIVISION, N. P. PORTLAND, OREGON							
DWORSHAK DAM AND RESERVOIR NORTH FORK CLEARWATER RIVER, IDAHO POWERHOUSE GENERATOR BAYS - CONCRETE OUTLINE LONGITUDINAL SECTION THRU BAY 3							
SUBMITTED:		APPROVED: FOR DIV. ENGINEER		DATE		1989 NOV 14	
CHIEF, HYDRO-ELECTRIC DESIGN BRANCH		CHIEF, ENGINEERING DIVISION		SCALE AS SHOWN		SPEC. NO.	
SHEET 72		DWP-1-4-4/27					



PLAN
Scale: $\frac{3}{4}$ " = 1'-0"



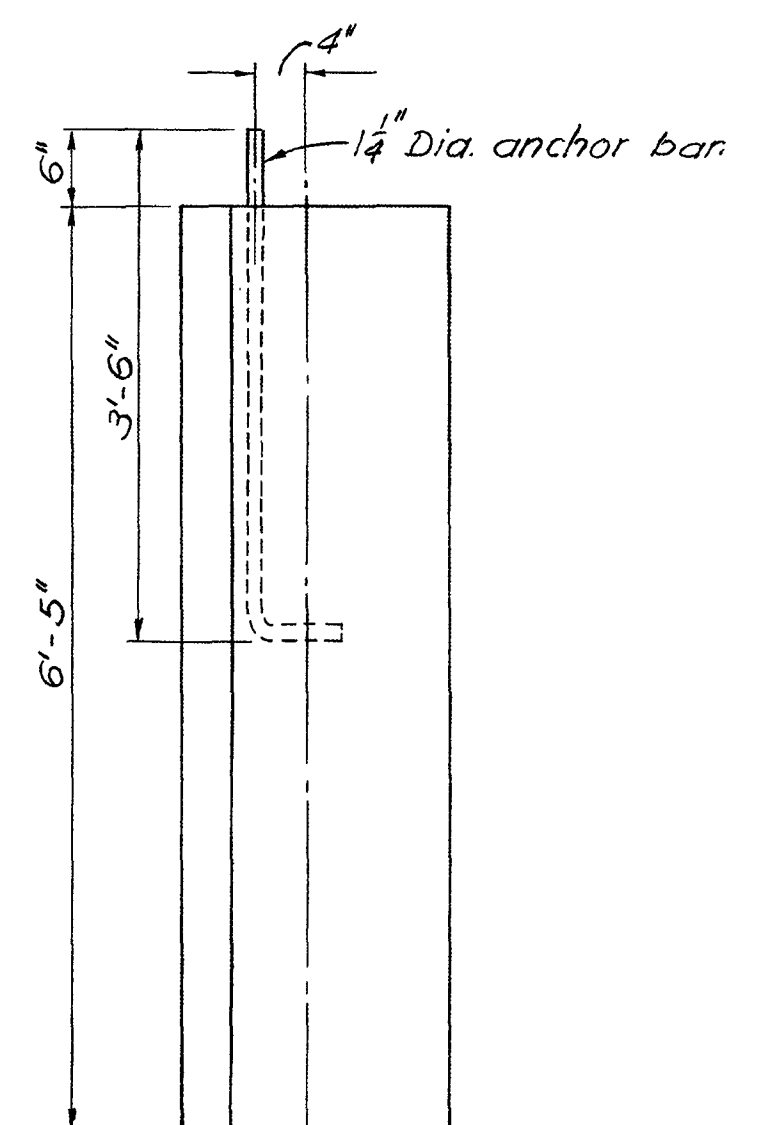
ELEVATION



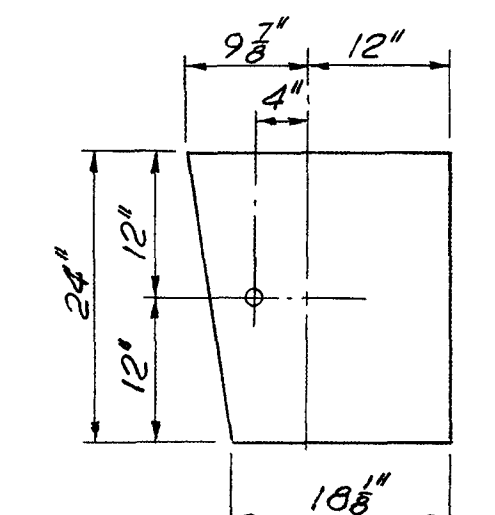
PIERS 1 THRU 59
Scale: $\frac{3}{4}$ " = 1'-0"

PIER No.	H	1/4" DIA. RODS
1	9'-10"	3'-6"
2	6'-1 1/2"	3'-6"
3	6'-1 1/2"	3'-6"
4	9'-10"	3'-6"
5	9'-8"	3'-6"
6	5'-7"	3'-6"
7	5'-7"	3'-6"
8	9'-8"	3'-6"
9	4'-4"	3'-6"
10	10"	14"
11	10"	14"
12	4'-4"	3'-6"
13	4'-4"	3'-6"
14	10"	14"
15	10"	14"
16	4'-4"	3'-6"
17	4'-4"	3'-6"
18	10"	14"
19	10"	14"
20	4'-4"	3'-6"
21	4'-3 1/2"	3'-6"
22	10"	14"
23	10"	14"
24	4'-3 1/2"	3'-6"
25	4'-4"	3'-6"
26	11 1/2"	14"
27	11 1/2"	14"
28	4'-5"	3'-6"
29	14"	14"
30	14"	14"

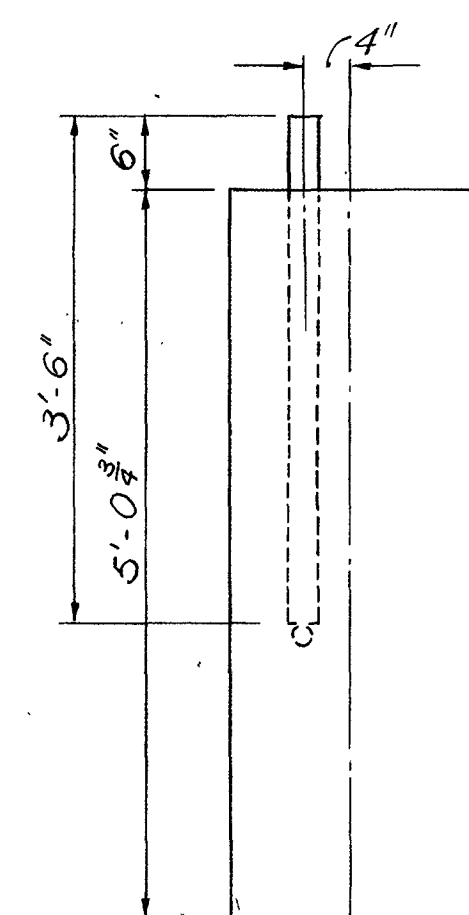
PIER No.	H	1/4" DIA. RODS
31	4'-7"	3'-6"
32	16"	14"
33	16"	14"
34	4'-7 1/2"	3'-6"
35	18"	24"
36	18"	24"
37	4'-9 1/2"	3'-6"
38	21 1/2"	24"
39	21 1/2"	24"
40	4'-9 1/2"	3'-6"
41	22 1/2"	24"
42	22 1/2"	24"
43	4'-10 1/2"	3'-6"
44	2'-1 1/2"	24"
45	2'-1 1/2"	24"
46	5'-0 1/2"	3'-6"
47	2'-5"	24"
48	2'-5"	24"
49	2'-2"	3'-6"
50	2'-7 1/2"	24"
51	2'-7 1/2"	24"
52	5'-4"	3'-6"
53	2'-9"	24"
54	5'-5"	3'-6"
55	2'-11 1/2"	3'-6"
56	5'-9"	3'-6"
57	3'-6"	3'-6"
58	6'-1"	3'-6"
59	3'-11"	3'-6"



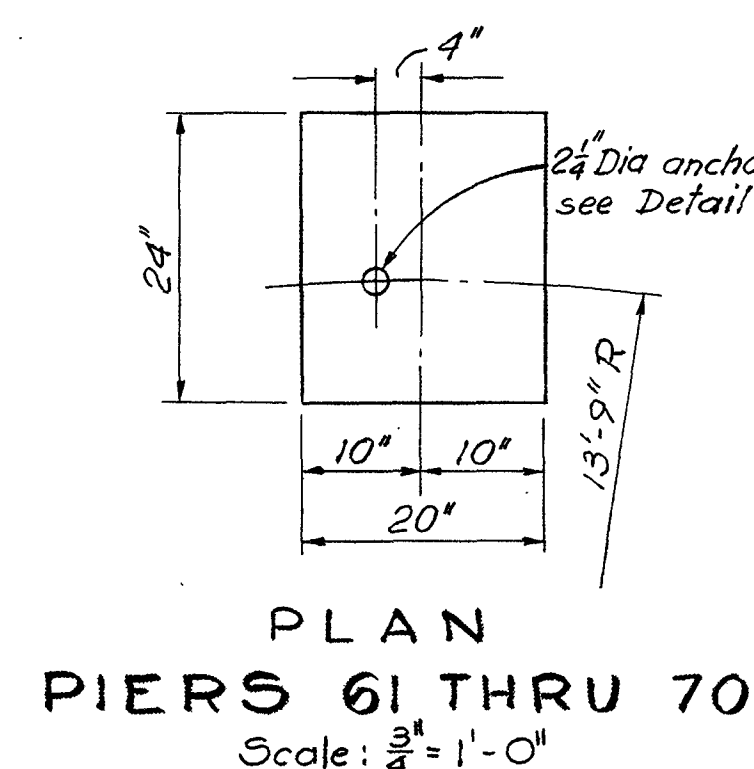
ELEVATION



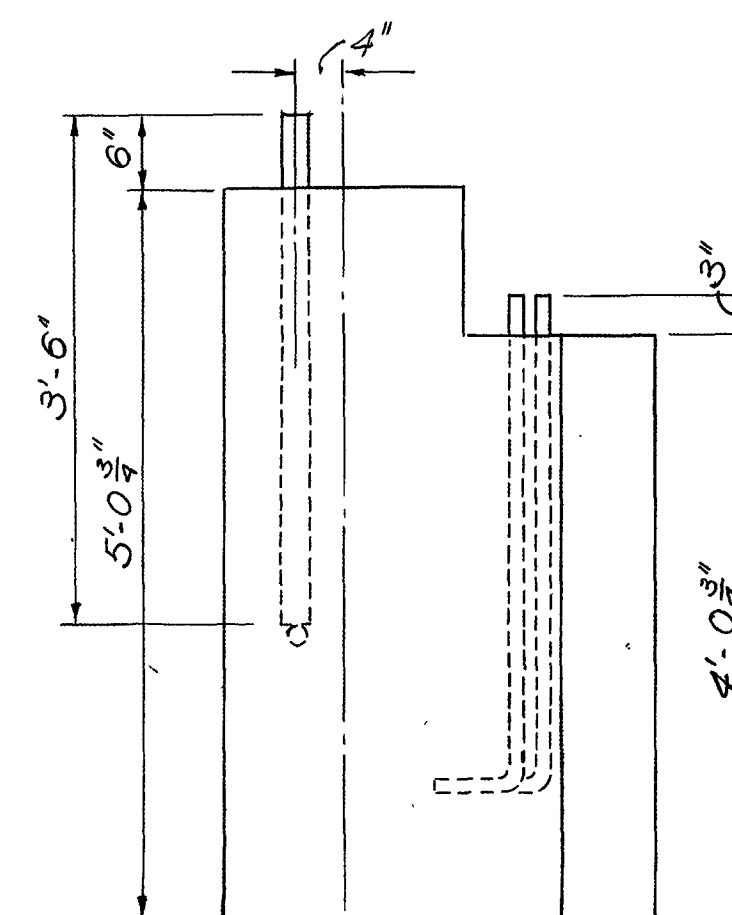
PLAN
PIER 60
Scale: $\frac{3}{4}$ " = 1'-0"



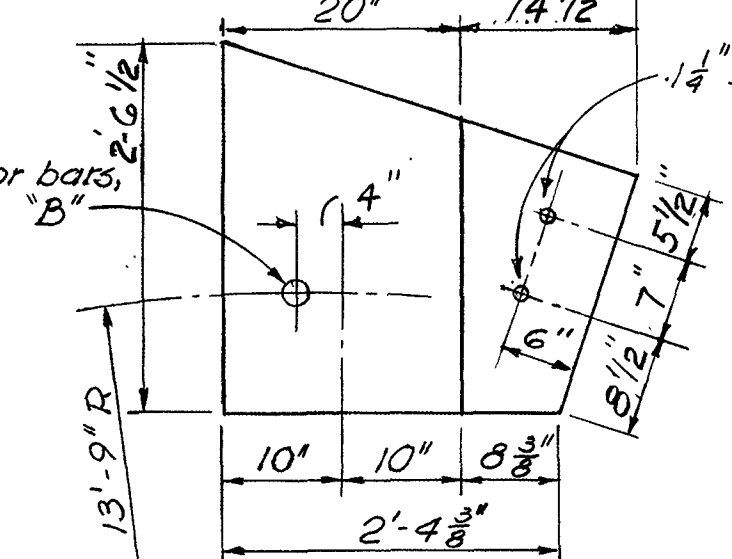
ELEVATION



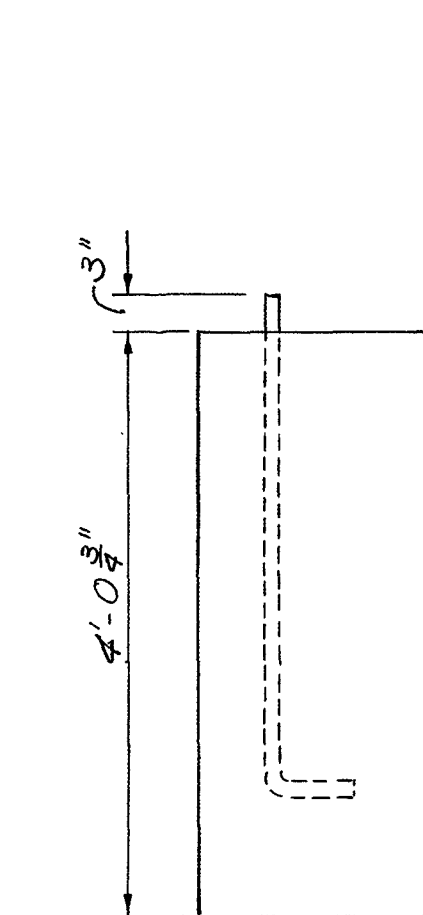
PLAN
PIERS 61 THRU 70
Scale: $\frac{3}{4}$ " = 1'-0"



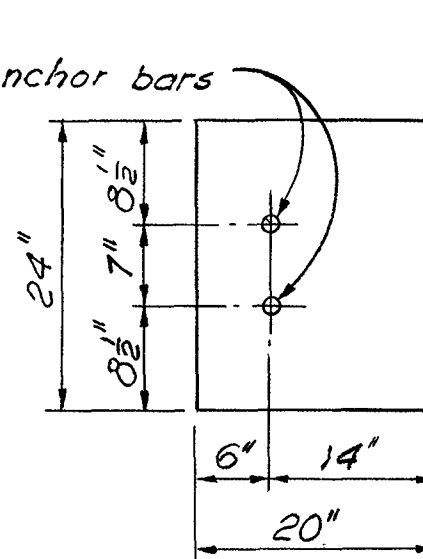
ELEVATION



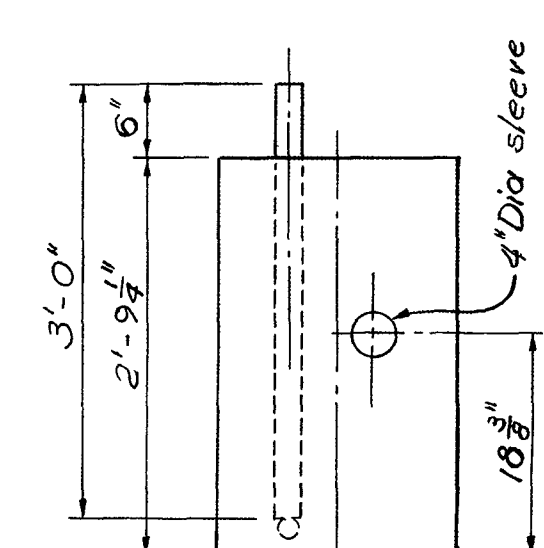
PLAN
PIER 71
Scale: $\frac{3}{4}$ " = 1'-0"



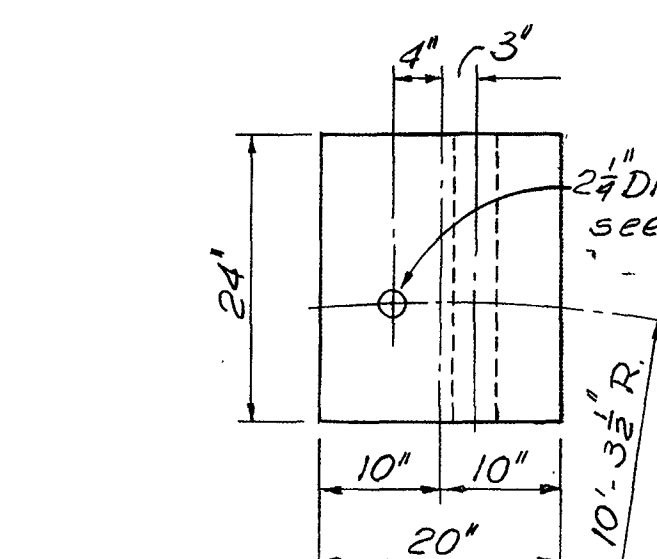
ELEVATION



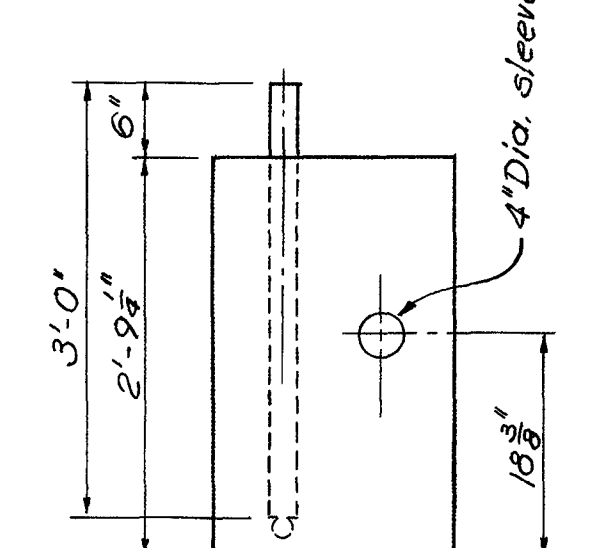
PLAN
PIER 72
Scale: $\frac{3}{4}$ " = 1'-0"



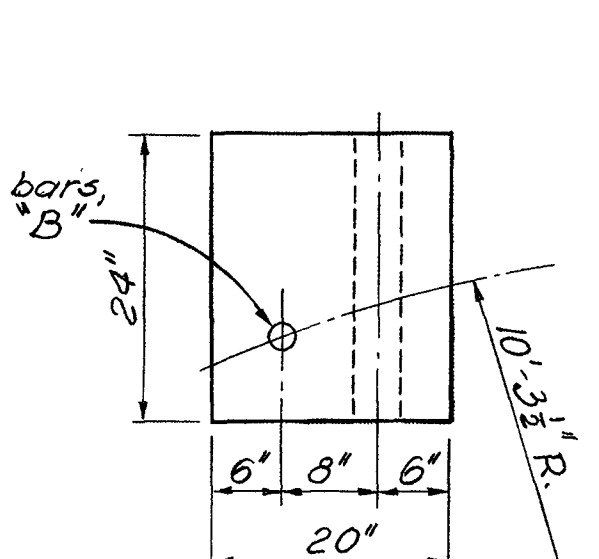
ELEVATION



PLAN
PIERS 73 THRU 81
Scale: $\frac{3}{4}$ " = 1'-0"



ELEVATION



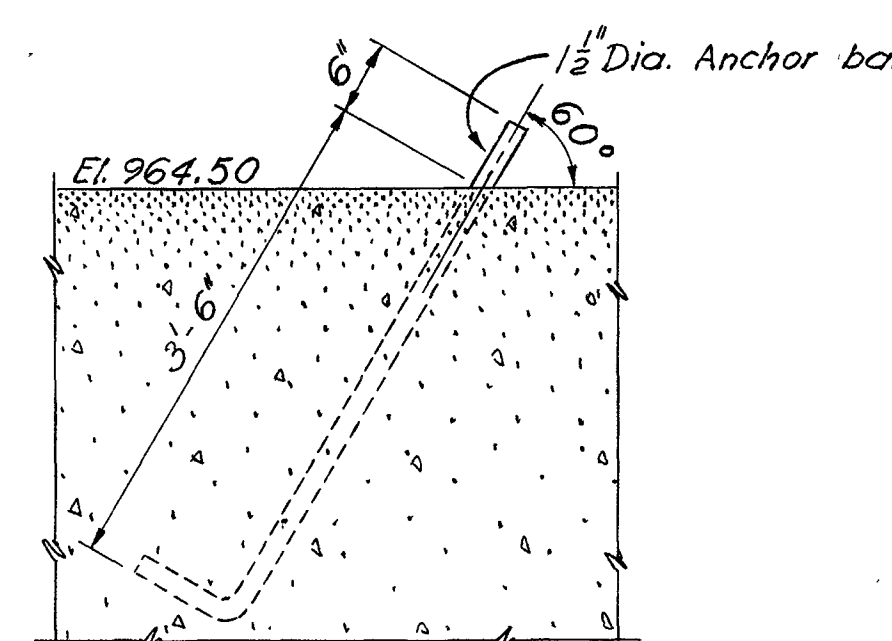
PLAN
PIER 82 AS SHOWN
PIER 83 OPP HAND
Scale: $\frac{3}{4}$ " = 1'-0"

NOTES.

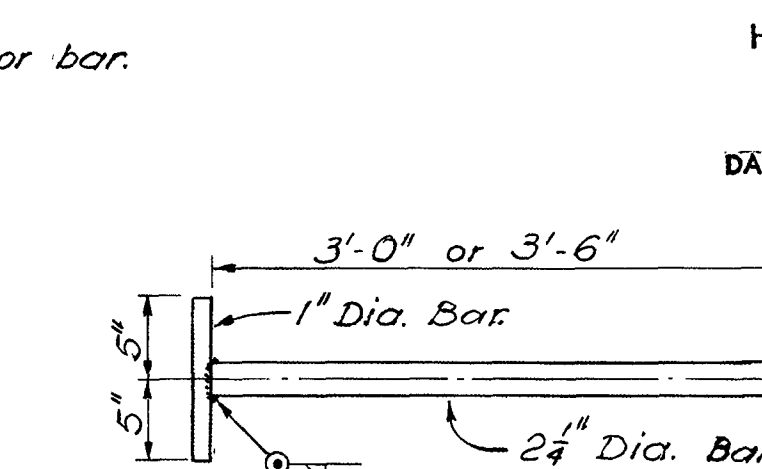
For general notes see Dwg. DWP-1-4-2/1
All anchor bars to be A-36 steel (60)
Layout details subject to change pending approval of manufacturers drawings.
Spiral case blackout waterstops to extend continuously from El. 964.00 to El. 987.75 Downstream waterstops to extend continuously from El. 964.00 to El. 977.75

AS CONSTRUCTED
HYDRO ELECTRIC DESIGN BRANCH, NPD

DATE 1975 JUL 81 BY *C.J.H.*

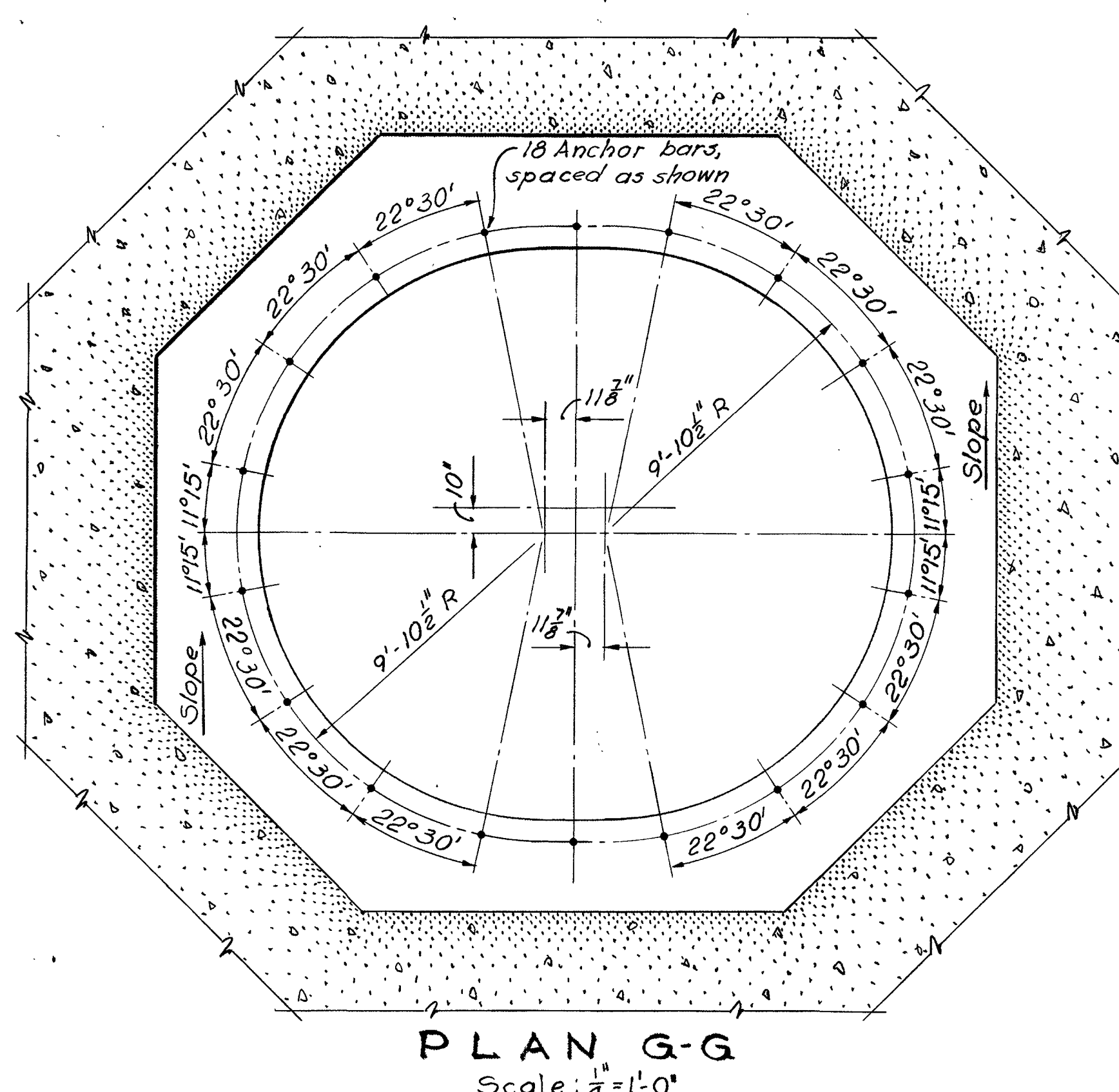
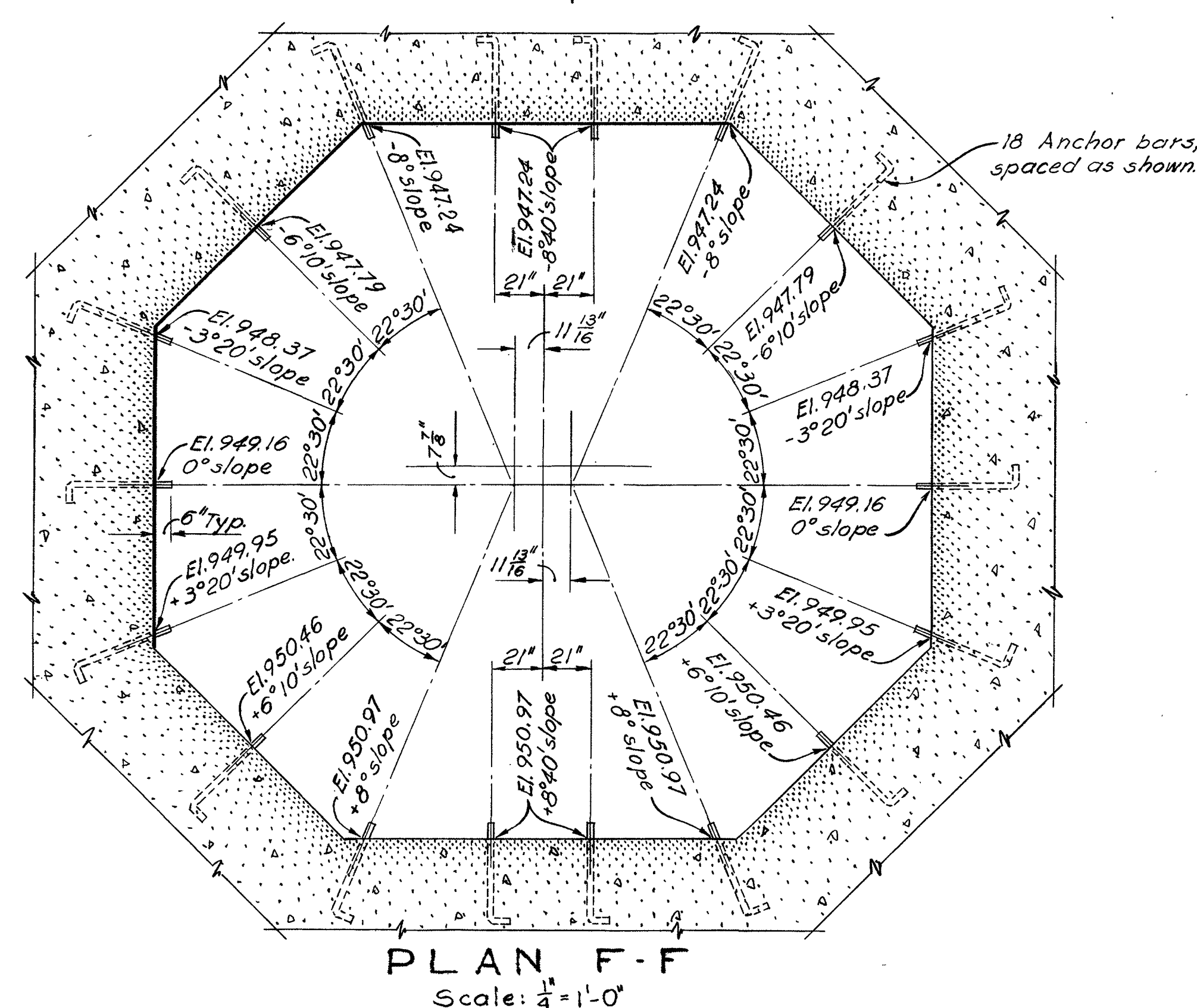
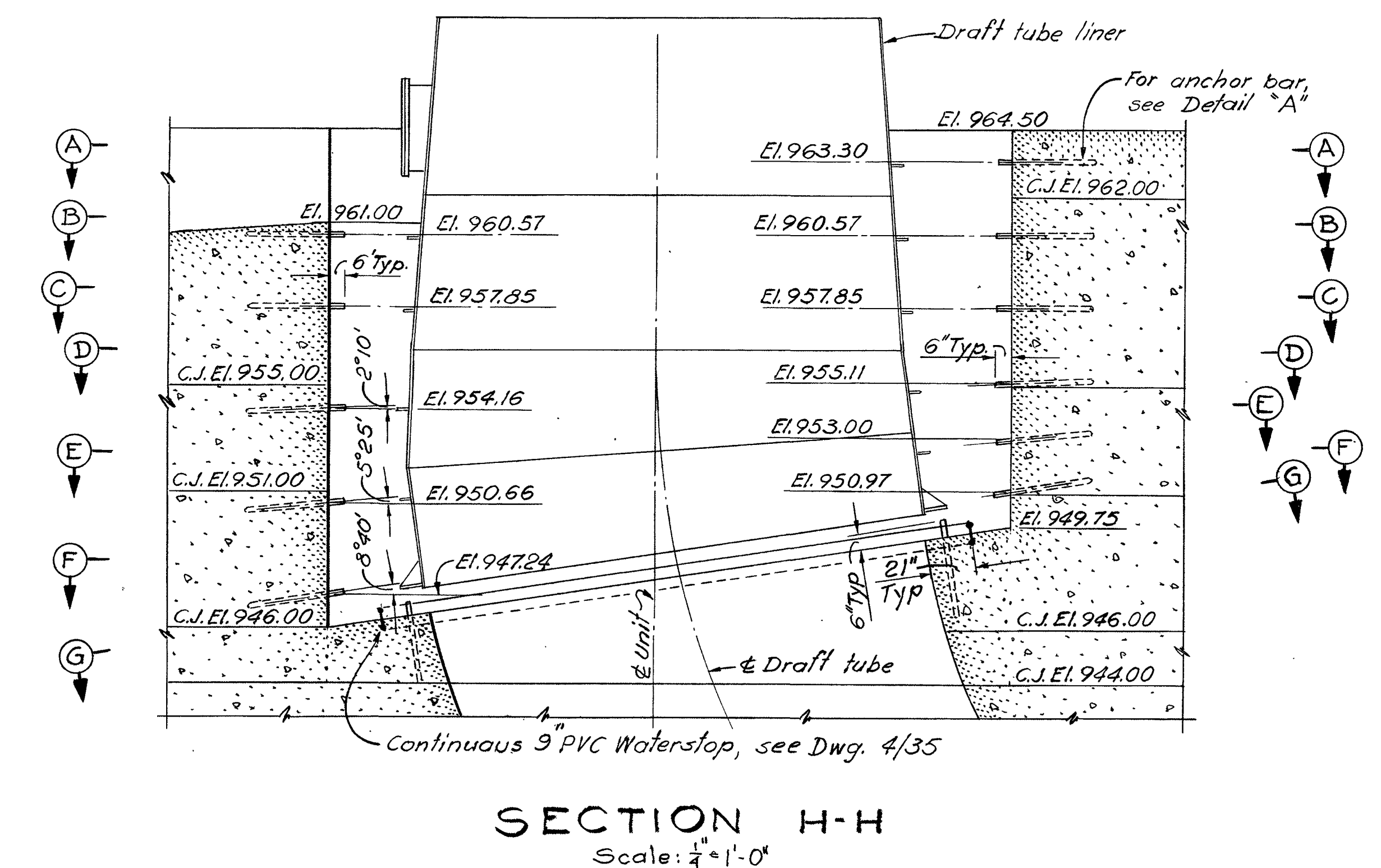
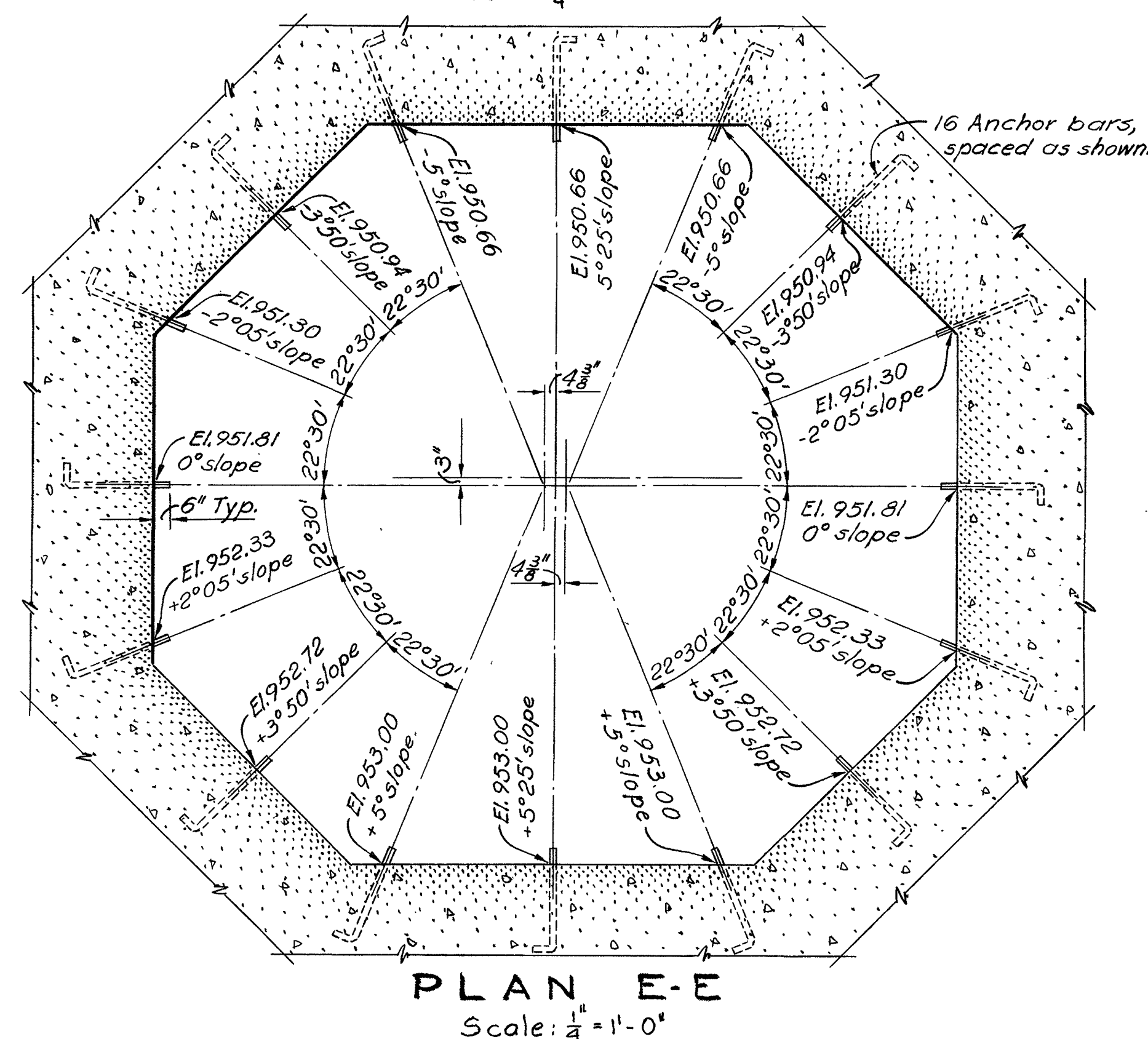
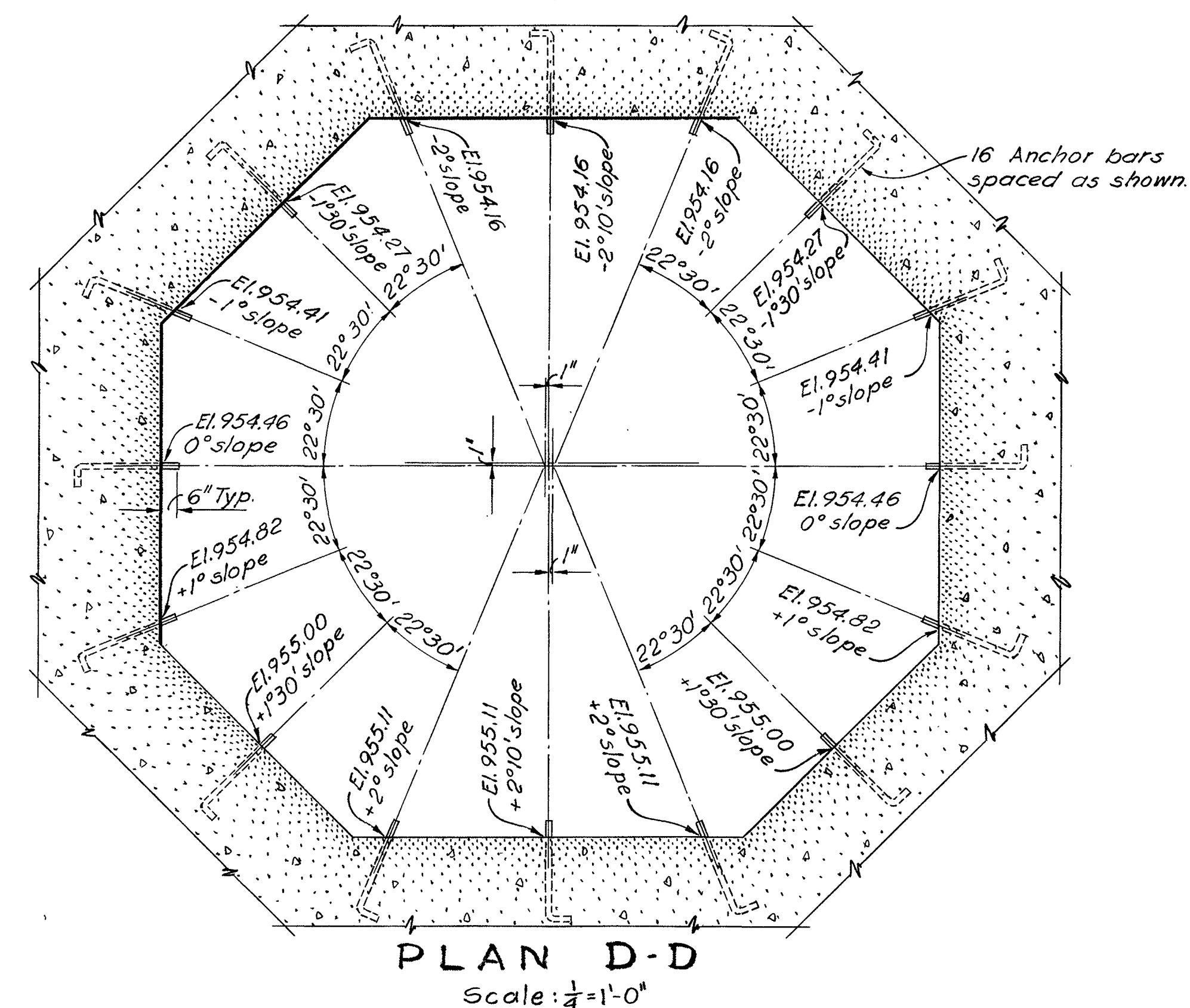
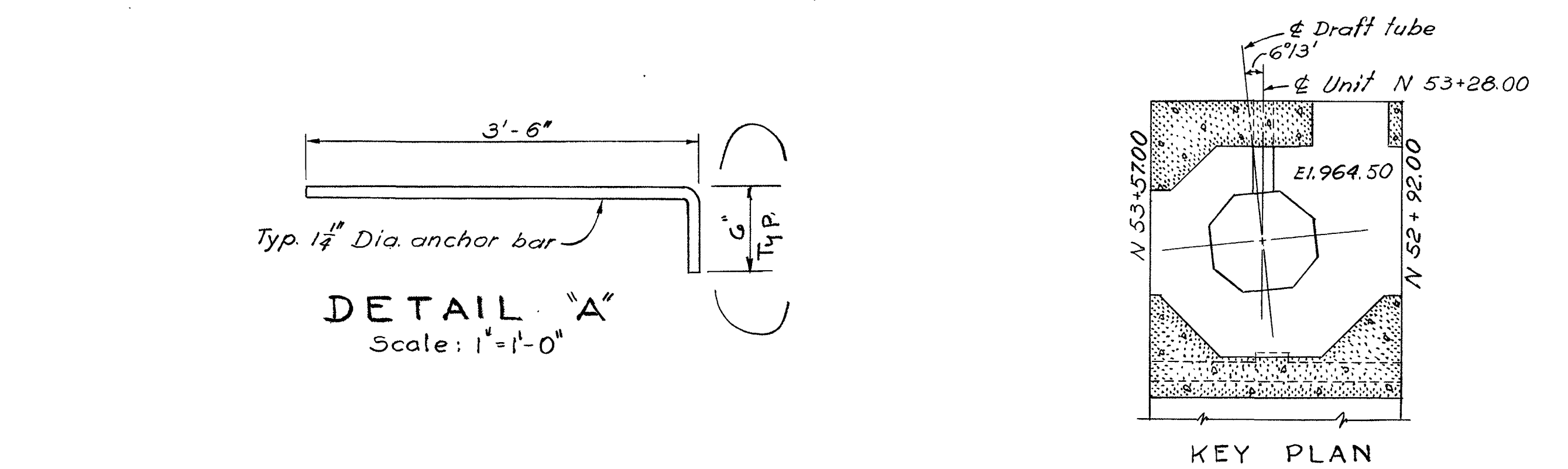
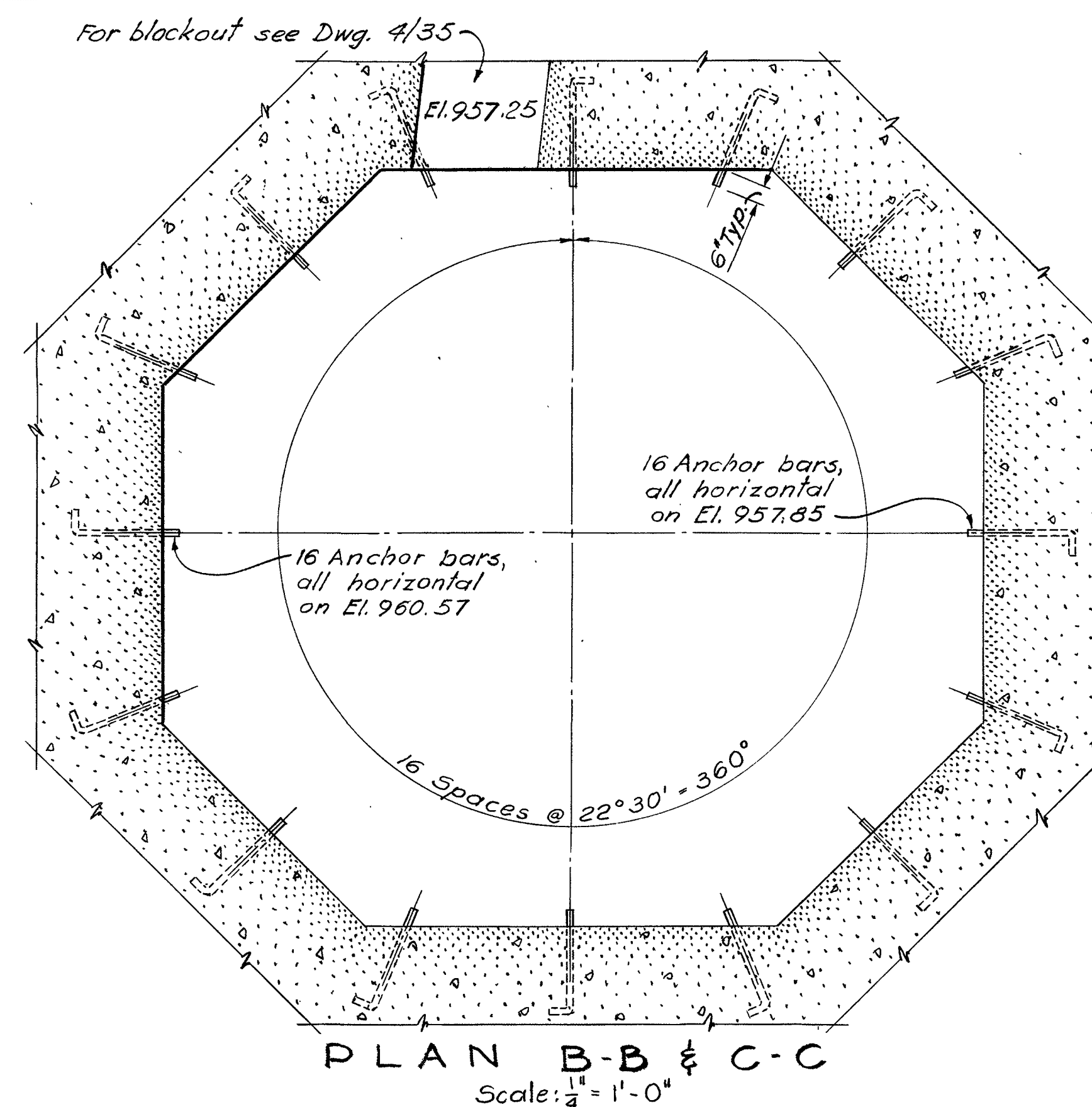
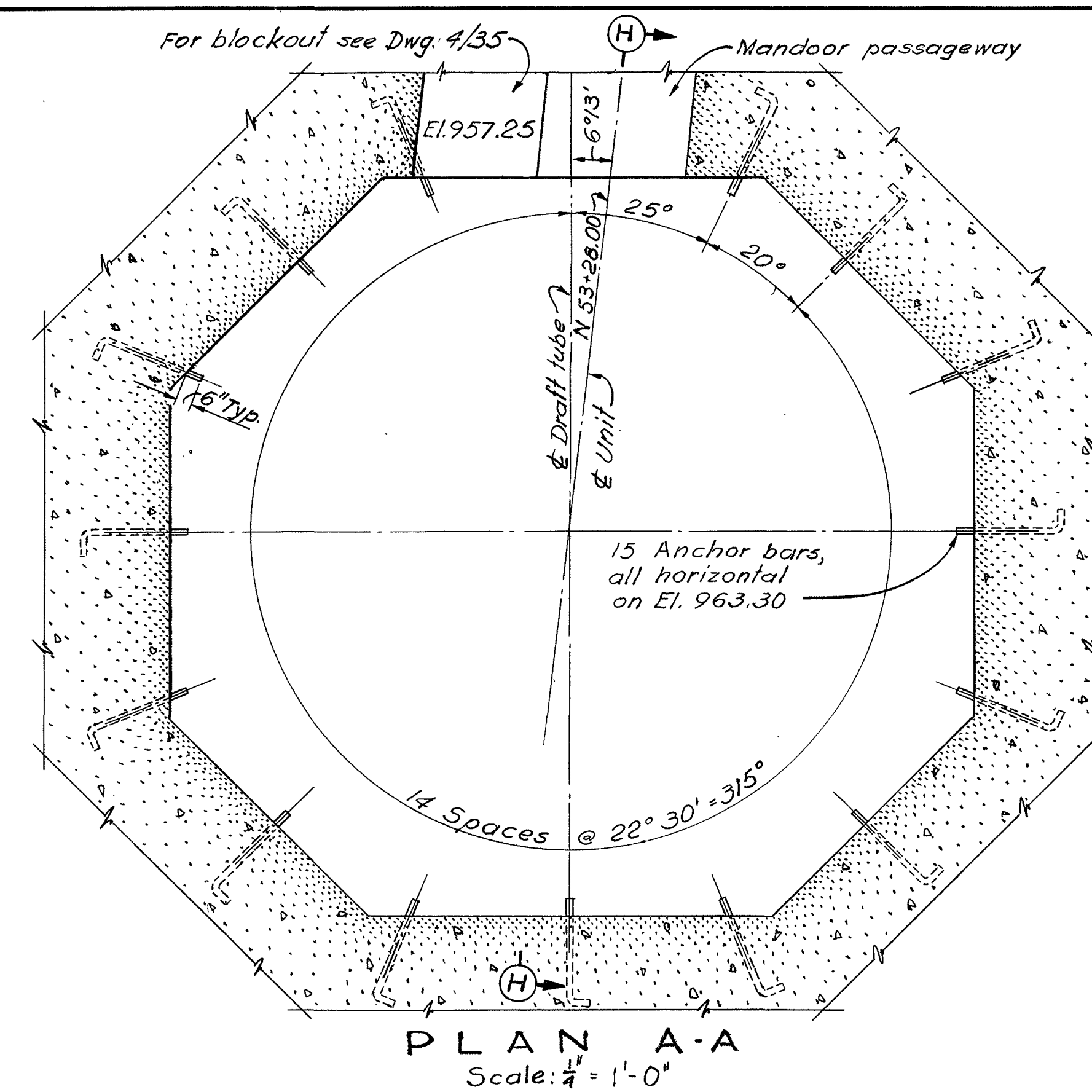


DETAIL "A"
Scale: $\frac{3}{4}$ " = 1'-0"



DETAIL "B"
Scale: 1" = 1'-0"

	<u>E</u>	75 July 31	As Constructed	C.J.M.
	<u>D</u>	70 Dec 23	Added waterstops	C.J.M.
	<u>C</u>	70 Aug 11	Revised Pier 71	C.J.M.
	<u>B</u>	70 Mar 4	Minor revision	C.J.M.
	<u>A</u>	70 Jan 30	Minor revisions	C.J.M.
REVISION	DATE	DESCRIPTION		BY
U. S. ARMY ENGINEER DIVISION, N. P. PORTLAND, OREGON				
DWORSHAK DAM AND RESERVOIR NORTH FORK CLEARWATER RIVER, IDAHO POWERHOUSE GENERATOR BAYS — CONCRETE OUTLINE TURBINE FOUNDATION DETAILS BAY 3				
DESIGNED BY: <u>A.C.</u>				
DRAWN BY: <u>V.S.</u>				
CHECKED BY: <u>C.J.H.</u>				
PREPARED BY: <u>Paul E. Drake</u>				
HEAD, STRUCTURAL SECTION				
SUBMITTED: <u>P.L. Drake</u>				
CHIEF, HYDRO-ELECTRIC DESIGN BRANCH				
		APPROVED: FOR DIV. ENGINEER <u>P.L. Drake</u> CHIEF, ENGINEERING DIVISION		
		DATE: 1968 NOV. 14		
		SCALE AS SHOWN SPEC. NO. _____		
		SHEET 70 DWP-1-4-4/34		



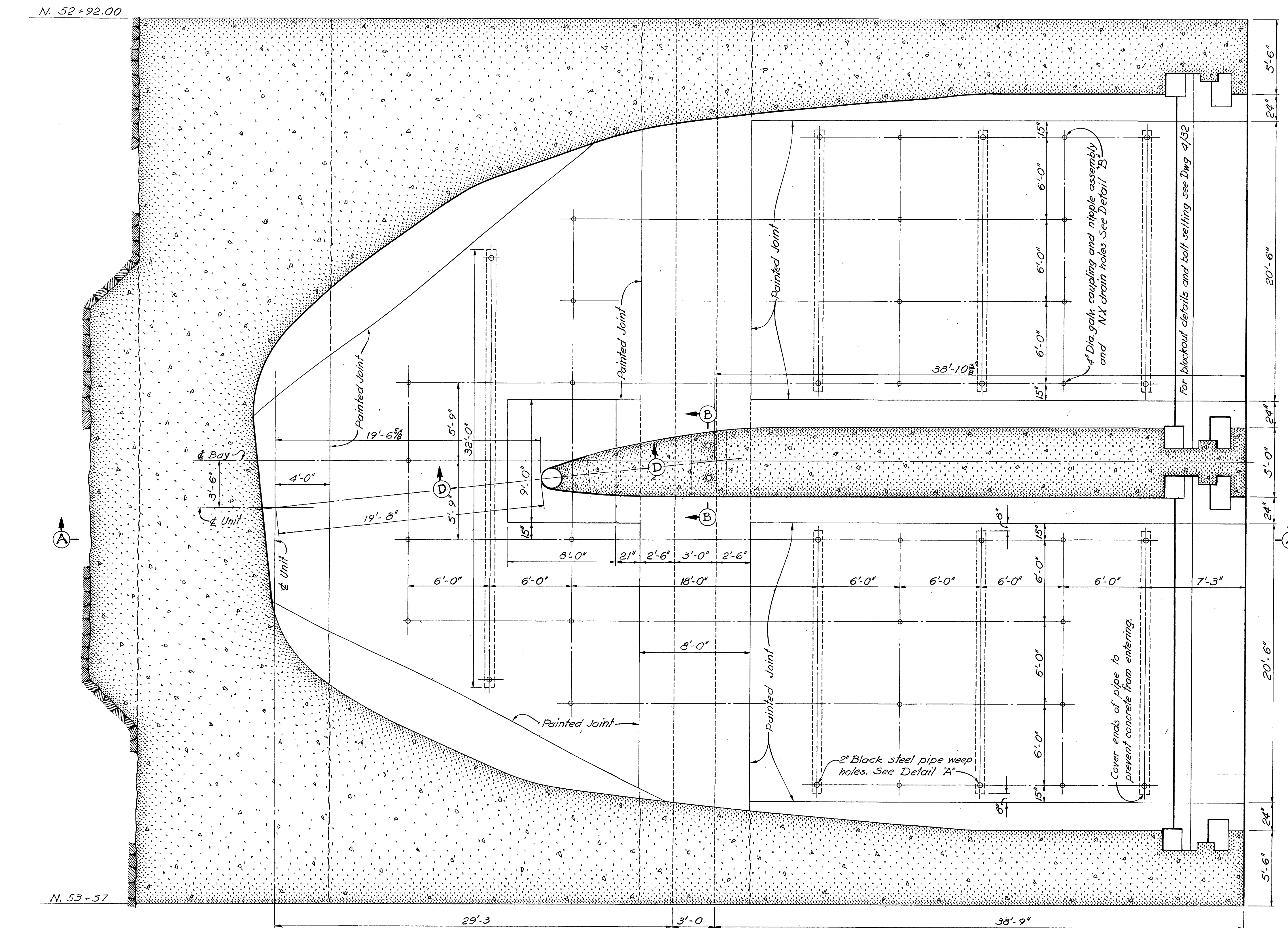
NOTES.

For general notes see Dwg. DWP-1-4-2/1
All sections shown are normal to vertical & of unit
except section G-G
For location and dimensions of blackout for draft
tube liner see Dwg. DWP-1-4-4/35
Draft tube liner jacks and tie rod assemblies furnished
by equipment manufacturer are subject to change
pending approval of manufacturers drawings.
All anchor bars to be A-36 steel. (60)

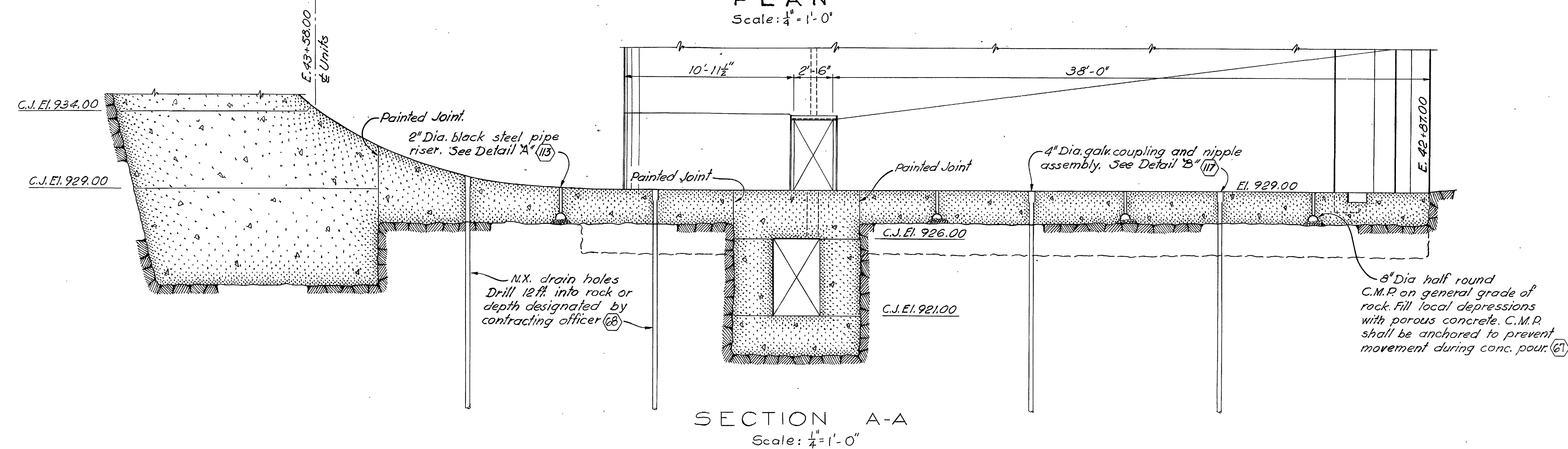
AS CONSTRUCTED
HYDRO ELECTRIC DESIGN BRANCH, NPD

DATE 1975 JUL 31 BY C. J. McLean

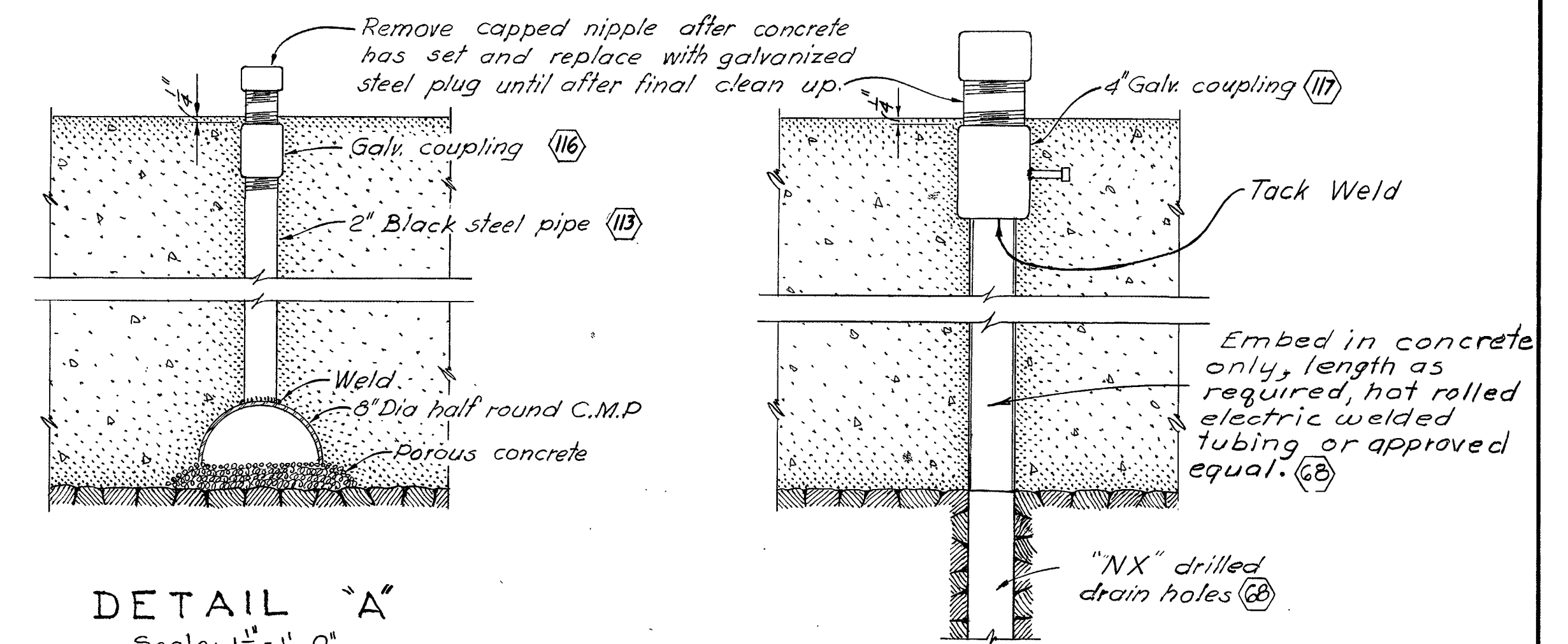
C	75 July 31	As Constructed	C.J.M.
A	70 Aug 11	Added blockout & PVC Waterslop	C.J.M.
A	70 Jan 30	Minor revisions	C.J.M.
REVISION	DATE	DESCRIPTION	BY
U. S. ARMY ENGINEER DIVISION, N. P. PORTLAND, OREGON DWORSHAK DAM AND RESERVOIR NORTH FORK CLEARWATER RIVER, IDAHO POWERHOUSE GENERATOR BAYS — CONCRETE OUTLINE DRAFT TUBE LINER BLOCKOUT & ANCHOR DETAILS BAY 3			
DESIGNED BY: <i>A.C.</i>			
DRAWN BY: <i>V.S.</i>			
CHECKED BY: <i>C.H.M.</i>			
PREPARED: <i>Paul E. Drake</i> HEAD, STRUCTURAL SECTION			
SUBMITTED: <i>L. E. Fisher</i>	APPROVED: FOR DIV. ENGINEER DATE: 1953 NOV 14		
CHIEF, HYDRO-ELECTRIC DESIGN BRANCH	CHIEF, ENGINEERING DIVISION <i>Wes Barnard</i> SCALE AS SHOWN SPEC. NO.		
SHEET 78		DWP-1-4-4/33	



PLAN
Scale: $\frac{1}{4}'' = 1'-0''$

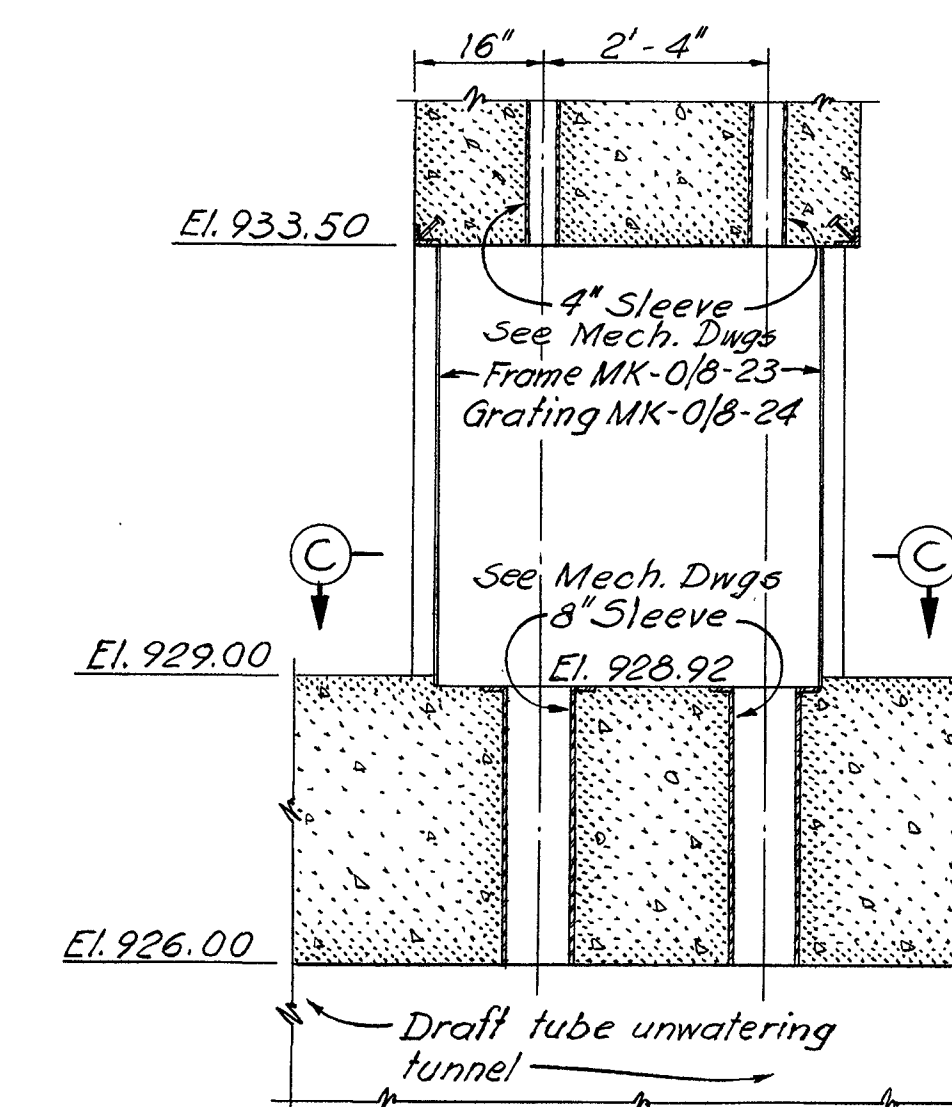


SECTION A-A
Scale: $\frac{1}{4}'' = 1'-0''$

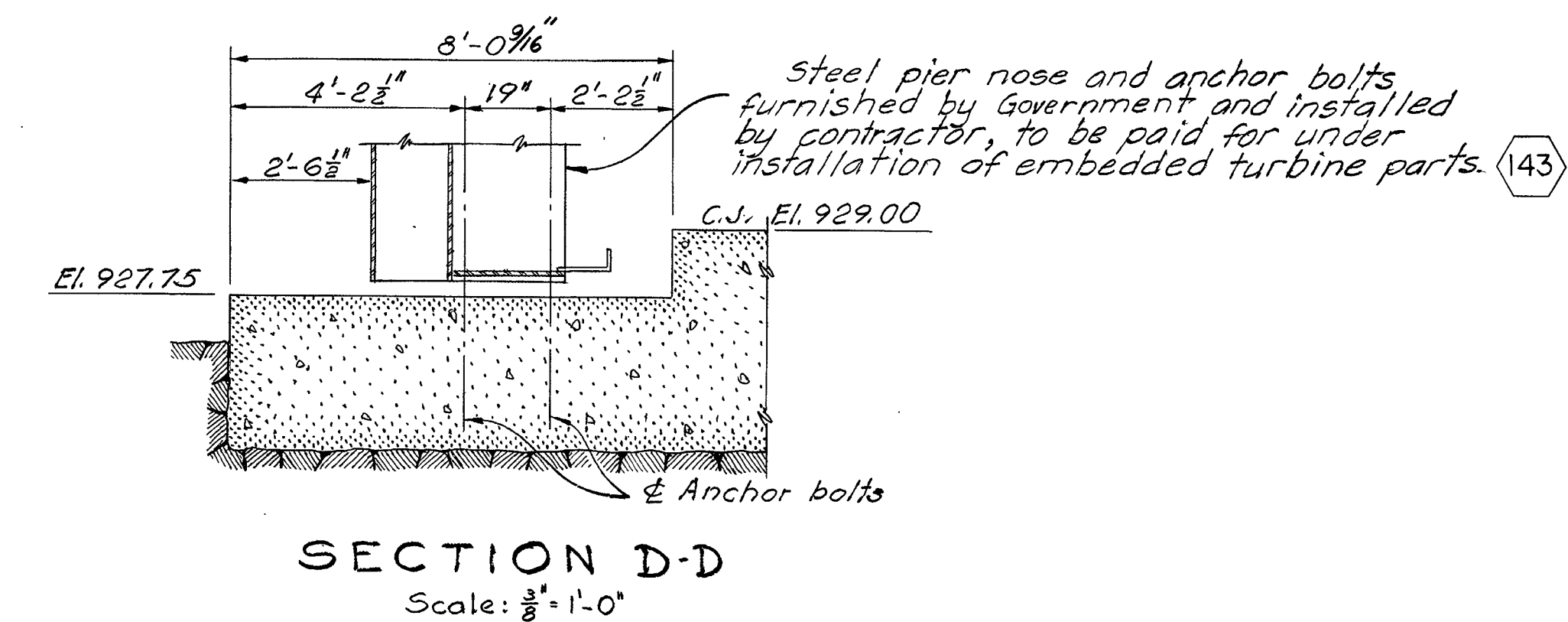


DETAIL "A"
Scale: $\frac{1}{2}'' = 1'-0''$

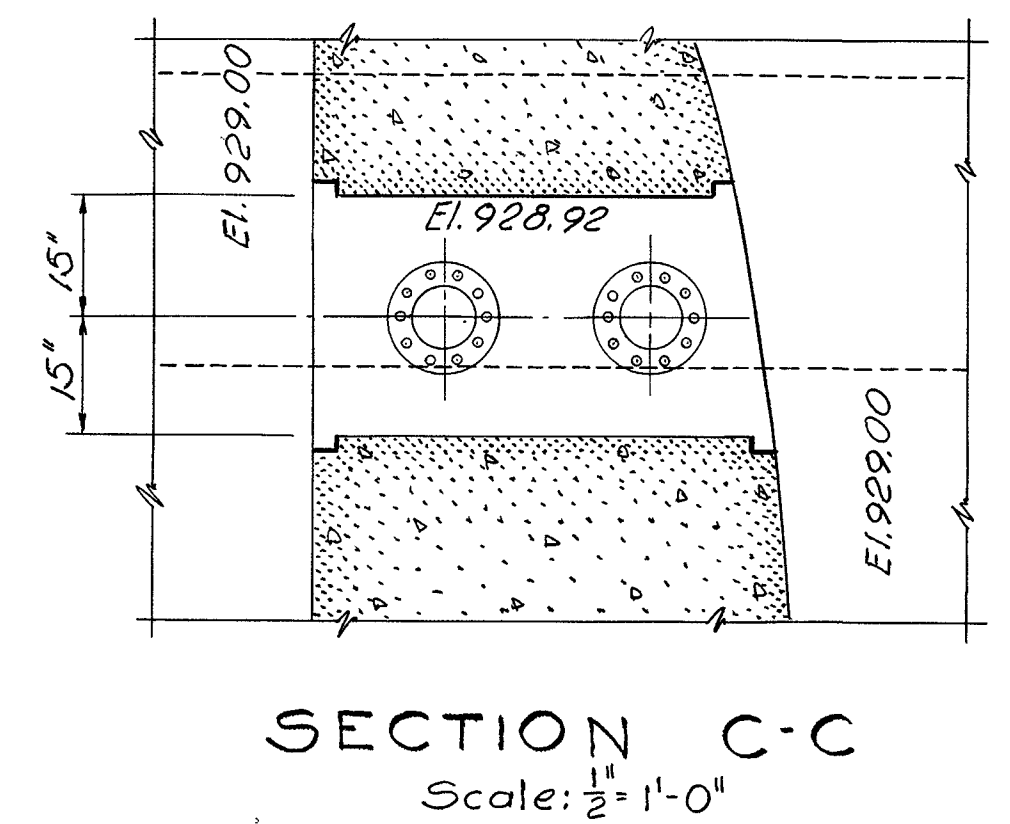
DETAIL "B"
Scale: $\frac{1}{2}'' = 1'-0''$



SECTION B-B
Scale: $\frac{1}{2}'' = 1'-0''$



SECTION D-D
Scale: $\frac{1}{2}'' = 1'-0''$



SECTION C-C
Scale: $\frac{1}{2}'' = 1'-0''$

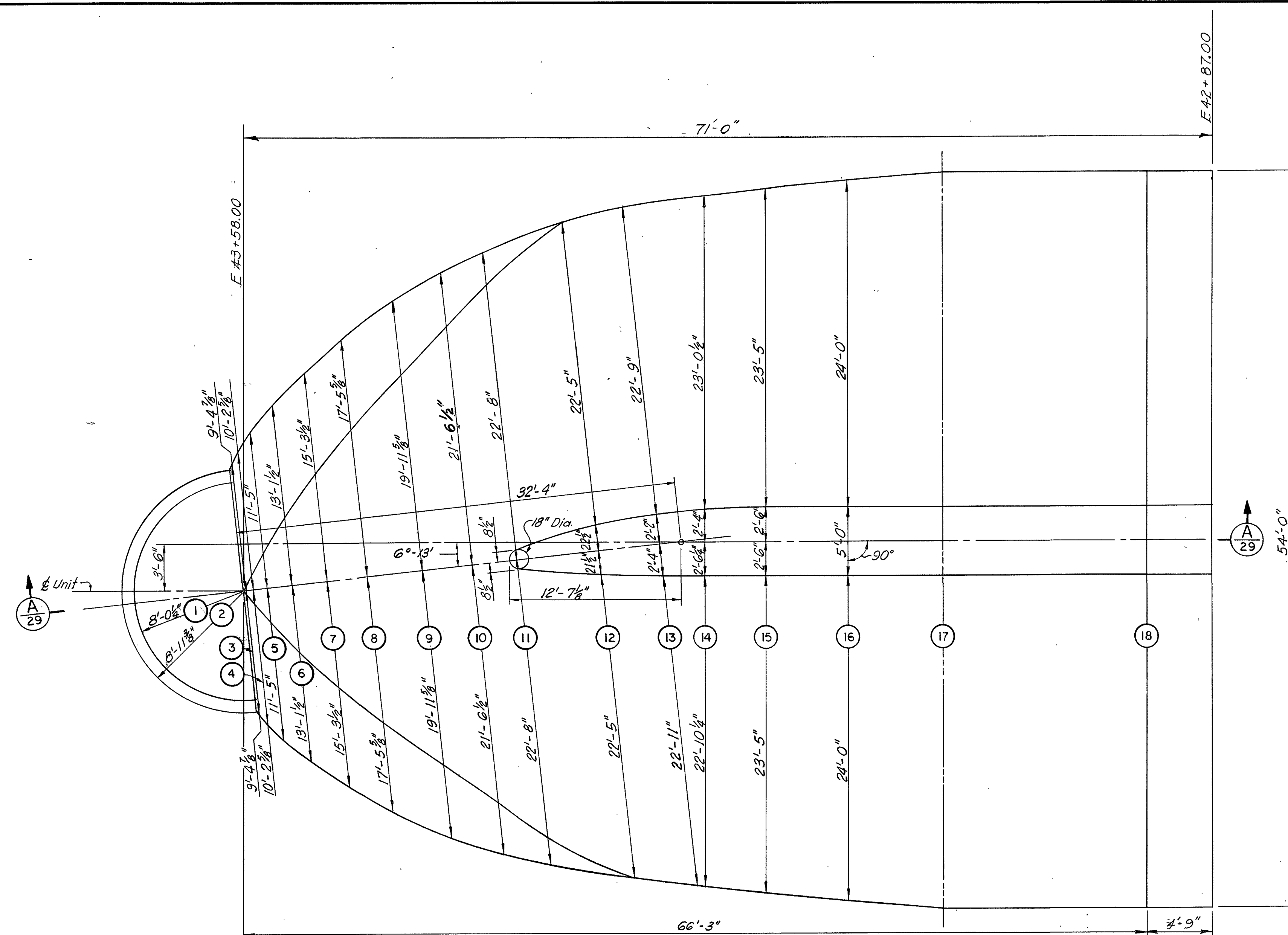
NOTES

For general notes see Dwg. DWP-1-4-2/1
Uniformed surfaces of draft tube to have float finish.
See Dwg. 4/91 for 3" diameter anchor holes.

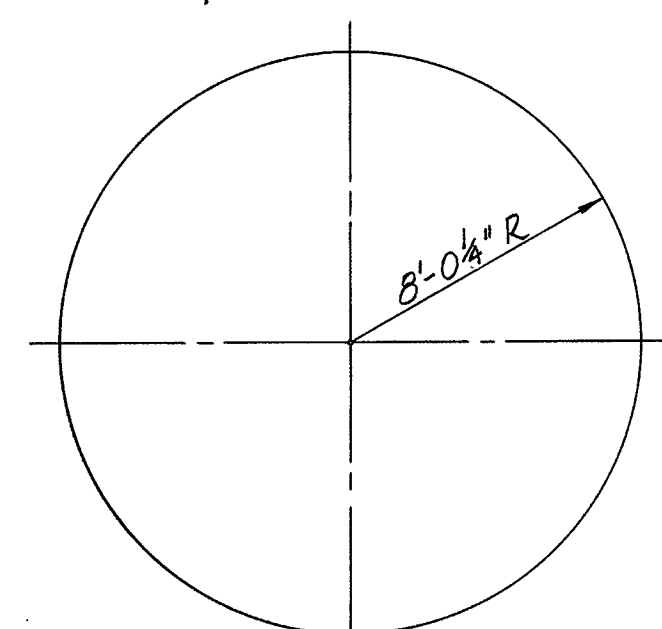
AS CONSTRUCTED
HYDRO ELECTRIC DESIGN BRANCH, NPD

DATE 1975 JUL 31 BY C.J. McQuinn

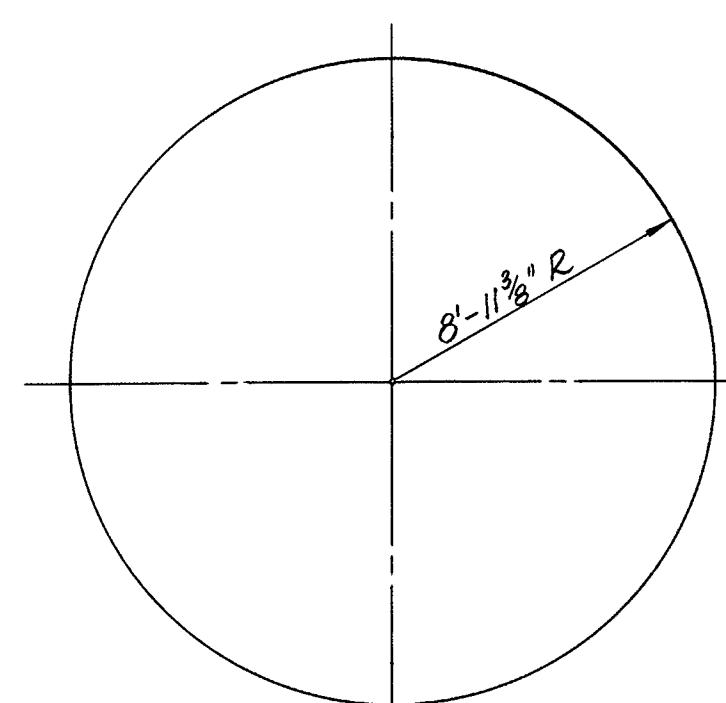
DESIGNED BY: M.S.	DRAWN BY: V.S.	CHECKED BY: G.M.	PREPARED BY: R.F. [Signature]	APPROVED: FOR DIV. ENGINEER: [Signature]	DATE: 1969 NOV 14
REVISION	DATE	DESCRIPTION	BY		
B	75 July 31	As Constructed	C.J.M.		
A	70 Mar 4	Added note - Detail "B"	C.J.M.		
U. S. ARMY ENGINEER DIVISION, N. P.					
PORTLAND, OREGON					
DWORSHAK DAM AND RESERVOIR					
NORTH FORK CLEARWATER RIVER, IDAHO					
POWERHOUSE					
GENERATOR BAYS—CONCRETE OUTLINE					
DRAFT TUBE BASE PLAN					
BAY 3					
SHEET 75				DWP-1-4-4/30	



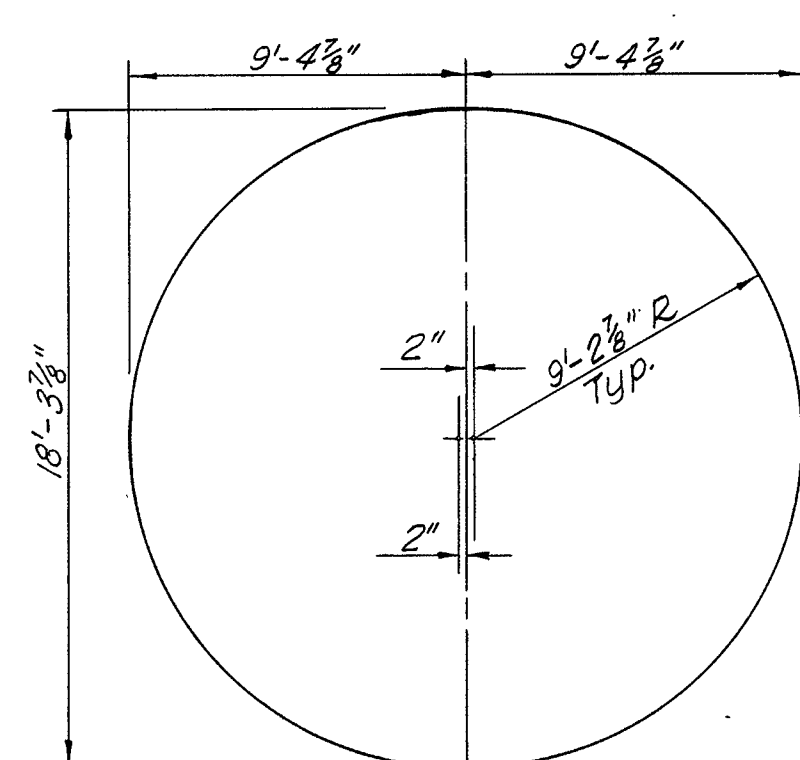
PLAN - DRAFT TUBE



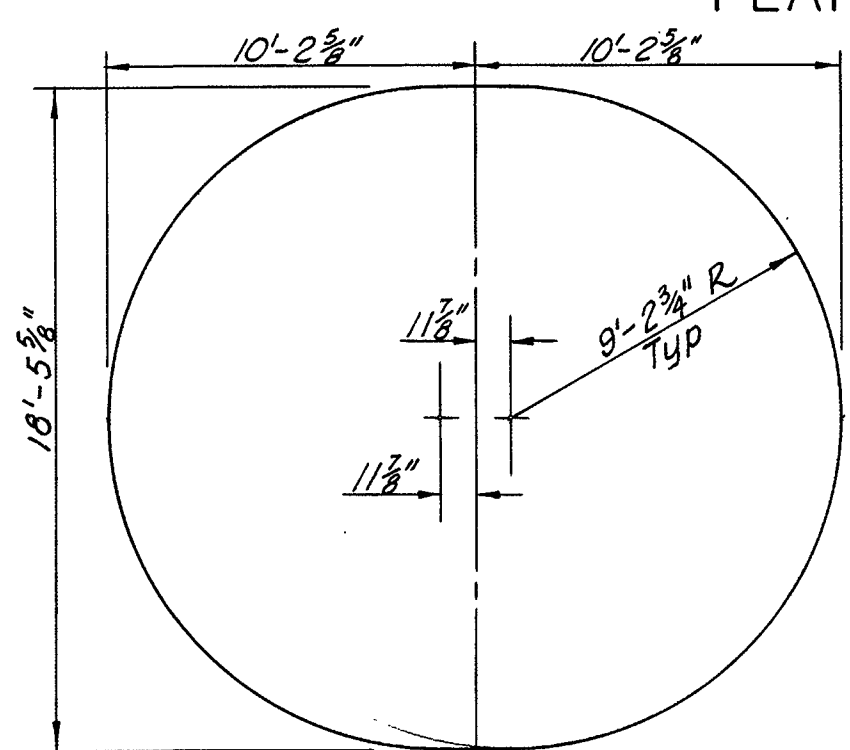
SECTION 1



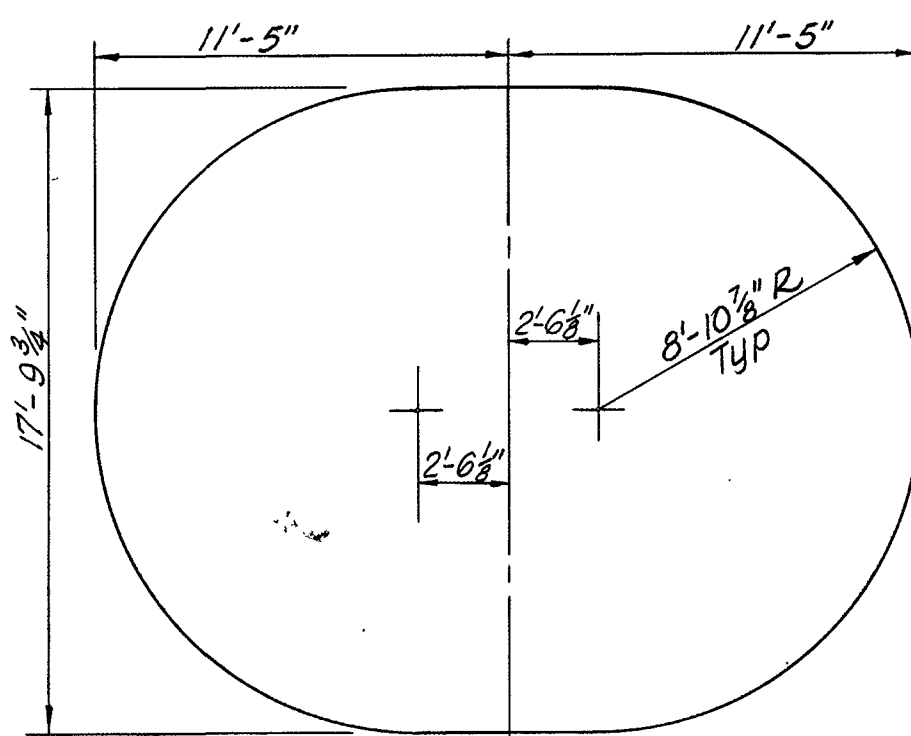
SECTION 2



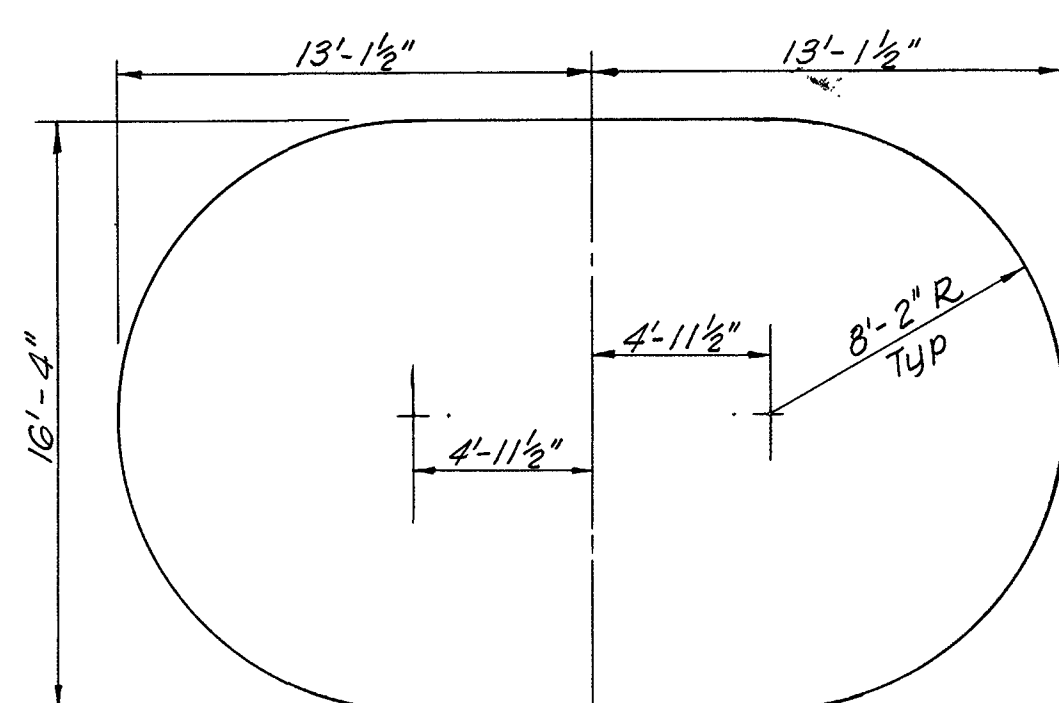
SECTION 3



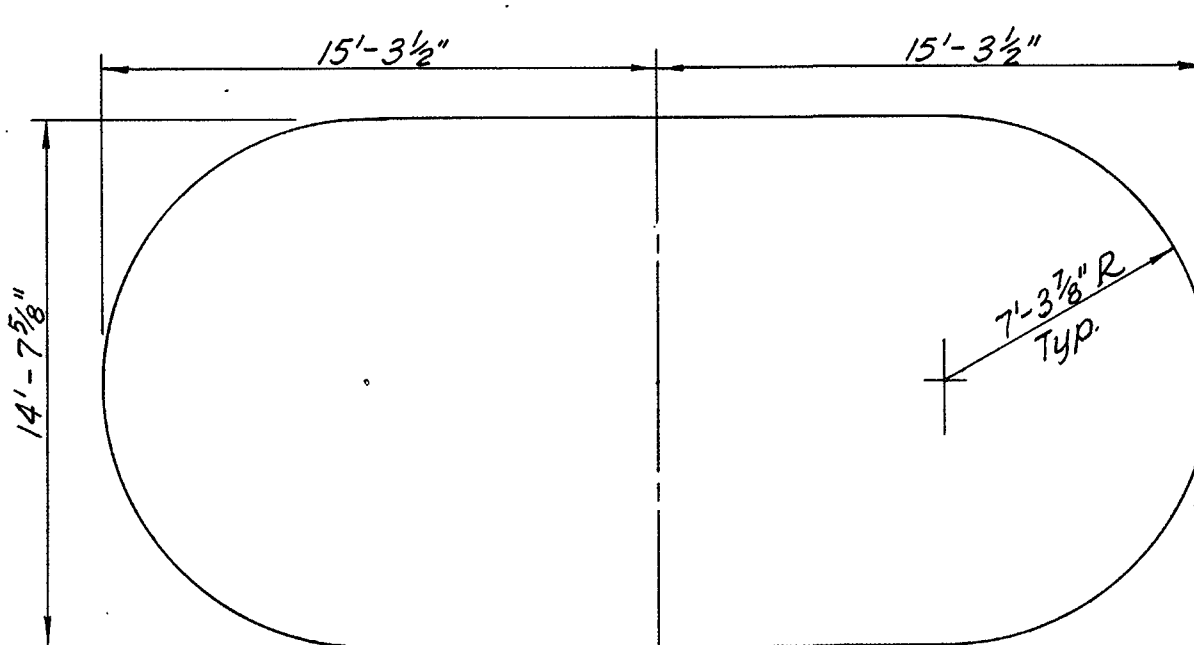
SECTION 4



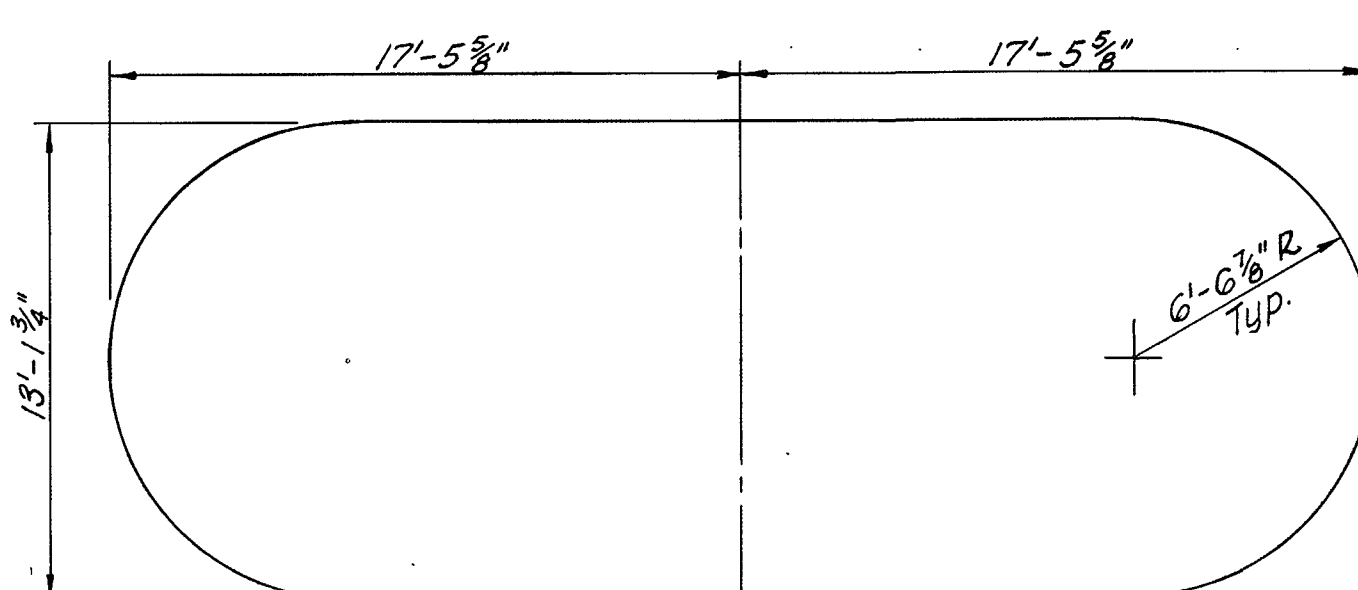
SECTION 5



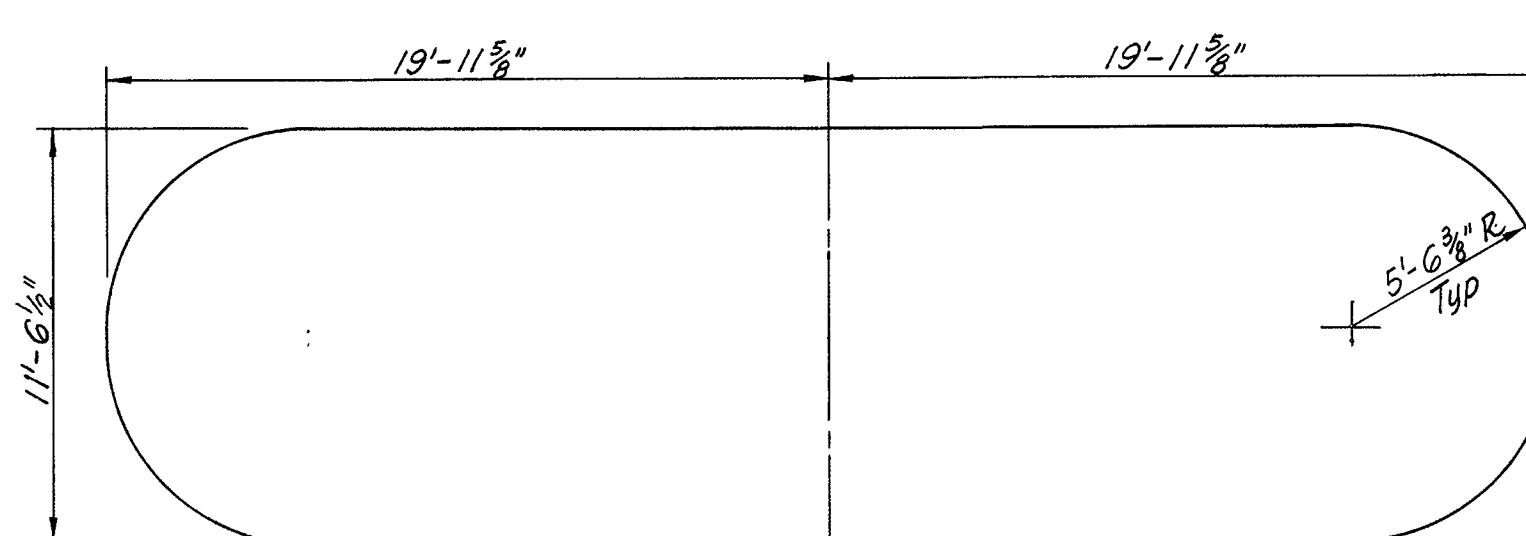
SECTION 6



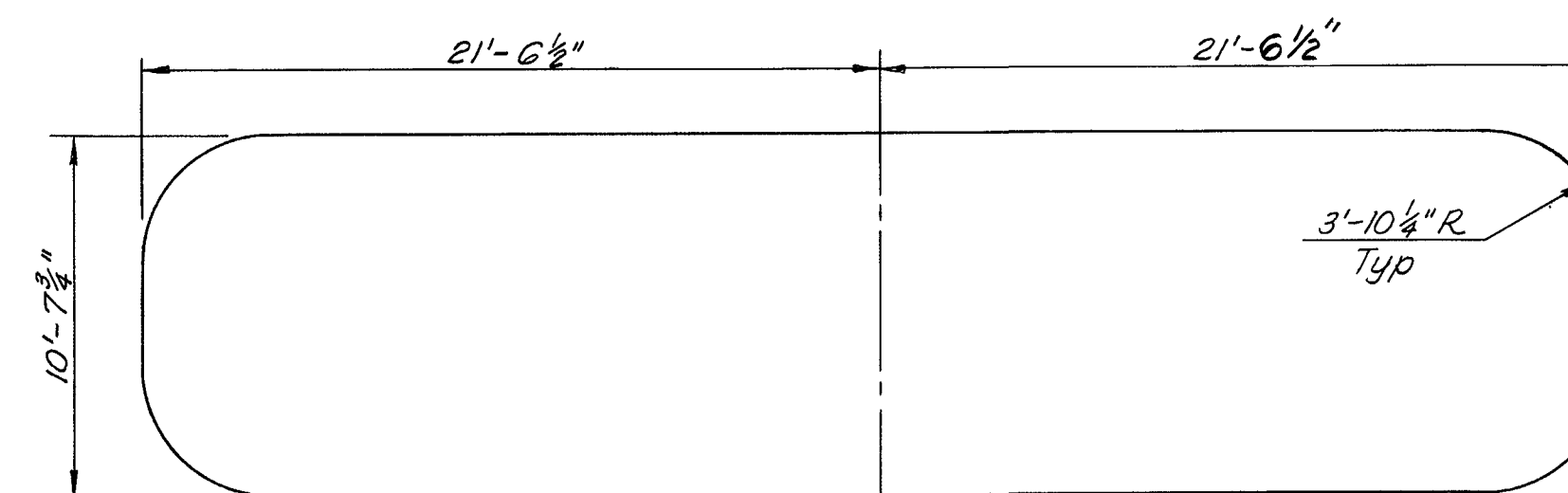
SECTION 7



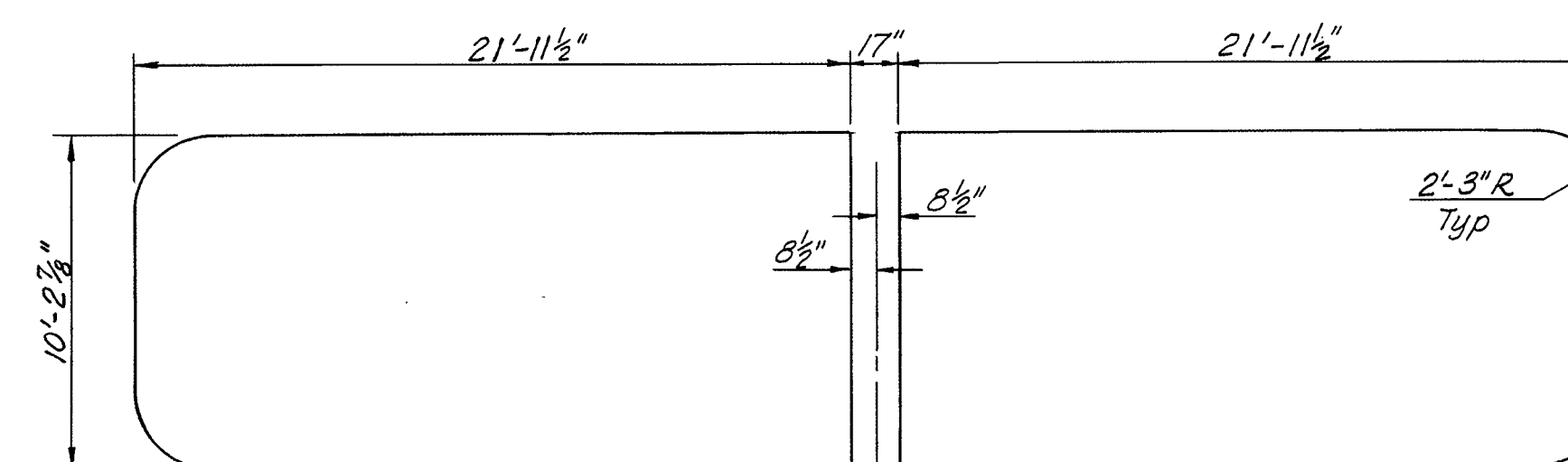
SECTION 8



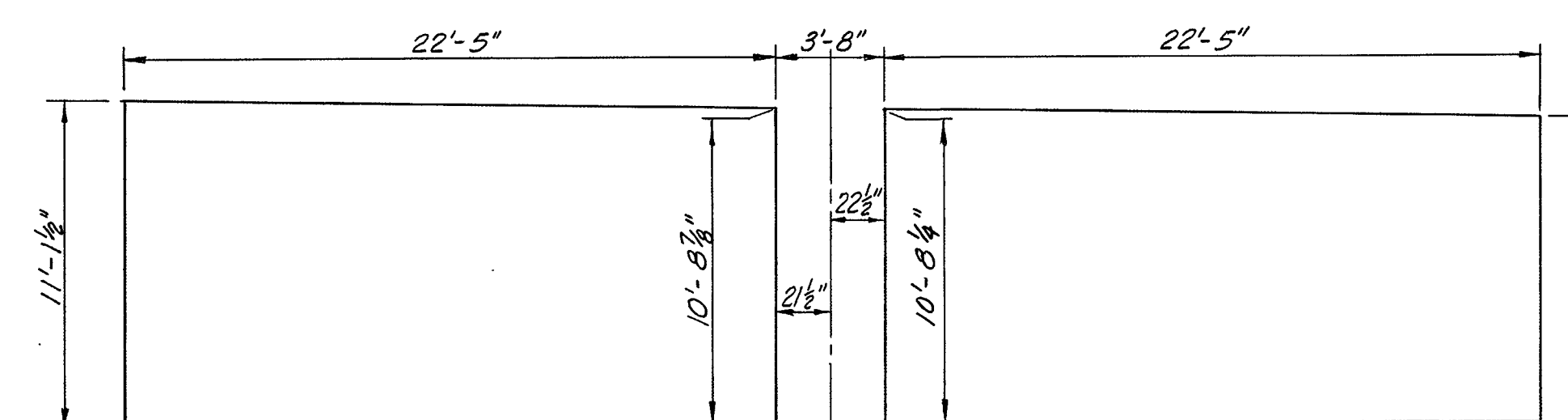
SECTION 9



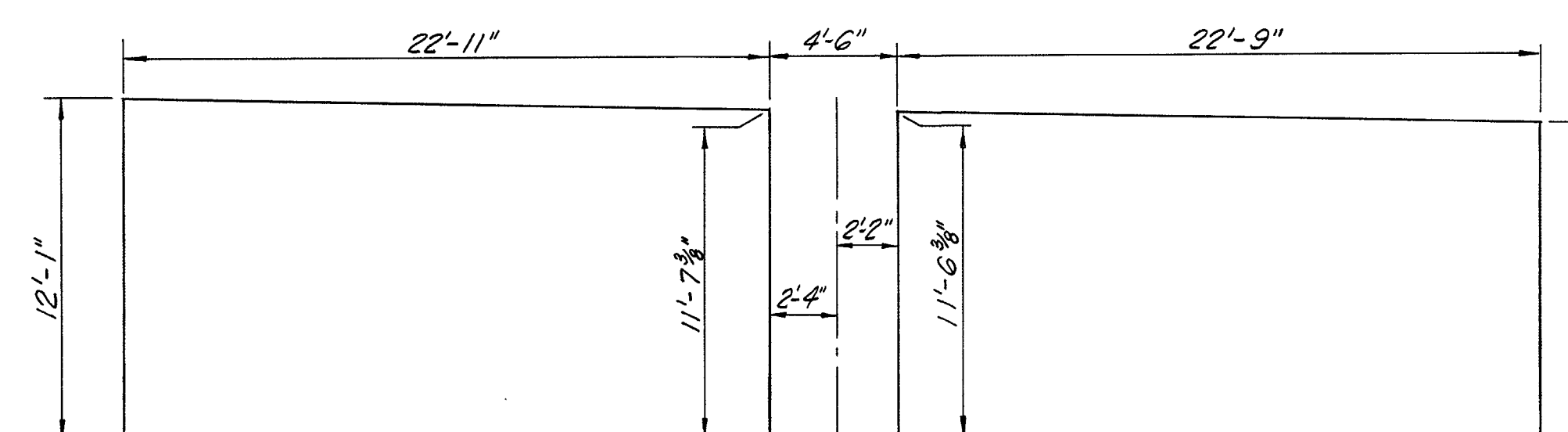
SECTION 10



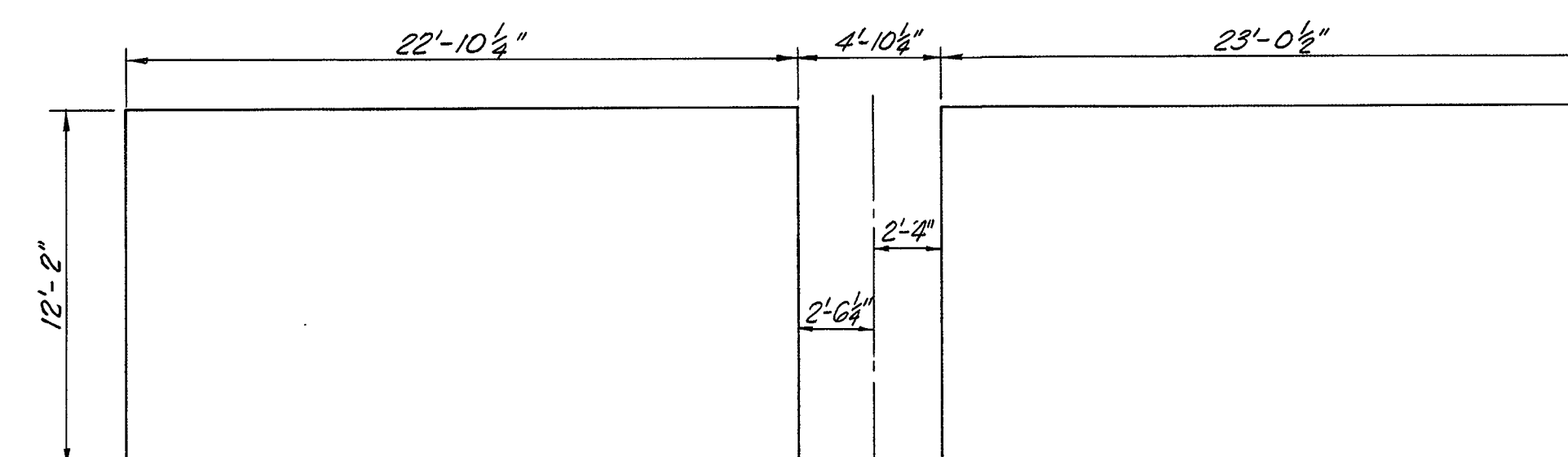
SECTION 11



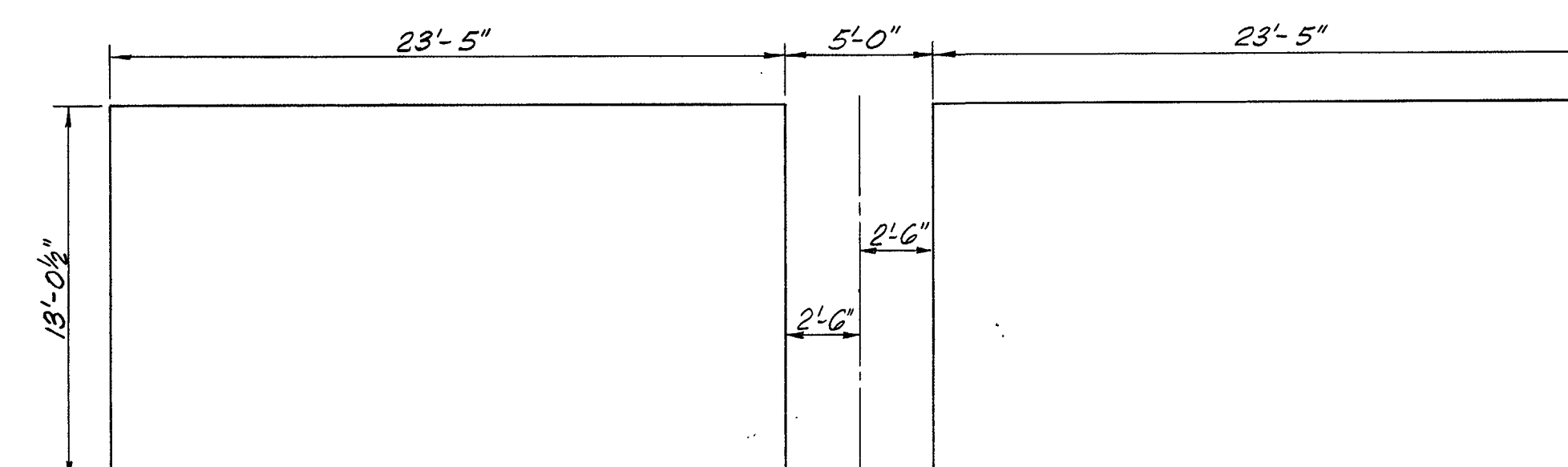
SECTION 12



SECTION 13



SECTION 14



SECTION 15

NOTES

For general notes see Dwg. DWP-1-4-2/1
All sections are shown with views looking upstream.

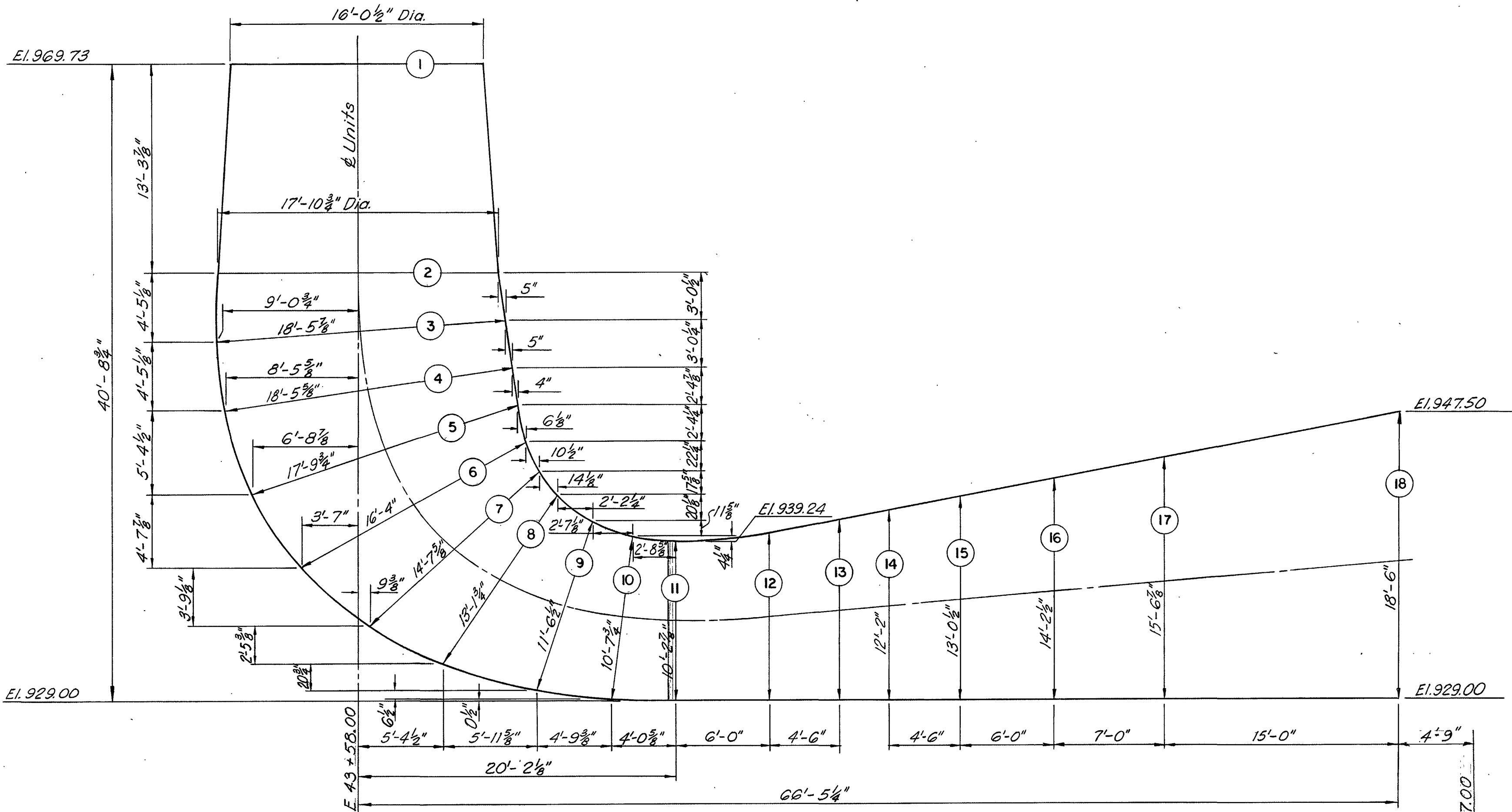
AS CONSTRUCTED
HYDRO ELECTRIC DESIGN BRANCH, NPD

DATE: 1975 JUL 31 BY: *McLennan*

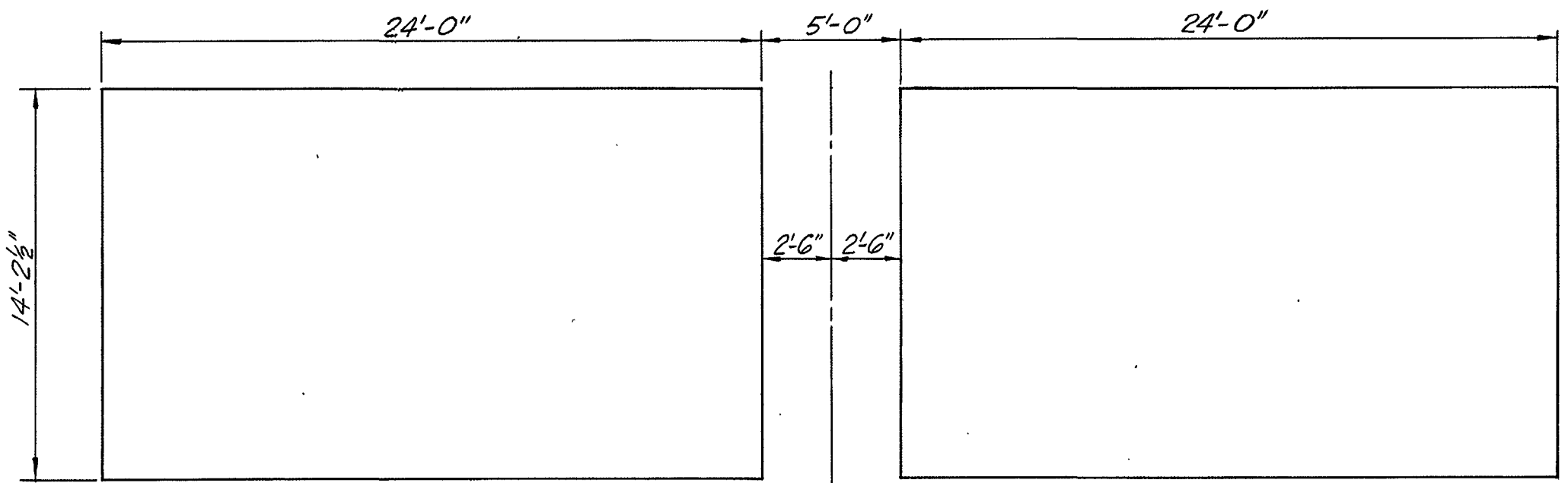
REVISION	DATE	DESCRIPTION	BY
A	75 July 31	As Constructed	CJM
U. S. ARMY ENGINEER DIVISION, N. P. PORTLAND, OREGON			
DESIGNED BY: AC			
DRAWN BY: JWN			
CHECKED BY: CJM			
PREPARED BY: <i>Kenneth H. Hinkle</i>			
SUBMITTED BY: <i>R. H. Fisher</i>			
APPROVED: FOR DIV. ENGINEER DATE: 1975 NOV 14 SCALE: 3/8" = 1'-0" SPEC. NO.			
SHEET 73			

DWORSHAK DAM AND RESERVOIR
NORTH FORK CLEARWATER RIVER, IDAHO
POWERHOUSE
GENERATOR BAYS - CONCRETE OUTLINE
DRAFT TUBE OUTLINE
BAYS 3 THRU 6 - SHEET 1

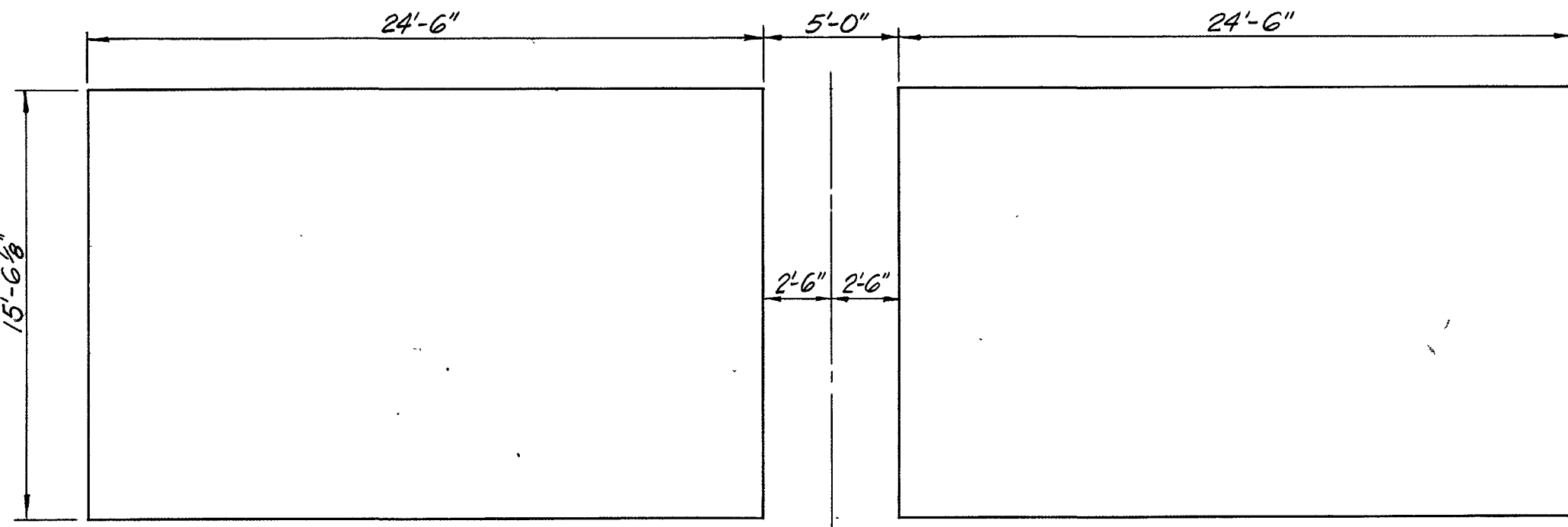
DWP-1-4-4/28



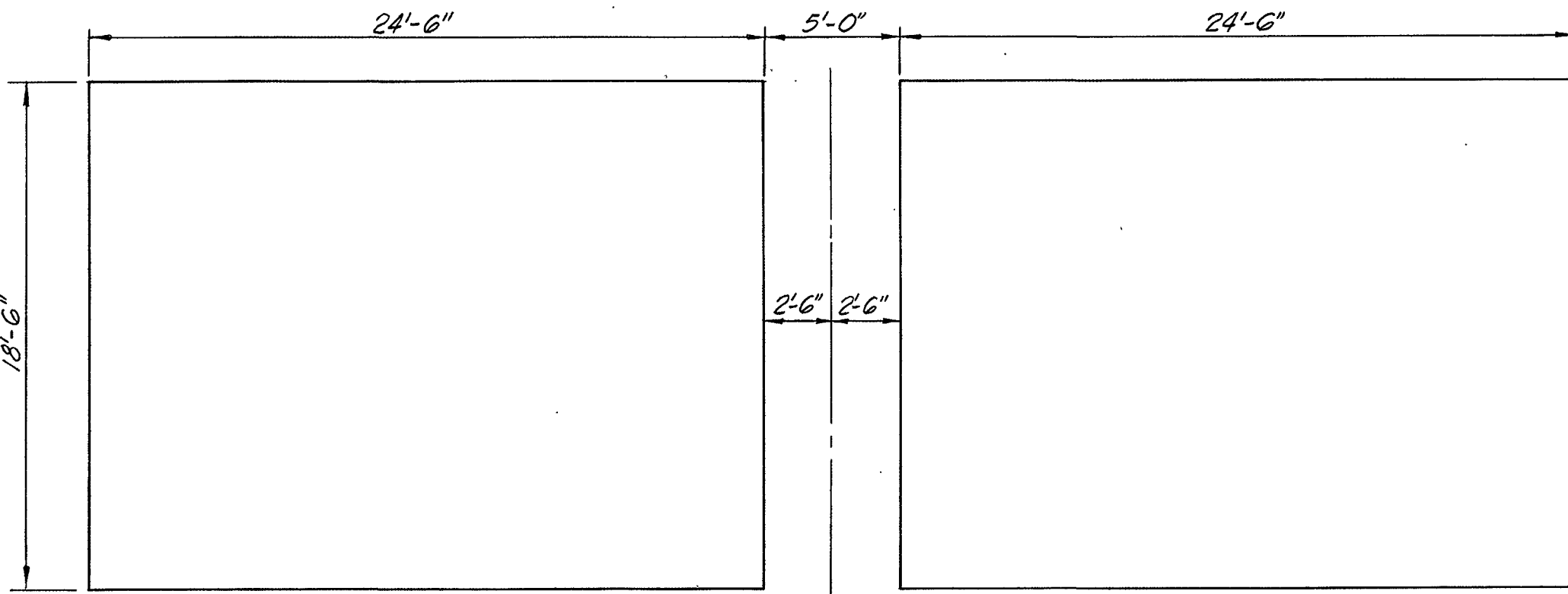
SECTION A-A



SECTION 16



SECTION 17



SECTION 18

NOTES

For general notes see Dwg. DWP-1-4-2/1
All sections are shown with views looking upstream.

AS CONSTRUCTED
HYDRO ELECTRIC DESIGN BRANCH, NPD

DATE 1975 JUL 31 BY C. J. McHown

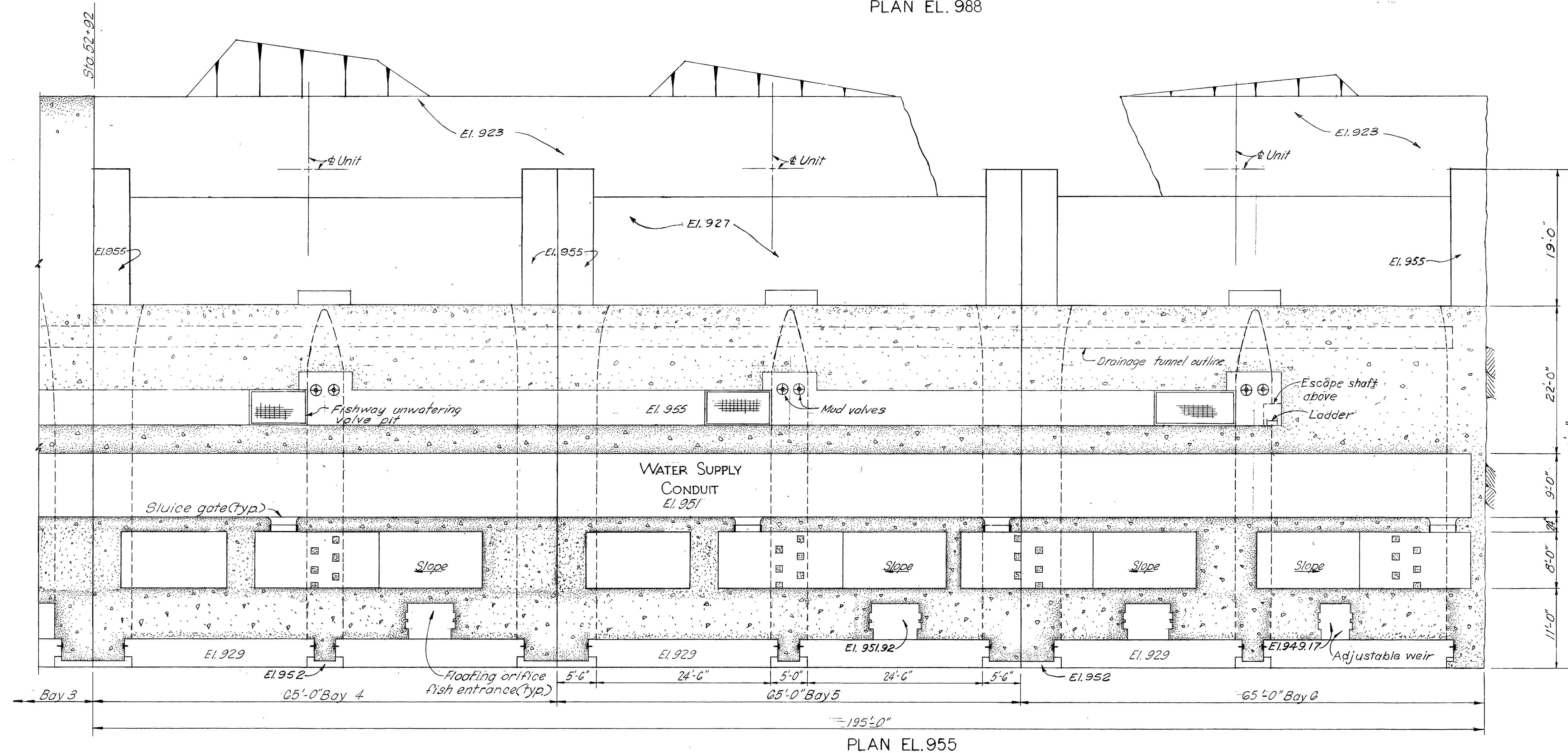
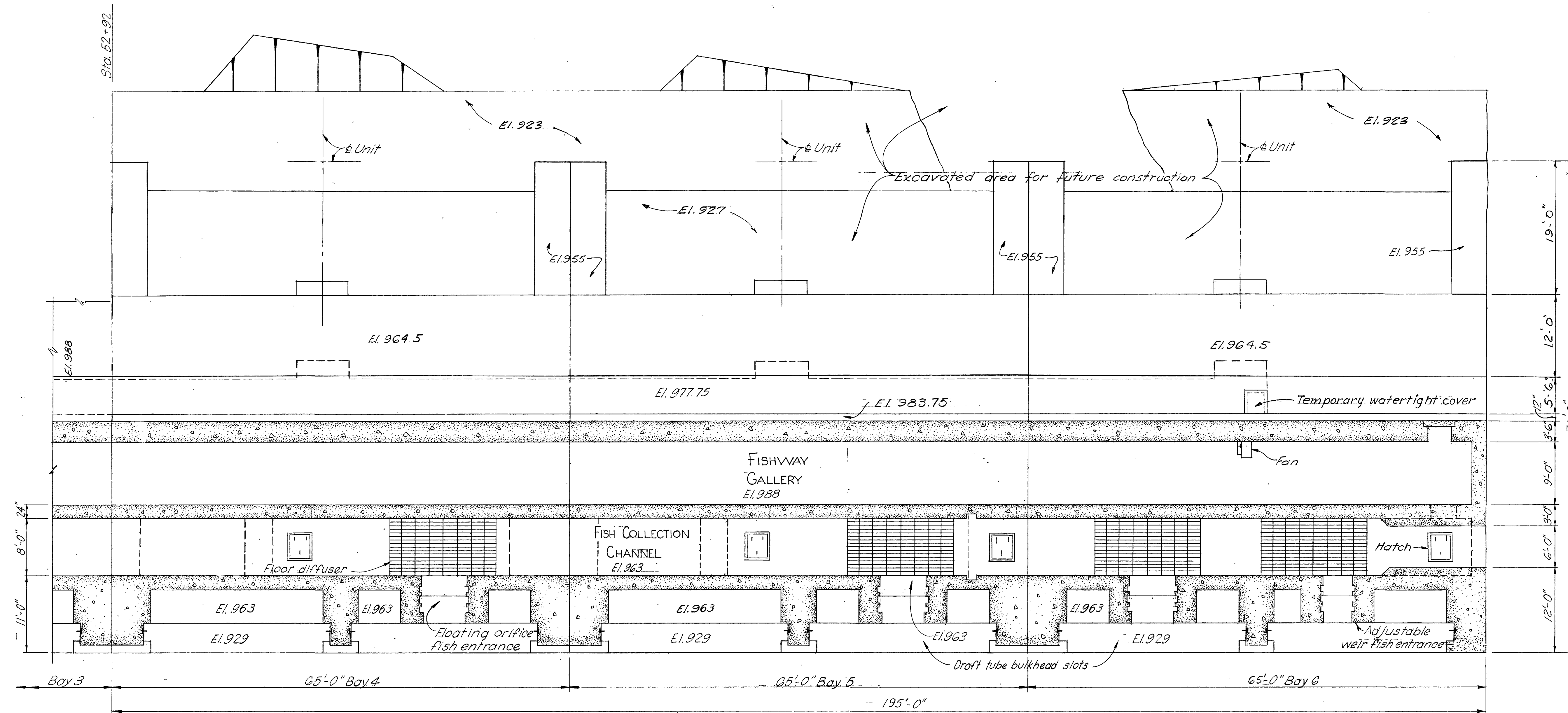
REVISION	DATE	DESCRIPTION	BY
A	25 July 31	As Constructed	CJM

U. S. ARMY ENGINEER DIVISION, N. P.
PORTLAND, OREGON

DESIGNED BY AC.....
DRAWN BY J.W.....
CHECKED BY C.J.M.....
PREPARED: Paul E. Drake
HEADQUARTERS SECTION

DWORSHAK DAM AND RESERVOIR
NORTH FORK CLEARWATER RIVER, IDAHO
POWERHOUSE
GENERATOR BAYS — CONCRETE OUTLINE
DRAFT TUBE OUTLINE
BAYS 3 THRU 6 — SHEET 2

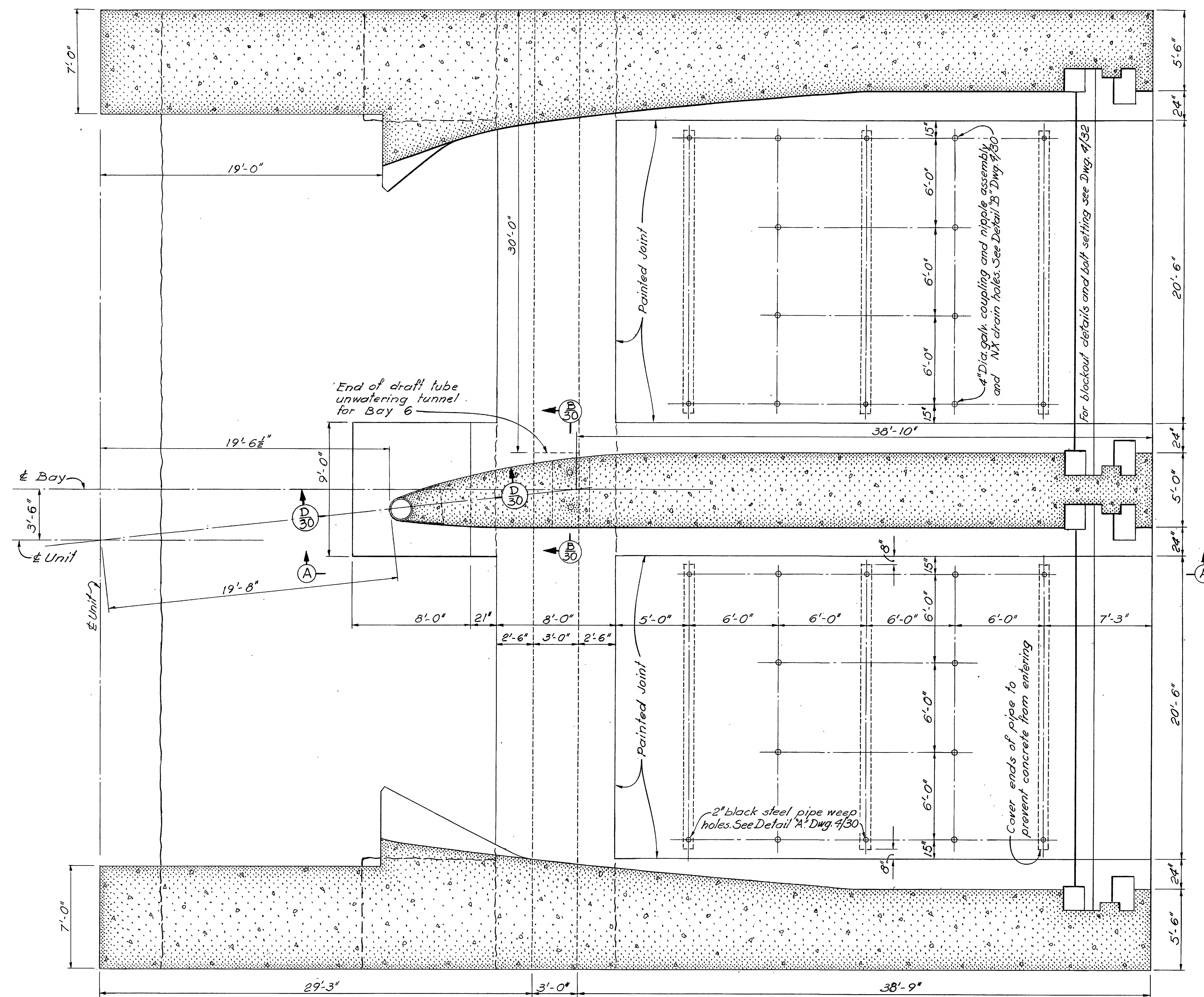
APPROVED: FOR DIV. ENGINEER DATE 30 NOV 14
CHIEF, ENGINEERING DIVISION
SCALE: 3/8" = 1'-0" SPEC. NO.
SHEET 74 DWP-1-4-4/29



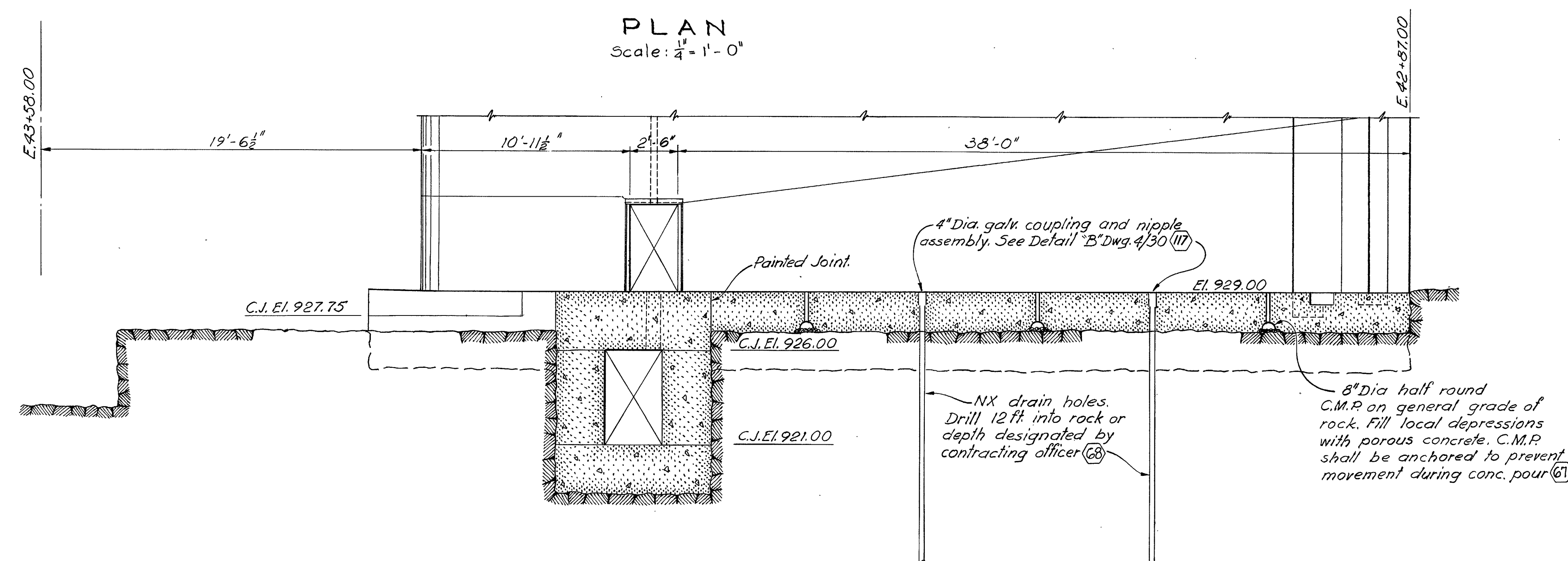
AS CONSTRUCTED
HYDRO ELECTRIC DESIGN BRANCH, NPD

DATE 1975 JUL 31 BY *C. J. McLean*

REVISION	DATE	DESCRIPTION	BY
A	75 July 31	As Constructed	C.J.M.
U. S. ARMY ENGINEER DIVISION, N. P. PORTLAND, OREGON			
DESIGNED BY <i>M.L.S.</i>			
DRAWN BY <i>PER</i>			
CHECKED BY <i>C.L.M.</i>			
PREPARED BY <i>Raul H. Diaz</i> CHIEF, HYDRO-ELECTRIC DESIGN BRANCH			
SUBMITTED BY <i>L. F. Fisher</i> CHIEF, HYDRO-ELECTRIC DESIGN BRANCH			
APPROVED FOR DIV. ENGINEER DATE 75 NOV 14 CHIEF, ENGINEERING DIVISION SCALE: 1/8" = 1'-0" SPEC. NO.			
SHEET 13 DWP-6-O-0/6			



PLAN
Scale: $\frac{1}{4}'' = 1' - 0''$



SECTION A-A
Scale: $\frac{1}{4}'' = 1' - 0''$

NOTES

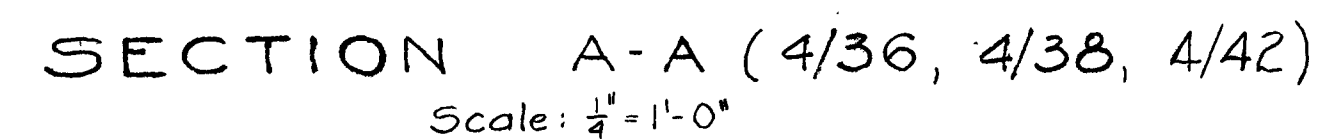
For general notes see Dwg. DWP-1-4-2/1
Slabs to be float finished.

AS CONSTRUCTED

HYDRO ELECTRIC DESIGN BRANCH, NPD

DATE 1975 JUL 31 BY C. J. M. Gower

REVISION	DATE	DESCRIPTION	BY
A	75 July 31	As Constructed	C.J.M.
U. S. ARMY ENGINEER DIVISION, N. P. PORTLAND, OREGON			
DESIGNED BY: MJS.			
DRAWN BY: K.S.			
CHECKED BY: C.J.M.			
PREPARED BY: R. E. V. (11)			
SUBMITTED BY: L. E. (11)			
CHIEF, HYDRO-ELECTRIC DESIGN BRANCH			
APPROVED FOR DIV. ENGINEER DATE 1969 NOV. 14			
CHIEF, ENGINEERING DIVISION			
SCALE AS SHOWN SPEC. NO.			
DWP-1-4-4/31			
SHEET 76			



For general notes see Dwg. DWP-1-4-2/1
Additional concrete required because of over excavation
and the areas enclosed by the symbol  will be paid
for under item (57) or (57b); all other concrete will be paid for
under item (56).
Draft tube pier nosings shall be painted system No. 4

DATE 1975 JUL 31 BY C. C. McGowan

	B	75 July 31 75 Jan 30	As Constructed Minor revision	C.J.M. C.J.M.
REVISION	DATE	DESCRIPTION		BY
U. S. ARMY ENGINEER DIVISION, N. P. PORTLAND, OREGON				
DESIGNED BY: M.W.S.		DWORSHAK DAM AND RESERVOIR		
DRAWN BY: V.S.		NORTH FORK CLEARWATER RIVER, IDAHO		
CHECKED BY: C.J.M.		POWERHOUSE		
PREPARED: <i>Pamela Deane</i>		GENERATOR BAYS - CONCRETE OUTLINE		
HEAD, STRUCTURAL SECTION		TRANSVERSE SECTION		
		BAYS 4, 5 & 6		
SUBMITTED:	<i>E. Fisher</i>		APPROVED FOR DIV. ENGINEER	DATE: 1959 NOV 18
CHIEF, HYDRO-ELECTRIC DESIGN BRANCH		CHIEF, ENGINEERING DIVISION		
		SCALE AS SHOWN	SPEC. NO.	
		SHEET 71 DWP-1-a-4-26		